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张宇考研数学系列·一

考研数学基础30讲
高等数学分册

核心计算通关讲义

主　编　张　宇
副主编　高昆轮

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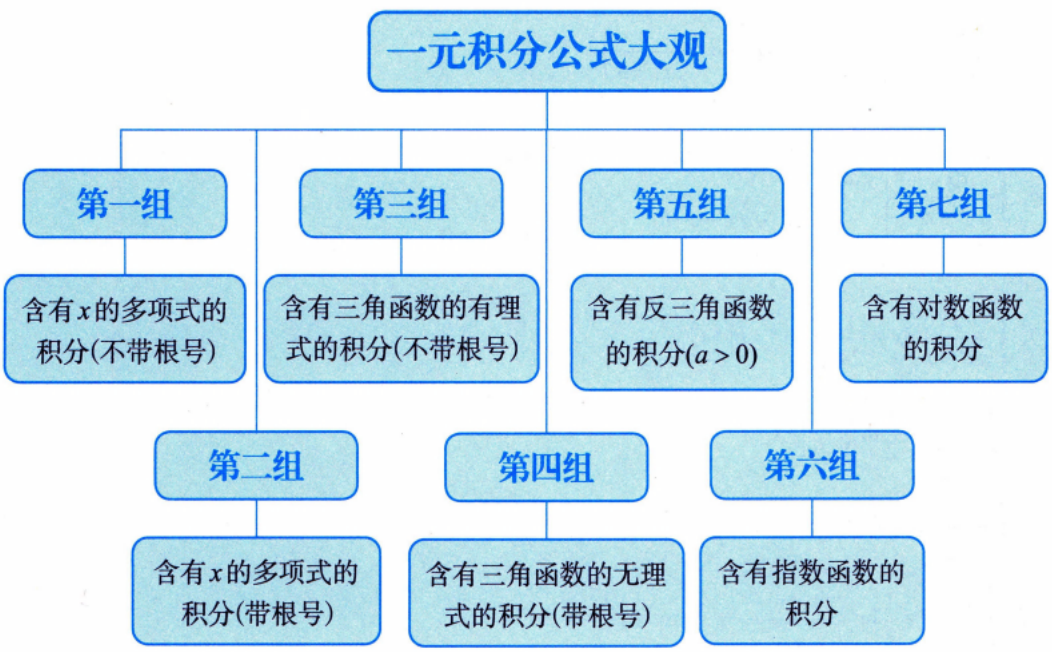

本部分又可称为“一元积分公式大观”，专门整理了可以手算的常用积分公式，有两个目的：

一、用于提高积分计算能力．这里几乎囊括了（部分需要经过简单变形或换元）积分题的积分方法和原函数，将这些积分全部演算一遍，可使积分计算能力达到炉火纯青、登峰造极的水平，这是数学计算能力的重要衡量指标．再往深走，基本都要借助其他方法，或者根本算不出初等函数形式的原函数了．

二、用于查找积分结果．随着现代技术的发展，寻找原函数早已不是主流的方法，今后的学习或者研究中，对于积分结果的获取，可以寻求两种帮助，一是查积分表，本部分内容只是一个有初等函数形式的简单积分表，国内外有不少积分结果汇编成册的资料，以后各位要用时，查表即可．二是用数值计算等方法，绕开原函数，甚至绕开分析学，都有很好的办法达到目的，这些方法就请考生在考研后自行研究了，那是更为广阔的天地．

所有积分公式的计算过程附于本部分最后，供各位参考，希望大家先不看答案，自行尝试解决．同样地，本部分的公式要常看，常练，以巩固掌握．

第一组

“含有 x 的多项式的积分（不带根号）”．

1. $\int \frac{\mathrm{d}x}{ax+b}=\frac{1}{a}\ln|ax+b|+C$．

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2. $\int (ax+b)^{\mu}\mathrm{d}x=\frac{1}{a(\mu+1)}(ax+b)^{\mu+1}+C,\ \mu\neq -1.$

3. $\int \frac{x}{ax+b}\mathrm{d}x=\frac{1}{a^2}(ax-b\ln|ax+b|)+C.$

4. $\int \frac{\mathrm{d}x}{x(ax+b)}=-\frac{1}{b}\ln\left|\frac{ax+b}{x}\right|+C.$

5. $\int \frac{x}{(ax+b)^2}\mathrm{d}x=\frac{1}{a^2}\left(\ln|ax+b|+\frac{b}{ax+b}\right)+C.$

6. $\int \frac{\mathrm{d}x}{ax^2+b}=\begin{cases}\frac{1}{\sqrt{ab}}\arctan\sqrt{\frac{a}{b}}x+C, & a>0,\ b>0,\\ \frac{1}{2\sqrt{-ab}}\ln\left|\frac{\sqrt{a}x-\sqrt{-b}}{\sqrt{a}x+\sqrt{-b}}\right|+C, & a>0,\ b<0.\end{cases}$

7. $\int \frac{x}{ax^2+b}\mathrm{d}x=\frac{1}{2a}\ln|ax^2+b|+C.$

8. $\int \frac{\mathrm{d}x}{1-x^2}=\frac{1}{2}\ln\left|\frac{1+x}{1-x}\right|+C.$

9. $\int \frac{x}{1+x^2}\mathrm{d}x=\frac{1}{2}\ln(1+x^2)+C.$

10. $\int \frac{\mathrm{d}x}{1+x^3}=\frac{1}{3}\ln\frac{|1+x|}{\sqrt{1-x+x^2}}+\frac{1}{\sqrt{3}}\arctan\frac{2x-1}{\sqrt{3}}+C.$

11. $\int \frac{\mathrm{d}x}{1-x^3}=\frac{1}{3}\ln\frac{\sqrt{1+x+x^2}}{|1-x|}+\frac{1}{\sqrt{3}}\arctan\frac{2x+1}{\sqrt{3}}+C.$

12. $\int \frac{\mathrm{d}x}{1+x^4}=\frac{\sqrt{2}}{4}\arctan\frac{\sqrt{2}(x^2-1)}{2x}-\frac{\sqrt{2}}{8}\ln\frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x+1}+C.$

13. $\int \frac{\mathrm{d}x}{1-x^4}=\frac{1}{4}\ln\left|\frac{1+x}{1-x}\right|+\frac{1}{2}\arctan x+C.$

14. $\int \frac{x}{1+x^3}\mathrm{d}x=\frac{1}{6}\ln\frac{x^2-x+1}{(1+x)^2}+\frac{1}{\sqrt{3}}\arctan\frac{2x-1}{\sqrt{3}}+C.$

15. $\int \frac{x}{1-x^3}\mathrm{d}x=\frac{1}{6}\ln\frac{1+x+x^2}{(1-x)^2}-\frac{1}{\sqrt{3}}\arctan\frac{2x+1}{\sqrt{3}}+C.$

16. $\int \frac{x}{1+x^4}\mathrm{d}x=\frac{1}{2}\arctan x^2+C.$

17. $\int \frac{x}{1-x^4}\mathrm{d}x=\frac{1}{4}\ln\left|\frac{1+x^2}{1-x^2}\right|+C.$

18. $\int \frac{\mathrm{d}x}{c^2+x^2}=\frac{1}{c}\arctan\frac{x}{c}+C.$

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19. $\int \frac{\mathrm{d}x}{(c^2+x^2)^2}=\frac{1}{2c^3}\left(\frac{cx}{c^2+x^2}+\arctan\frac{x}{c}\right)+C$.

20. $\int \frac{\mathrm{d}x}{(c^2+x^2)^3}=\frac{1}{8c^5}\left(\frac{cx}{c^2+x^2}+3\arctan\frac{x}{c}\right)-\frac{1}{4x(c^2+x^2)^2}+\frac{1}{4c^4x}+C$.

21. $\int \frac{x}{c^2+x^2}\mathrm{d}x=\frac{1}{2}\ln(c^2+x^2)+C$.

22. $\int \frac{x}{(c^2+x^2)^2}\mathrm{d}x=-\frac{1}{2(c^2+x^2)}+C$.

23. $\int \frac{x^2}{c^2+x^2}\mathrm{d}x=x-c\arctan\frac{x}{c}+C$.

一手+V：kaoyan33311

24. $\int \frac{x^2}{(c^2+x^2)^2}\mathrm{d}x=-\frac{x}{2(c^2+x^2)}+\frac{1}{2c}\arctan\frac{x}{c}+C$.

25. $\int \frac{x^3}{c^2+x^2}\mathrm{d}x=\frac{x^2}{2}-\frac{c^2}{2}\ln(c^2+x^2)+C$.

26. $\int \frac{x^3}{(c^2+x^2)^2}\mathrm{d}x=\frac{c^2}{2(c^2+x^2)}+\frac{1}{2}\ln(c^2+x^2)+C$.

27. $\int \frac{\mathrm{d}x}{x(c^2+x^2)}=\frac{1}{2c^2}\ln\frac{x^2}{c^2+x^2}+C$.

28. $\int \frac{\mathrm{d}x}{x(c^2+x^2)^2}=\frac{1}{2c^2(c^2+x^2)}+\frac{1}{2c^4}\ln\frac{x^2}{c^2+x^2}+C$.

29. $\int \frac{\mathrm{d}x}{x^2(c^2+x^2)}=-\frac{1}{c^2x}-\frac{1}{c^3}\arctan\frac{x}{c}+C$.

30. $\int \frac{\mathrm{d}x}{x^2(c^2+x^2)^2}=-\frac{1}{c^4x}-\frac{x}{2c^4(c^2+x^2)}-\frac{3}{2c^5}\arctan\frac{x}{c}+C$.

31. $\int \frac{\mathrm{d}x}{c^2-x^2}=\frac{1}{2c}\ln\left|\frac{c+x}{c-x}\right|+C$.

32. $\int \frac{\mathrm{d}x}{(c^2-x^2)^2}=\frac{x}{2c^2(c^2-x^2)}+\frac{1}{4c^3}\ln\left|\frac{c+x}{c-x}\right|+C$.

33. $\int \frac{x^2}{c^2-x^2}\mathrm{d}x=-x+\frac{c}{2}\ln\left|\frac{c+x}{c-x}\right|+C$.

34. $\int \frac{x^2}{(c^2-x^2)^2}\mathrm{d}x=\frac{x}{2(c^2-x^2)}-\frac{1}{4c}\ln\left|\frac{c+x}{c-x}\right|+C$.

35. $\int \frac{\mathrm{d}x}{x^2(c^2-x^2)}=-\frac{1}{c^2x}+\frac{1}{2c^3}\ln\left|\frac{c+x}{c-x}\right|+C$.

36. $\int \frac{\mathrm{d}x}{x^2-c^2}=\frac{1}{2c}\ln\left|\frac{x-c}{x+c}\right|+C$.

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37. $\int \frac{x}{x^2-c^2}\mathrm{d}x=\frac{1}{2}\ln\left|x^2-c^2\right|+C$.

38. $\int \frac{a+x}{a^3+x^3}\mathrm{d}x=\frac{2}{\sqrt{3}a}\arctan\frac{2x-a}{\sqrt{3}a}+C$.

39. $\int \frac{a+x}{a^3-x^3}\mathrm{d}x=-\frac{1}{3a}\ln\frac{(a-x)^2}{x^2+ax+a^2}+C$.

40. $\int \frac{a-x}{a^3+x^3}\mathrm{d}x=\frac{1}{3a}\ln\frac{(a+x)^2}{x^2-ax+a^2}+C$.

41. $\int \frac{a-x}{a^3-x^3}\mathrm{d}x=\frac{2}{\sqrt{3}a}\arctan\frac{2x+a}{\sqrt{3}a}+C$.

42. $\int \frac{\mathrm{d}x}{ax^2+bx+c}=\begin{cases}\frac{1}{\sqrt{b^2-4ac}}\ln\left|\frac{2ax+b-\sqrt{b^2-4ac}}{2ax+b+\sqrt{b^2-4ac}}\right|+C, & b^2>4ac,\\ \frac{2}{\sqrt{4ac-b^2}}\arctan\frac{2ax+b}{\sqrt{4ac-b^2}}+C, & b^2<4ac,\\ -\frac{2}{2ax+b}+C, & b^2=4ac.\end{cases}$

第二组

“含有 x 的多项式的积分（带根号）”.

在以下含有 $\sqrt{a^2\pm x^2}$，$\sqrt{x^2\pm a^2}$ 类似的式子中，均默认 $a>0$.

1. $\int \sqrt{ax+b}\,\mathrm{d}x=\frac{2}{3a}\sqrt{(ax+b)^3}+C$.

2. $\int x\sqrt{ax+b}\,\mathrm{d}x=\frac{2}{15a^2}(3ax-2b)\sqrt{(ax+b)^3}+C$.

3. $\int \frac{\mathrm{d}x}{x\sqrt{ax+b}}=\begin{cases}\frac{1}{\sqrt{b}}\ln\left|\frac{\sqrt{ax+b}-\sqrt{b}}{\sqrt{ax+b}+\sqrt{b}}\right|+C, & b>0,\\ \frac{2}{\sqrt{-b}}\arctan\sqrt{\frac{ax+b}{-b}}+C, & b<0.\end{cases}$

4. $\int \frac{\sqrt{ax+b}}{x^2}\mathrm{d}x=-\frac{\sqrt{ax+b}}{x}+\frac{a}{2}\int \frac{\mathrm{d}x}{x\sqrt{ax+b}}$.

5. $\int \frac{x}{\sqrt{x^2-a^2}}\mathrm{d}x=\sqrt{x^2-a^2}+C$.

6. $\int \frac{x}{\sqrt{(x^2-a^2)^3}}\mathrm{d}x=-\frac{1}{\sqrt{x^2-a^2}}+C$.

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7. $\int \frac{x^2}{\sqrt{x^2-a^2}}\mathrm{d}x=\frac{x}{2}\sqrt{x^2-a^2}+\frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right|+C$.

8. $\int \frac{\mathrm{d}x}{x\sqrt{x^2-a^2}}=\frac{1}{a}\arccos\frac{a}{|x|}+C$.

9. $\int \frac{\mathrm{d}x}{x^2\sqrt{x^2-a^2}}=\frac{\sqrt{x^2-a^2}}{a^2x}+C$.

10. $\int x\sqrt{x^2-a^2}\,\mathrm{d}x=\frac{1}{3}\sqrt{(x^2-a^2)^3}+C$.

11. $\int \frac{\sqrt{x^2-a^2}}{x}\mathrm{d}x=\sqrt{x^2-a^2}-a\arccos\frac{a}{|x|}+C$.

12. $\int \frac{\sqrt{x^2-a^2}}{x^2}\mathrm{d}x=-\frac{\sqrt{x^2-a^2}}{x}+\ln\left|x+\sqrt{x^2-a^2}\right|+C$.

13. $\int \frac{\mathrm{d}x}{\sqrt{(a^2-x^2)^3}}=\frac{x}{a^2\sqrt{a^2-x^2}}+C$.

14. $\int \frac{x}{\sqrt{a^2-x^2}}\mathrm{d}x=-\sqrt{a^2-x^2}+C$.

15. $\int \frac{x}{\sqrt{(a^2-x^2)^3}}\mathrm{d}x=\frac{1}{\sqrt{a^2-x^2}}+C$.

16. $\int \frac{x^2}{\sqrt{a^2-x^2}}\mathrm{d}x=-\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\arcsin\frac{x}{a}+C$.

17. $\int \frac{x^2}{\sqrt{(a^2-x^2)^3}}\mathrm{d}x=\frac{x}{\sqrt{a^2-x^2}}-\arcsin\frac{x}{a}+C$.

18. $\int \frac{\mathrm{d}x}{x^2\sqrt{a^2-x^2}}=-\frac{\sqrt{a^2-x^2}}{a^2x}+C$.

19. $\int x\sqrt{a^2-x^2}\,\mathrm{d}x=-\frac{1}{3}\sqrt{(a^2-x^2)^3}+C$.

20. $\int \frac{\sqrt{a^2-x^2}}{x^2}\mathrm{d}x=-\frac{\sqrt{a^2-x^2}}{x}-\arcsin\frac{x}{a}+C$.

21. $\int \sqrt{(a^2-x^2)^3}\mathrm{d}x=\frac{1}{4}\left[x\sqrt{(a^2-x^2)^3}+\frac{3a^2x}{2}\sqrt{a^2-x^2}+\frac{3a^4}{2}\arcsin\frac{x}{a}\right]+C$.

22. $\int x^2\sqrt{a^2-x^2}\,\mathrm{d}x=-\frac{x}{4}\sqrt{(a^2-x^2)^3}+\frac{a^2}{8}\left(x\sqrt{a^2-x^2}+a^2\arcsin\frac{x}{a}\right)+C$.

23. $\int \sqrt{2ax-x^2}\mathrm{d}x=\frac{1}{2}\left[(x-a)\sqrt{2ax-x^2}+a^2\arcsin\frac{x-a}{a}\right]+C$.

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24. $\int \frac{\mathrm{d}x}{\sqrt{2ax-x^2}} = -\arccos\frac{x-a}{a} + C = \arcsin\frac{x-a}{a} + C$.

25. $\int \frac{\mathrm{d}x}{\sqrt{2ax+x^2}} = \ln\left|x+a+\sqrt{2ax+x^2}\right| + C$.

26. $\int \frac{\mathrm{d}x}{x\sqrt{x^n+a^2}} = \frac{2}{na}\ln\frac{\sqrt{x^n+a^2}-a}{\sqrt{x^n}} + C$.

27. $\int \frac{\mathrm{d}x}{x\sqrt{x^n-a^2}} = -\frac{2}{na}\arcsin\frac{a}{\sqrt{x^n}} + C$.

28. $\int \sqrt{\frac{x}{a^3-x^3}}\,\mathrm{d}x = \frac{2}{3}\arcsin\left(\frac{x}{a}\right)^{\frac{3}{2}} + C$.

29. $\int \sqrt{\frac{a+bx}{a-bx}}\,\mathrm{d}x = -\frac{1}{b}\sqrt{(a+bx)(a-bx)} + \frac{a}{b}\arcsin\frac{bx}{a} + C$.

30. $\int \sqrt{\frac{a-bx}{a+bx}}\,\mathrm{d}x = \frac{1}{b}\sqrt{(a+bx)(a-bx)} + \frac{a}{b}\arcsin\frac{bx}{a} + C$.

31. $\int \sqrt{\frac{1+x}{1-x}}\,\mathrm{d}x = \arcsin x - \sqrt{1-x^2} + C$.

32. $\int \frac{\mathrm{d}x}{\sqrt{x^2+a^2}} = \ln(x+\sqrt{x^2+a^2}) + C$.

33. $\int \frac{\mathrm{d}x}{\sqrt{(x^2+a^2)^3}} = \frac{x}{a^2\sqrt{x^2+a^2}} + C$.

34. $\int \frac{x}{\sqrt{x^2+a^2}}\,\mathrm{d}x = \sqrt{x^2+a^2} + C$.

35. $\int \frac{x}{\sqrt{(x^2+a^2)^3}}\,\mathrm{d}x = -\frac{1}{\sqrt{x^2+a^2}} + C$.

36. $\int \frac{x^2}{\sqrt{x^2+a^2}}\,\mathrm{d}x = \frac{x}{2}\sqrt{x^2+a^2} - \frac{a^2}{2}\ln(x+\sqrt{x^2+a^2}) + C$.

37. $\int \frac{\mathrm{d}x}{x^2\sqrt{x^2+a^2}} = -\frac{\sqrt{x^2+a^2}}{a^2x} + C$.

38. $\int \sqrt{x^2+a^2}\,\mathrm{d}x = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\ln(x+\sqrt{x^2+a^2}) + C$.

39. $\int x\sqrt{x^2+a^2}\,\mathrm{d}x = \frac{1}{3}\sqrt{(x^2+a^2)^3} + C$.

40. $\int \frac{\sqrt{x^2+a^2}}{x^2}\,\mathrm{d}x = -\frac{\sqrt{x^2+a^2}}{x} + \ln(x+\sqrt{x^2+a^2}) + C$.

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41. $\int \frac{\mathrm{d}x}{\sqrt{ax^2+bx+c}}=\begin{cases}\frac{1}{\sqrt{a}}\ln\left|2ax+b+2\sqrt{a}\sqrt{ax^2+bx+c}\right|+C, & b^2-4ac\neq 0,\\ \frac{1}{\sqrt{a}}\ln\left|x+\frac{b}{2a}\right|+C, & b^2-4ac=0.\end{cases}$

42. $\int \frac{\mathrm{d}x}{\sqrt{c+bx-ax^2}}=\frac{1}{\sqrt{a}}\arcsin\frac{2ax-b}{\sqrt{b^2+4ac}}+C$.

43. $\int\sqrt{\frac{x-a}{x-b}}\,\mathrm{d}x=(x-b)\sqrt{\frac{x-a}{x-b}}+(b-a)\ln(\sqrt{|x-a|}+\sqrt{|x-b|})+C$.

44. $\int\sqrt{\frac{x-a}{b-x}}\,\mathrm{d}x=(x-b)\sqrt{\frac{x-a}{b-x}}+(b-a)\arctan\sqrt{\frac{x-a}{b-x}}+C$.

45. $\int\frac{\mathrm{d}x}{\sqrt{(x-a)(b-x)}}=2\arcsin\sqrt{\frac{x-a}{b-a}}+C,\ a<b$.

46. $\int\sqrt{(x-a)(b-x)}\mathrm{d}x=\frac{2x-a-b}{4}\cdot\sqrt{(x-a)(b-x)}+\frac{(b-a)^2}{4}\arcsin\sqrt{\frac{x-a}{b-a}}+C,\ a<b$.

第三组

“含有三角函数的有理式的积分（不带根号）”.

1. $\int\cos^3 ax\mathrm{d}x=\frac{\sin ax}{a}-\frac{\sin^3 ax}{3a}+C$.

2. $\int\tan^2 ax\mathrm{d}x=\frac{1}{a}\tan ax-x+C$.

3. $\int\tan^3 ax\mathrm{d}x=\frac{1}{2a}\tan^2 ax+\frac{1}{a}\ln|\cos ax|+C$.

4. $\int\cot^2 ax\mathrm{d}x=-\frac{1}{a}\cot ax-x+C$.

5. $\int\cot^3 ax\mathrm{d}x=-\frac{1}{2a}\cot^2 ax-\frac{1}{a}\ln|\sin ax|+C$.

6. $\int\frac{\mathrm{d}x}{\sin^3 ax}=-\frac{\cos ax}{2a\sin^2 ax}+\frac{1}{2a}\ln\left|\tan\frac{ax}{2}\right|+C$.

7. $\int\frac{\mathrm{d}x}{\cos^3 ax}=\frac{\sin ax}{2a\cos^2 ax}+\frac{1}{2a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right|+C$.

8. $\int\sin mx\sin nx\mathrm{d}x=\frac{\sin(m-n)x}{2(m-n)}-\frac{\sin(m+n)x}{2(m+n)}+C,\ m^2\neq n^2$.

9. $\int\sin mx\cos nx\mathrm{d}x=-\frac{\cos(m-n)x}{2(m-n)}-\frac{\cos(m+n)x}{2(m+n)}+C,\ m^2\neq n^2$.

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10. $\int \cos mx\cos nx\mathrm{d}x=\dfrac{\sin(m-n)x}{2(m-n)}+\dfrac{\sin(m+n)x}{2(m+n)}+C$，$m^2\neq n^2$.

11. $\int \sin ax\cos ax\mathrm{d}x=\dfrac{1}{2a}\sin^2 ax+C$.

12. $\int \sin^2 ax\cos ax\mathrm{d}x=\dfrac{\sin^3 ax}{3a}+C$.

13. $\int \sin^2 ax\cos^2 ax\mathrm{d}x=-\dfrac{\sin 4ax}{32a}+\dfrac{x}{8}+C$.

14. $\int \sin^2 ax\cos^3 ax\mathrm{d}x=\dfrac{1}{3a}\sin^3 ax-\dfrac{1}{5a}\sin^5 ax+C$.

15. $\int \sin^3 ax\cos ax\mathrm{d}x=\dfrac{\sin^4 ax}{4a}+C$.

16. $\int \sin^3 ax\cos^2 ax\mathrm{d}x=-\dfrac{\cos^3 ax}{3a}+\dfrac{\cos^5 ax}{5a}+C$.

17. $\int \sin^3 ax\cos^3 ax\mathrm{d}x=-\dfrac{1}{16a}\cos 2ax+\dfrac{1}{48a}\cos^3 2ax+C$.

18. $\int \dfrac{\sin ax}{\cos^2 ax}\mathrm{d}x=\dfrac{1}{a\cos ax}+C=\dfrac{\sec ax}{a}+C$.

19. $\int \dfrac{\sin^2 ax}{\cos^3 ax}\mathrm{d}x=\dfrac{\sin ax}{2a\cos^2 ax}-\dfrac{1}{2a}\ln\left|\tan\left(\dfrac{\pi}{4}+\dfrac{ax}{2}\right)\right|+C$.

20. $\int \dfrac{\sin^2 ax}{\cos^4 ax}\mathrm{d}x=\dfrac{\tan^3 ax}{3a}+C$.

21. $\int \dfrac{\sin^3 ax}{\cos ax}\mathrm{d}x=-\dfrac{\sin^2 ax}{2a}-\dfrac{1}{a}\ln|\cos ax|+C$.

22. $\int \dfrac{\sin^3 ax}{\cos^2 ax}\mathrm{d}x=\dfrac{\cos ax}{a}+\dfrac{1}{a\cos ax}+C$.

23. $\int \dfrac{\sin^3 ax}{\cos^6 ax}\mathrm{d}x=\dfrac{1}{5a\cos^5 ax}-\dfrac{1}{3a\cos^3 ax}+C$.

24. $\int \dfrac{\cos^2 ax}{\sin ax}\mathrm{d}x=\dfrac{\cos ax}{a}+\dfrac{1}{a}\ln\left|\tan\dfrac{ax}{2}\right|+C$.

25. $\int \dfrac{\cos^2 ax}{\sin^3 ax}\mathrm{d}x=-\dfrac{\cos ax}{2a\sin^2 ax}-\dfrac{1}{2a}\ln\left|\tan\dfrac{ax}{2}\right|+C$.

26. $\int \dfrac{\cos^2 ax}{\sin^4 ax}\mathrm{d}x=-\dfrac{\cot^3 ax}{3a}+C$.

27. $\int \dfrac{\cos^3 ax}{\sin ax}\mathrm{d}x=\dfrac{\cos^2 ax}{2a}+\dfrac{1}{a}\ln|\sin ax|+C$.

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28. $\int \frac{\cos^3 ax}{\sin^2 ax}\mathrm{d}x = -\frac{\sin ax}{a} - \frac{1}{a\sin ax} + C$.

29. $\int \frac{\cos^3 ax}{\sin^4 ax}\mathrm{d}x = -\frac{1}{3a\sin^3 ax} + \frac{1}{a\sin ax} + C$.

30. $\int \frac{\mathrm{d}x}{\sin ax\cos ax} = \frac{1}{a}\ln|\tan ax| + C$.

31. $\int \frac{\mathrm{d}x}{\sin ax\cos^2 ax} = \frac{1}{a}\left(\sec ax + \ln\left|\tan\frac{ax}{2}\right|\right) + C$.

32. $\int \frac{\mathrm{d}x}{\sin^2 ax\cos ax} = -\frac{1}{a}\csc ax + \frac{1}{a}\ln\left|\tan\left(\frac{\pi}{4} + \frac{ax}{2}\right)\right| + C$.

33. $\int \frac{\mathrm{d}x}{\sin^2 ax\cos^2 ax} = -\frac{2}{a}\cot 2ax + C$.

34. $\int \frac{\mathrm{d}x}{\sin^2 ax\cos^3 ax} = \frac{1}{2a\sin ax}\left(\frac{1}{\cos^2 ax} - 3\right) + \frac{3}{2a}\ln\left|\tan\left(\frac{\pi}{4} + \frac{ax}{2}\right)\right| + C$.

35. $\int \frac{\mathrm{d}x}{\sin^2 ax\cos^4 ax} = \frac{\tan^3 ax}{3a} + \frac{\tan ax}{a} - \frac{2}{a}\cot 2ax + C$.

36. $\int \frac{\mathrm{d}x}{\sin^3 ax\cos ax} = -\frac{1}{2a\sin^2 ax} + \frac{1}{a}\ln|\tan ax| + C$.

37. $\int \frac{\mathrm{d}x}{\sin^3 ax\cos^2 ax} = \frac{1}{a\cos ax} - \frac{\cos ax}{2a\sin^2 ax} + \frac{3}{2a}\ln\left|\tan\frac{ax}{2}\right| + C$.

38. $\int \frac{\mathrm{d}x}{\sin^3 ax\cos^3 ax} = -\frac{2\cos 2ax}{a\sin^2 2ax} + \frac{2}{a}\ln|\tan ax| + C$.

39. $\int \frac{\mathrm{d}x}{\sin^3 ax\cos^4 ax} = \frac{2}{a\cos ax} + \frac{1}{3a\cos^3 ax} - \frac{\cos ax}{2a\sin^2 ax} + \frac{5}{2a}\ln\left|\tan\frac{ax}{2}\right| + C$.

40. $\int \frac{\mathrm{d}x}{1\pm\sin ax} = \mp\frac{1}{a}\tan\left(\frac{\pi}{4} \mp \frac{ax}{2}\right) + C$.

41. $\int \frac{x}{1+\sin ax}\mathrm{d}x = -\frac{x}{a}\tan\left(\frac{\pi}{4} - \frac{ax}{2}\right) + \frac{1}{a^2}\ln|1+\sin ax| + C$.

42. $\int \frac{x}{1-\sin ax}\mathrm{d}x = \frac{x}{a}\tan\left(\frac{\pi}{4} + \frac{ax}{2}\right) + \frac{1}{a^2}\ln|1-\sin ax| + C$.

43. $\int \frac{\sin ax}{1\pm\sin ax}\mathrm{d}x = \pm x + \frac{1}{a}\tan\left(\frac{\pi}{4} \mp \frac{ax}{2}\right) + C$.

44. $\int \frac{\cos ax}{1\pm\sin ax}\mathrm{d}x = \pm\frac{1}{a}\ln|1\pm\sin ax| + C$.

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45. $\int \frac{\mathrm{d}x}{\sin ax(1\pm\sin ax)} = \frac{1}{a}\tan\left(\frac{\pi}{4}\mp\frac{ax}{2}\right)+\frac{1}{a}\ln\left|\tan\frac{ax}{2}\right|+C$.

46. $\int \frac{\mathrm{d}x}{\sin ax(1\pm\cos ax)} = \pm\frac{1}{2a(1\pm\cos ax)}+\frac{1}{2a}\ln\left|\tan\frac{ax}{2}\right|+C$.

47. $\int \frac{\mathrm{d}x}{\cos ax(1\pm\sin ax)} = \mp\frac{1}{2a(1\pm\sin ax)}+\frac{1}{2a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right|+C$.

48. $\int \frac{\sin ax}{\cos ax(1\pm\sin ax)}\mathrm{d}x = \frac{1}{2a(1\pm\sin ax)}\pm\frac{1}{2a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right|+C$.

49. $\int \frac{\sin ax}{\cos ax(1\pm\cos ax)}\mathrm{d}x = \frac{1}{a}\ln|\sec ax\pm 1|+C$.

50. $\int \frac{\cos ax}{\sin ax(1\pm\sin ax)}\mathrm{d}x = -\frac{1}{a}\ln|\csc ax\pm 1|+C$.

51. $\int \frac{\cos ax}{\sin ax(1\pm\cos ax)}\mathrm{d}x = -\frac{1}{2a(1\pm\cos ax)}\pm\frac{1}{2a}\ln\left|\tan\frac{ax}{2}\right|+C$.

52. $\int \frac{\mathrm{d}x}{(1+\sin ax)^2} = -\frac{1}{2a}\cot\left(\frac{\pi}{4}+\frac{ax}{2}\right)-\frac{1}{6a}\cot^3\left(\frac{\pi}{4}+\frac{ax}{2}\right)+C$.

53. $\int \frac{\mathrm{d}x}{(1-\sin ax)^2} = -\frac{1}{2a}\cot\left(\frac{ax}{2}-\frac{\pi}{4}\right)-\frac{1}{6a}\cot^3\left(\frac{ax}{2}-\frac{\pi}{4}\right)+C$.

54. $\int \frac{\sin ax}{(1+\sin ax)^2}\mathrm{d}x = -\frac{2}{a}\frac{1}{\left(\tan\frac{ax}{2}+1\right)^2}+\frac{4}{3a}\frac{1}{\left(\tan\frac{ax}{2}+1\right)^3}+C$.

55. $\int \frac{\sin ax}{(1-\sin ax)^2}\mathrm{d}x = -\frac{2}{a}\frac{1}{\left(\tan\frac{ax}{2}-1\right)^2}-\frac{4}{3a}\frac{1}{\left(\tan\frac{ax}{2}-1\right)^3}+C$.

56. $\int \frac{\mathrm{d}x}{1+\cos ax} = \frac{1}{a}\tan\frac{ax}{2}+C$.

57. $\int \frac{\mathrm{d}x}{1-\cos ax} = -\frac{1}{a}\cot\frac{ax}{2}+C$.

58. $\int \frac{x}{1+\cos ax}\mathrm{d}x = \frac{x}{a}\tan\frac{ax}{2}+\frac{2}{a^2}\ln\left|\cos\frac{ax}{2}\right|+C$.

59. $\int \frac{x}{1-\cos ax}\mathrm{d}x = -\frac{x}{a}\cot\frac{ax}{2}+\frac{2}{a^2}\ln\left|\sin\frac{ax}{2}\right|+C$.

60. $\int \frac{\sin ax}{1\pm\cos ax}\mathrm{d}x = \mp\frac{1}{a}\ln(1\pm\cos ax)+C$.

61. $\int \frac{\cos ax}{1+\cos ax}\mathrm{d}x = -\frac{1}{a}\tan\frac{ax}{2}+x+C$.

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62. $\int \frac{\cos ax}{1-\cos ax}\mathrm{d}x=-\frac{1}{a}\cot\frac{ax}{2}-x+C$.

63. $\int \frac{\mathrm{d}x}{\cos ax(1+\cos ax)}=-\frac{1}{a}\tan\frac{ax}{2}-\frac{1}{a}\ln\left|\frac{\tan\frac{ax}{2}-1}{\tan\frac{ax}{2}+1}\right|+C$.

64. $\int \frac{\mathrm{d}x}{\cos ax(1-\cos ax)}=-\frac{1}{a}\cot\frac{ax}{2}-\frac{1}{a}\ln\left|\frac{\tan\frac{ax}{2}-1}{\tan\frac{ax}{2}+1}\right|+C$.

65. $\int \frac{\mathrm{d}x}{(1+\cos ax)^2}=\frac{1}{2a}\tan\frac{ax}{2}+\frac{1}{6a}\tan^3\frac{ax}{2}+C$.

66. $\int \frac{\mathrm{d}x}{(1-\cos ax)^2}=-\frac{1}{2a}\cot\frac{ax}{2}-\frac{1}{6a}\cot^3\frac{ax}{2}+C$.

67. $\int \frac{\cos ax}{(1+\cos ax)^2}\mathrm{d}x=\frac{1}{2a}\tan\frac{ax}{2}-\frac{1}{6a}\tan^3\frac{ax}{2}+C$.

68. $\int \frac{\cos ax}{(1-\cos ax)^2}\mathrm{d}x=\frac{1}{2a}\cot\frac{ax}{2}-\frac{1}{6a}\cot^3\frac{ax}{2}+C$.

69. $\int \frac{\mathrm{d}x}{1+\cos ax+\sin ax}=\frac{1}{a}\ln\left|1+\tan\frac{ax}{2}\right|+C$.

70. $\int \frac{\mathrm{d}x}{1+\sin^2 ax}=\frac{1}{\sqrt{2}a}\arctan(\sqrt{2}\tan ax)+C$.

71. $\int \frac{\mathrm{d}x}{1+b\sin^2 ax}=\frac{1}{a\sqrt{1+b}}\arctan(\sqrt{1+b}\tan ax)+C,\quad b>-1$.

72. $\int \frac{\mathrm{d}x}{1-\sin^2 ax}=\frac{1}{a}\tan ax+C$.

73. $\int \frac{\mathrm{d}x}{1-b\sin^2 ax}=\begin{cases}\frac{1}{a\sqrt{1-b}}\arctan(\sqrt{1-b}\tan ax)+C, & 0<b<1,\\ \frac{1}{2a\sqrt{b-1}}\ln\left|\frac{\sqrt{b-1}\tan ax+1}{\sqrt{b-1}\tan ax-1}\right|+C, & b>1.\end{cases}$

74. $\int \frac{\mathrm{d}x}{(1-b\sin^2 ax)^2}=-\frac{b}{2a(1-b)^2}\frac{\tan ax}{\tan^2 ax+\frac{1}{1-b}}+\frac{2-b}{2a\sqrt{1-b}(1-b)}\arctan(\sqrt{1-b}\tan ax)+C,\ b<1$.

75. $\int \frac{\sin ax\cos ax}{1\pm b\sin^2 ax}\mathrm{d}x=\pm\frac{1}{2ab}\ln|1\pm b\sin^2 ax|+C,\ b\neq 0$.

76. $\int \frac{\mathrm{d}x}{1+b\cos^2 ax}=\frac{1}{a\sqrt{1+b}}\arctan\frac{\tan ax}{\sqrt{1+b}}+C,\ b>-1$.

77. $\int \frac{\mathrm{d}x}{1-\cos^2 ax}=-\frac{1}{a}\cot ax+C$.

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78. $$\int \frac{\mathrm{d}x}{1-b\cos^2 ax}=\begin{cases}\dfrac{1}{a\sqrt{1-b}}\arctan\dfrac{\tan ax}{\sqrt{1-b}}+C, & 0<b<1,\\ \dfrac{1}{2a\sqrt{b-1}}\ln\left|\dfrac{\tan ax-\sqrt{b-1}}{\tan ax+\sqrt{b-1}}\right|+C, & b>1.\end{cases}$$

79. $$\int \frac{\mathrm{d}x}{(1-b\cos^2 ax)^2}=\frac{b\sin 2ax}{4a(1-b)(1-b\cos^2 ax)}+\frac{2-b}{2a(1-b)}\cdot\begin{cases}\dfrac{1}{\sqrt{1-b}}\arctan\dfrac{\tan ax}{\sqrt{1-b}}+C, & 0<b<1,\\ \dfrac{1}{2\sqrt{b-1}}\ln\left|\dfrac{\tan ax-\sqrt{b-1}}{\tan ax+\sqrt{b-1}}\right|+C, & b>1.\end{cases}$$

80. $$\int \frac{\sin^2 ax}{1+b\cos^2 ax}\mathrm{d}x=-\frac{x}{b}+\frac{\sqrt{1+b}}{ab}\arctan\frac{\tan ax}{\sqrt{1+b}}+C,\ b>-1,\ \text{且}\ b\neq 0.$$

81. $$\int \frac{\cos^2 ax}{1+b\cos^2 ax}\mathrm{d}x=\frac{x}{b}-\frac{1}{ab\sqrt{1+b}}\arctan\frac{\tan ax}{\sqrt{1+b}}+C,\ b>-1,\ \text{且}\ b\neq 0.$$

82. $$\int \frac{\sin ax\cos ax}{1\pm b\cos^2 ax}\mathrm{d}x=\mp\frac{1}{ab}\ln\sqrt{|1\pm b\cos^2 ax|}+C,\ b\neq 0.$$

83. $$\int \frac{\sin^2 ax}{1-b\cos^2 ax}\mathrm{d}x=\begin{cases}-\dfrac{\sqrt{1-b}}{ab}\arctan\dfrac{\tan ax}{\sqrt{1-b}}+\dfrac{x}{b}+C, & 0<b<1,\\ \dfrac{\sqrt{b-1}}{2ab}\ln\left|\dfrac{\tan ax-\sqrt{b-1}}{\tan ax+\sqrt{b-1}}\right|+\dfrac{x}{b}+C, & b>1.\end{cases}$$

84. $$\int \frac{\cos^2 ax}{1-b\cos^2 ax}\mathrm{d}x=\begin{cases}\dfrac{1}{ab\sqrt{1-b}}\arctan\dfrac{\tan ax}{\sqrt{1-b}}-\dfrac{x}{b}+C, & 0<b<1,\\ \dfrac{1}{2ab\sqrt{b-1}}\ln\left|\dfrac{\tan ax-\sqrt{b-1}}{\tan ax+\sqrt{b-1}}\right|-\dfrac{x}{b}+C, & b>1.\end{cases}$$

85. $$\int \frac{\mathrm{d}x}{a+b\sin x}=\begin{cases}\dfrac{2}{\sqrt{a^2-b^2}}\arctan\dfrac{a\tan\dfrac{x}{2}+b}{\sqrt{a^2-b^2}}+C, & a^2>b^2,\\ \dfrac{1}{\sqrt{b^2-a^2}}\ln\left|\dfrac{a\tan\dfrac{x}{2}+b-\sqrt{b^2-a^2}}{a\tan\dfrac{x}{2}+b+\sqrt{b^2-a^2}}\right|+C, & a^2<b^2,\end{cases}\quad a>0.$$

86. $$\int \frac{\mathrm{d}x}{a+b\cos x}=\begin{cases}\dfrac{2\operatorname{sgn}(a+b)}{\sqrt{a^2-b^2}}\arctan\dfrac{\sqrt{a^2-b^2}\tan\dfrac{x}{2}}{|a+b|}+C, & a^2>b^2,\\ \dfrac{1}{a}\tan\dfrac{x}{2}+C, & a=b,\\ -\dfrac{1}{a}\cot\dfrac{x}{2}+C, & a=-b,\\ \dfrac{1}{\sqrt{b^2-a^2}}\ln\left|\dfrac{\sqrt{b^2-a^2}\tan\dfrac{x}{2}+|a+b|}{\sqrt{b^2-a^2}\tan\dfrac{x}{2}-|a+b|}\right|+C, & a^2<b^2.\end{cases}$$

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87. $\int \frac{\mathrm{d}x}{(a+b\sin x)^2}=\begin{cases}\frac{b\cos x}{(a^2-b^2)(a+b\sin x)}+\frac{2a}{(a^2-b^2)^{\frac{3}{2}}}\arctan\frac{a\tan\frac{x}{2}+b}{\sqrt{a^2-b^2}}+C, & a^2>b^2,\\ \frac{b\cos x}{(a^2-b^2)(a+b\sin x)}-\frac{a}{(b^2-a^2)^{\frac{3}{2}}}\ln\left|\frac{a\tan\frac{x}{2}+b-\sqrt{b^2-a^2}}{a\tan\frac{x}{2}+b+\sqrt{b^2-a^2}}\right|+C, & a^2<b^2,\end{cases}$ $a>0.$

88. $\int \frac{\mathrm{d}x}{(a+b\cos x)^2}=\frac{b\sin x}{(b^2-a^2)(a+b\cos x)}-\frac{a}{b^2-a^2}\int\frac{\mathrm{d}x}{a+b\cos x}$.

89. $\int \frac{\mathrm{d}x}{\sin x(a+b\sin x)}=\frac{1}{a}\ln\left|\tan\frac{x}{2}\right|-\frac{b}{a}\int\frac{\mathrm{d}x}{a+b\sin x}$.

90. $\int \frac{\mathrm{d}x}{\cos x(a+b\cos x)}=\frac{1}{a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right|-\frac{b}{a}\int\frac{\mathrm{d}x}{a+b\cos x}$.

91. $\int \frac{\sin x}{a+b\sin x}\mathrm{d}x=\frac{x}{b}-\frac{a}{b}\int\frac{\mathrm{d}x}{a+b\sin x}$.

92. $\int \frac{\cos x}{a+b\cos x}\mathrm{d}x=\frac{x}{b}-\frac{a}{b}\int\frac{\mathrm{d}x}{a+b\cos x}$.

93. $\int \frac{\cos x}{(a+b\cos x)^2}\mathrm{d}x=\frac{a\sin x}{(a^2-b^2)(a+b\cos x)}-\frac{b}{a^2-b^2}\int\frac{\mathrm{d}x}{a+b\cos x}$.

94. $\int \frac{\mathrm{d}x}{a^2\cos^2 x+b^2\sin^2 x}=\frac{1}{ab}\arctan\frac{b\tan x}{a}+C$.

95. $\int \frac{\sin cx}{a\cos cx+b\sin cx}\mathrm{d}x=\frac{1}{c(a^2+b^2)}(bcx-a\ln|a\cos cx+b\sin cx|)+C$.

96. $\int \frac{\cos cx}{a\cos cx+b\sin cx}\mathrm{d}x=\frac{1}{c(a^2+b^2)}(acx+b\ln|a\cos cx+b\sin cx|)+C$.

97. $\int \frac{\sin cx\cos cx}{a\cos^2 cx+b\sin^2 cx}\mathrm{d}x=\frac{1}{2c(b-a)}\ln\left|a\cos^2 cx+b\sin^2 cx\right|+C$.

98. $\int \frac{\mathrm{d}x}{a^2+b^2\sin^2 cx}=\frac{1}{ac\sqrt{a^2+b^2}}\arctan\frac{\sqrt{a^2+b^2}\tan cx}{a}+C$.

99. $\int \frac{\cos^2 cx}{a^2+b^2\sin^2 cx}\mathrm{d}x=\frac{\sqrt{a^2+b^2}}{ab^2c}\arctan\frac{\sqrt{a^2+b^2}\tan cx}{a}-\frac{x}{b^2}+C$.

100. $\int \frac{\mathrm{d}x}{a^2-b^2\sin^2 cx}=\begin{cases}\frac{1}{ac\sqrt{a^2-b^2}}\arctan\frac{\sqrt{a^2-b^2}\tan cx}{a}+C, & a^2>b^2,\\ \frac{1}{2ac\sqrt{b^2-a^2}}\ln\left|\frac{\sqrt{b^2-a^2}\tan cx+a}{\sqrt{b^2-a^2}\tan cx-a}\right|+C, & a^2<b^2,\end{cases}$ $a>0.$

101. $\int \frac{\mathrm{d}x}{a^2+b^2\cos^2 cx}=\frac{1}{ac\sqrt{a^2+b^2}}\arctan\frac{a\tan cx}{\sqrt{a^2+b^2}}+C$.

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102. $\int \frac{\mathrm{d}x}{a^2-b^2\cos^2 cx}=\begin{cases}\frac{1}{ac\sqrt{a^2-b^2}}\arctan\frac{a\tan cx}{\sqrt{a^2-b^2}}+C, & a^2>b^2,\\ \frac{1}{2ac\sqrt{b^2-a^2}}\ln\left|\frac{a\tan cx-\sqrt{b^2-a^2}}{a\tan cx+\sqrt{b^2-a^2}}\right|+C, & a^2<b^2,\end{cases}$ $a,\ b>0.$

103. $\int \frac{\mathrm{d}x}{a^2+b^2-2ab\cos cx}=\frac{2}{c(a^2-b^2)}\arctan\left(\frac{a+b}{a-b}\tan\frac{cx}{2}\right)+C.$

104. $\int \frac{x+\sin x}{1+\cos x}\mathrm{d}x=x\tan\frac{x}{2}+C.$

105. $\int \frac{x-\sin x}{1-\cos x}\mathrm{d}x=-x\cot\frac{x}{2}+C.$

106. $\int \frac{\mathrm{d}x}{\sin ax\pm\cos ax}=\frac{1}{\sqrt{2}a}\ln\left|\tan\left(\frac{ax}{2}\pm\frac{\pi}{8}\right)\right|+C.$

107. $\int \frac{\mathrm{d}x}{(\sin ax\pm\cos ax)^2}=\frac{1}{2a}\tan\left(ax\mp\frac{\pi}{4}\right)+C.$

108. $\int \frac{\sin ax}{\sin ax\pm\cos ax}\mathrm{d}x=\frac{1}{2a}(ax\mp\ln|\sin ax\pm\cos ax|)+C.$

109. $\int \frac{\cos ax}{\sin ax\pm\cos ax}\mathrm{d}x=\frac{1}{2a}(\ln|\sin ax\pm\cos ax|\pm ax)+C.$

110. $\int \frac{\mathrm{d}x}{a\cos x+b\sin x}=\frac{\ln\left|\tan\left(\frac{x}{2}+\frac{1}{2}\arctan\frac{a}{b}\right)\right|}{\sqrt{a^2+b^2}}+C.$

111. $\int \frac{\mathrm{d}x}{a^2\cos^2 x+b^2\sin^2 x}=\frac{1}{ab}\arctan\left(\frac{b}{a}\tan x\right)+C.$

112. $\int \frac{\mathrm{d}x}{a^2\cos^2 x-b^2\sin^2 x}=\frac{1}{2ab}\ln\left|\frac{b\tan x+a}{b\tan x-a}\right|+C.$

113. $\int x\sin ax\mathrm{d}x=\frac{1}{a^2}\sin ax-\frac{1}{a}x\cos ax+C.$

114. $\int x^2\sin ax\mathrm{d}x=-\frac{1}{a}x^2\cos ax+\frac{2}{a^2}x\sin ax+\frac{2}{a^3}\cos ax+C.$

115. $\int x\cos ax\mathrm{d}x=\frac{1}{a^2}\cos ax+\frac{1}{a}x\sin ax+C.$

116. $\int x^2\cos ax\mathrm{d}x=\frac{1}{a}x^2\sin ax+\frac{2}{a^2}x\cos ax-\frac{2}{a^3}\sin ax+C.$

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第四组

“含有三角函数的无理式的积分（带根号）”.

1. $\int \sqrt{1+\sin ax}\mathrm{d}x=\mp\frac{2\sqrt{2}}{a}\cos\left(\frac{\pi}{4}+\frac{ax}{2}\right)+C$.

2. $\int \sqrt{1-\sin ax}\mathrm{d}x=\mp\frac{2\sqrt{2}}{a}\sin\left(\frac{\pi}{4}+\frac{ax}{2}\right)+C$.

3. $\int \frac{\mathrm{d}x}{\sqrt{1+\sin ax}}=\pm\frac{\sqrt{2}}{a}\ln\left|\tan\left(\frac{ax}{4}+\frac{\pi}{8}\right)\right|+C$.

4. $\int \frac{\mathrm{d}x}{\sqrt{1-\sin ax}}=\pm\frac{\sqrt{2}}{a}\ln\left|\tan\left(\frac{ax}{4}-\frac{\pi}{8}\right)\right|+C$.

第五组

“含有反三角函数的积分 $(a>0)$”.

1. $\int \arcsin\frac{x}{a}\mathrm{d}x=x\arcsin\frac{x}{a}+\sqrt{a^2-x^2}+C$.

2. $\int x\arcsin\frac{x}{a}\mathrm{d}x=\left(\frac{x^2}{2}-\frac{a^2}{4}\right)\arcsin\frac{x}{a}+\frac{x}{4}\sqrt{a^2-x^2}+C$.

3. $\int \arccos\frac{x}{a}\mathrm{d}x=x\arccos\frac{x}{a}-\sqrt{a^2-x^2}+C$.

4. $\int x\arccos\frac{x}{a}\mathrm{d}x=\left(\frac{x^2}{2}-\frac{a^2}{4}\right)\arccos\frac{x}{a}-\frac{x}{4}\sqrt{a^2-x^2}+C$.

5. $\int \arctan\frac{x}{a}\mathrm{d}x=x\arctan\frac{x}{a}-\frac{a}{2}\ln\left(a^2+x^2\right)+C$.

6. $\int x\arctan\frac{x}{a}\mathrm{d}x=\frac{1}{2}(a^2+x^2)\arctan\frac{x}{a}-\frac{a}{2}x+C$.

7. $\int \arcsin ax\mathrm{d}x=x\arcsin ax+\frac{\sqrt{1-a^2x^2}}{a}+C$.

8. $\int \arccos ax\mathrm{d}x=x\arccos ax-\frac{\sqrt{1-a^2x^2}}{a}+C$.

9. $\int \arctan ax\mathrm{d}x=x\arctan ax-\frac{1}{2a}\ln(1+a^2x^2)+C$.

10. $\int \operatorname{arccot} ax\mathrm{d}x=x\operatorname{arccot} ax+\frac{1}{2a}\ln\left(1+a^2x^2\right)+C$.

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11. $\int \operatorname{arcsec} ax\mathrm{d}x = x\operatorname{arcsec} ax - \frac{1}{a}\ln\left|ax+\sqrt{a^2x^2-1}\right| + C(x>0)$.

12. $\int \operatorname{arccsc} ax\mathrm{d}x = x\operatorname{arccsc} ax + \frac{1}{a}\ln\left|ax+\sqrt{a^2x^2-1}\right| + C(x>0)$.

13. $\int x\arcsin ax\mathrm{d}x = \frac{1}{4a^2}\left[(2a^2x^2-1)\arcsin ax + ax\sqrt{1-a^2x^2}\right] + C$.

14. $\int x\arccos ax\mathrm{d}x = \frac{1}{4a^2}\left[(2a^2x^2-1)\arccos ax - ax\sqrt{1-a^2x^2}\right] + C$.

15. $\int x\arctan ax\mathrm{d}x = \frac{1+a^2x^2}{2a^2}\arctan ax - \frac{x}{2a} + C$.

16. $\int x\operatorname{arccot} ax\mathrm{d}x = \frac{1+a^2x^2}{2a^2}\operatorname{arccot} ax + \frac{x}{2a} + C$.

17. $\int x\operatorname{arcsec} ax\mathrm{d}x = \frac{x^2}{2}\operatorname{arcsec} ax - \frac{1}{2a^2}\sqrt{a^2x^2-1} + C(x>0)$.

18. $\int x\operatorname{arccsc} ax\mathrm{d}x = \frac{x^2}{2}\operatorname{arccsc} ax + \frac{1}{2a^2}\sqrt{a^2x^2-1} + C(x>0)$.

19. $\int (\arcsin ax)^2\mathrm{d}x = x(\arcsin ax)^2 - 2x + \frac{2\sqrt{1-a^2x^2}}{a}\arcsin ax + C$.

20. $\int (\arccos ax)^2\mathrm{d}x = x(\arccos ax)^2 - 2x - \frac{2\sqrt{1-a^2x^2}}{a}\arccos ax + C$.

21. $\int \frac{\arcsin ax}{x^2}\mathrm{d}x = -\frac{1}{x}\arcsin ax + a\ln\left|\frac{1-\sqrt{1-a^2x^2}}{ax}\right| + C$.

22. $\int \frac{\arccos ax}{x^2}\mathrm{d}x = -\frac{1}{x}\arccos ax + a\ln\left|\frac{1+\sqrt{1-a^2x^2}}{ax}\right| + C$.

23. $\int \frac{\arctan ax}{x^2}\mathrm{d}x = -\frac{1}{x}\arctan ax - \frac{a}{2}\ln\frac{1+a^2x^2}{a^2x^2} + C$.

24. $\int \frac{\operatorname{arccot} ax}{x^2}\mathrm{d}x = -\frac{1}{x}\operatorname{arccot} ax - \frac{a}{2}\ln\frac{a^2x^2}{1+a^2x^2} + C$.

25. $\int \frac{\operatorname{arcsec} ax}{x^2}\mathrm{d}x = -\frac{1}{x}\operatorname{arcsec} ax + \frac{\sqrt{a^2x^2-1}}{x} + C$.

26. $\int \frac{\operatorname{arccsc} ax}{x^2}\mathrm{d}x = -\frac{1}{x}\operatorname{arccsc} ax - \frac{\sqrt{a^2x^2-1}}{x} + C$.

27. $\int \frac{\arcsin ax}{\sqrt{1-a^2x^2}}\mathrm{d}x = \frac{1}{2a}(\arcsin ax)^2 + C$.

28. $\int \frac{\arccos ax}{\sqrt{1-a^2x^2}}\mathrm{d}x = -\frac{1}{2a}(\arccos ax)^2 + C$.

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29. $\int \frac{\arctan ax}{1+a^2x^2}\mathrm{d}x=\frac{1}{2a}(\arctan ax)^2+C$.

30. $\int \frac{\operatorname{arccot} ax}{1+a^2x^2}\mathrm{d}x=-\frac{1}{2a}(\operatorname{arccot} ax)^2+C$.

31. $\int \arcsin\frac{x}{a}\mathrm{d}x=x\arcsin\frac{x}{a}+\sqrt{a^2-x^2}+C$.

32. $\int\left(\arcsin\frac{x}{a}\right)^2\mathrm{d}x=x\left(\arcsin\frac{x}{a}\right)^2+2\sqrt{a^2-x^2}\arcsin\frac{x}{a}-2x+C$.

33. $\int\left(\arcsin\frac{x}{a}\right)^3\mathrm{d}x=x\left(\arcsin\frac{x}{a}\right)^3+3\sqrt{a^2-x^2}\left(\arcsin\frac{x}{a}\right)^2-6x\arcsin\frac{x}{a}-6\sqrt{a^2-x^2}+C$.

34. $\int x\arcsin\frac{x}{a}\mathrm{d}x=\left(\frac{x^2}{2}-\frac{a^2}{4}\right)\arcsin\frac{x}{a}+\frac{x}{4}\sqrt{a^2-x^2}+C$.

35. $\int x^2\arcsin\frac{x}{a}\mathrm{d}x=\frac{x^3}{3}\arcsin\frac{x}{a}+\frac{x^2+2a^2}{9}\sqrt{a^2-x^2}+C$.

36. $\int \frac{x\arcsin x}{\sqrt{1-x^2}}\mathrm{d}x=x-\sqrt{1-x^2}\arcsin x+C$.

37. $\int \frac{x^2\arcsin x}{\sqrt{1-x^2}}\mathrm{d}x=\frac{x^2}{4}-\frac{x}{2}\sqrt{1-x^2}\arcsin x+\frac{1}{4}(\arcsin x)^2+C$.

38. $\int \frac{x^3\arcsin x}{\sqrt{1-x^2}}\mathrm{d}x=\frac{\arcsin x\cdot(1-x^2)^{\frac{3}{2}}}{3}+\frac{x^3}{9}-\arcsin x\cdot\sqrt{1-x^2}+\frac{2}{3}x+C$.

39. $\int \frac{\arcsin x}{\sqrt{(1-x^2)^3}}\mathrm{d}x=\frac{x\arcsin x}{\sqrt{1-x^2}}+\frac{1}{2}\ln\left|1-x^2\right|+C$.

40. $\int \frac{x\arcsin x}{\sqrt{(1-x^2)^3}}\mathrm{d}x=\frac{\arcsin x}{\sqrt{1-x^2}}+\frac{1}{2}\ln\left|\frac{1-x}{1+x}\right|+C$.

41. $\int \arccos\frac{x}{a}\mathrm{d}x=x\arccos\frac{x}{a}-\sqrt{a^2-x^2}+C$.

42. $\int x\arccos\frac{x}{a}\mathrm{d}x=\frac{x^2}{2}\arccos\frac{x}{a}-\frac{x}{4}\sqrt{a^2-x^2}+\frac{a^2}{4}\arcsin\frac{x}{a}+C$.

43. $\int x^2\arccos\frac{x}{a}\mathrm{d}x=\frac{x^3}{3}\arccos\frac{x}{a}-\frac{x^2+2a^2}{9}\sqrt{a^2-x^2}+C$.

44. $\int x\arctan\frac{x}{a}\mathrm{d}x=\frac{a^2+x^2}{2}\arctan\frac{x}{a}-\frac{ax}{2}+C$.

45. $\int x^2\arctan\frac{x}{a}\mathrm{d}x=\frac{x^3}{3}\arctan\frac{x}{a}+\frac{a^3}{6}\ln(a^2+x^2)-\frac{ax^2}{6}+C$.

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46. $\int \frac{\arctan\frac{x}{a}}{x^2}\mathrm{d}x=-\frac{1}{x}\arctan\frac{x}{a}-\frac{1}{a}\ln\left|\frac{a}{x}\sqrt{1+\frac{x^2}{a^2}}\right|+C$.

47. $\int \operatorname{arccot}\frac{x}{a}\mathrm{d}x=x\operatorname{arccot}\frac{x}{a}+\frac{a}{2}\ln(a^2+x^2)+C$.

48. $\int x\operatorname{arccot}\frac{x}{a}\mathrm{d}x=\frac{a^2+x^2}{2}\operatorname{arccot}\frac{x}{a}+\frac{ax}{2}+C$.

49. $\int x^2\operatorname{arccot}\frac{x}{a}\mathrm{d}x=\frac{x^3}{3}\operatorname{arccot}\frac{x}{a}-\frac{a^3}{6}\ln(a^2+x^2)+\frac{ax^2}{6}+C$.

50. $\int \frac{\operatorname{arccot}\frac{x}{a}}{x^2}\mathrm{d}x=-\frac{1}{x}\operatorname{arccot}\frac{x}{a}+\frac{1}{a}\ln\left|\frac{a}{x}\sqrt{1+\frac{x^2}{a^2}}\right|+C$.

第六组

“含有指数函数的积分”.

1. $\int \mathrm{e}^{ax}\mathrm{d}x=\frac{1}{a}\mathrm{e}^{ax}+C$.

2. $\int x\mathrm{e}^{ax}\mathrm{d}x=\frac{(ax-1)\mathrm{e}^{ax}}{a^2}+C$.

3. $\int xa^x\mathrm{d}x=\frac{x}{\ln a}a^x-\frac{a^x}{(\ln a)^2}+C$.

4. $\int \frac{\mathrm{d}x}{1+\mathrm{e}^x}=x-\ln(1+\mathrm{e}^x)+C=\ln\frac{\mathrm{e}^x}{1+\mathrm{e}^x}+C$.

5. $\int \frac{\mathrm{d}x}{a+b\mathrm{e}^{px}}=\frac{x}{a}-\frac{1}{ap}\ln(a+b\mathrm{e}^{px})+C$.

6. $\int \frac{\mathrm{d}x}{\sqrt{a+b\mathrm{e}^{\beta x}}}=\begin{cases}\frac{1}{\beta\sqrt{a}}\ln\frac{\sqrt{a+b\mathrm{e}^{\beta x}}-\sqrt{a}}{\sqrt{a+b\mathrm{e}^{\beta x}}+\sqrt{a}}+C, & a>0,b>0,\\ \frac{2}{\beta\sqrt{-a}}\arctan\frac{\sqrt{a+b\mathrm{e}^{\beta x}}}{\sqrt{-a}}+C, & a<0,b>0.\end{cases}$

7. $\int \frac{\mathrm{d}x}{a\mathrm{e}^{mx}+b\mathrm{e}^{-mx}}=\frac{1}{m\sqrt{ab}}\arctan\left(\mathrm{e}^{mx}\sqrt{\frac{a}{b}}\right)+C,\ a>0,\ b>0$.

8. $\int (a^x-a^{-x})\mathrm{d}x=\frac{a^x+a^{-x}}{\ln a}+C$.

9. $\int a^{px}\mathrm{d}x=\frac{a^{px}}{p\ln a}+C$.

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第七组

“含有对数函数的积分”.

1. $\int \ln x\mathrm{d}x = x\ln x - x + C$.

2. $\int \frac{\mathrm{d}x}{x\ln x} = \ln|\ln x| + C$.

3. $\int x^n \ln x\mathrm{d}x = \frac{1}{n+1}x^{n+1}\left(\ln x - \frac{1}{n+1}\right) + C$.

4. $\int (\ln x)^2\mathrm{d}x = x(\ln x)^2 - 2x\ln x + 2x + C$.

5. $\int x\ln x\mathrm{d}x = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$.

6. $\int x^2\ln x\mathrm{d}x = \frac{x^3}{3}\ln x - \frac{x^3}{9} + C$.

7. $\int \frac{\ln x}{(ax+b)^2}\mathrm{d}x = -\frac{\ln x}{a(ax+b)} + \frac{1}{ab}\ln\left|\frac{x}{ax+b}\right| + C$.

8. $\int \ln\frac{x+a}{x-a}\mathrm{d}x = (x+a)\ln|x+a| - (x-a)\ln|x-a| + C$.

9. $\int \frac{1}{x^2}\ln\frac{x+a}{x-a}\mathrm{d}x = \frac{1}{x}\ln\frac{x-a}{x+a} - \frac{1}{a}\ln\frac{x^2-a^2}{x^2} + C$.

10. $\int \ln(x^2+a^2)\mathrm{d}x = x\ln(x^2+a^2) - 2x + 2a\arctan\frac{x}{a} + C$.

11. $\int x\ln(x^2+a^2)\mathrm{d}x = \frac{1}{2}(x^2+a^2)\ln(x^2+a^2) - \frac{1}{2}x^2 + C$.

12. $\int x^2\ln(x^2+a^2)\mathrm{d}x = \frac{1}{3}\left[x^3\ln(x^2+a^2) - \frac{2}{3}x^3 + 2a^2x - 2a^3\arctan\frac{x}{a}\right] + C$.

13. $\int x^{2n}\ln(x^2+a^2)\mathrm{d}x = \frac{1}{2n+1}\left[x^{2n+1}\ln(x^2+a^2) + (-1)^n 2a^{2n+1}\arctan\frac{x}{a} - 2\sum_{k=0}^{n}\frac{(-1)^{n-k}}{2k+1}a^{2n-2k}x^{2k+1}\right] + C$.

14. $\int \ln(x^2-a^2)\mathrm{d}x = x\ln(x^2-a^2) - 2x + a\ln\left|\frac{x+a}{x-a}\right| + C$.

15. $\int \ln|x^2-a^2|\mathrm{d}x = x\ln|x^2-a^2| - 2x + a\ln\left|\frac{x+a}{x-a}\right| + C$.

16. $\int x\ln|x^2-a^2|\mathrm{d}x = \frac{1}{2}\left[(x^2-a^2)\ln|x^2-a^2| - x^2\right] + C$.

17. $\int x^2\ln|x^2-a^2|\mathrm{d}x = \frac{1}{3}\left(x^3\ln|x^2-a^2| - \frac{2}{3}x^3 - 2a^2x + a^3\ln\left|\frac{x+a}{x-a}\right|\right) + C$.

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18. $\int \sin(\ln x)\mathrm{d}x=\frac{x}{2}[\sin(\ln x)-\cos(\ln x)]+C$.

19. $\int \cos(\ln x)\mathrm{d}x=\frac{x}{2}[\cos(\ln x)+\sin(\ln x)]+C$.

20. $\int x^p\cos(b\ln x)\mathrm{d}x=\frac{x^{p+1}}{(p+1)^2+b^2}[(p+1)\cos(b\ln x)+b\sin(b\ln x)]+C$.

21. $\int x^p\sin(b\ln x)\mathrm{d}x=\frac{x^{p+1}}{(p+1)^2+b^2}[(p+1)\sin(b\ln x)-b\cos(b\ln x)]+C$.

一元积分公式大观求解过程

第一组

1. 解 $\int\frac{1}{ax+b}\mathrm{d}x=\frac{1}{a}\int\frac{1}{ax+b}\mathrm{d}(ax+b)=\frac{1}{a}\ln|ax+b|+C$.

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2. 解 $\int(ax+b)^\mu\mathrm{d}x=\frac{1}{a}\int(ax+b)^\mu\mathrm{d}(ax+b)=\frac{1}{a(\mu+1)}(ax+b)^{\mu+1}+C$，$\mu\neq-1$.

3. 解 $\int\frac{x}{ax+b}\mathrm{d}x=\frac{1}{a}\int\frac{ax+b-b}{ax+b}\mathrm{d}x=\frac{1}{a}\int\left(1-\frac{b}{ax+b}\right)\mathrm{d}x=\frac{1}{a}\int\mathrm{d}x-\frac{b}{a}\int\frac{1}{ax+b}\mathrm{d}x$

$$=\frac{x}{a}-\frac{b}{a}\cdot\frac{1}{a}\int\frac{1}{ax+b}\mathrm{d}(ax+b)$$
$$=\frac{x}{a}-\frac{b}{a^2}\ln|ax+b|+C$$
$$=\frac{1}{a^2}(ax-b\ln|ax+b|)+C.$$

4. 解 $\int\frac{\mathrm{d}x}{x(ax+b)}=\frac{1}{b}\int\left(\frac{1}{x}-\frac{a}{ax+b}\right)\mathrm{d}x=\frac{1}{b}(\ln|x|-\ln|ax+b|)+C=\frac{1}{b}\ln\left|\frac{x}{ax+b}\right|+C=-\frac{1}{b}\ln\left|\frac{ax+b}{x}\right|+C$.

5. 解 $\int\frac{x}{(ax+b)^2}\mathrm{d}x=\frac{1}{a}\int\frac{ax+b-b}{(ax+b)^2}\mathrm{d}x=\frac{1}{a}\int\left[\frac{1}{ax+b}-\frac{b}{(ax+b)^2}\right]\mathrm{d}x$

$$=\frac{1}{a}\cdot\frac{1}{a}\int\left[\frac{1}{ax+b}-\frac{b}{(ax+b)^2}\right]\mathrm{d}(ax+b)=\frac{1}{a^2}\left(\ln|ax+b|+\frac{b}{ax+b}\right)+C.$$

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6. 解　当 $a>0$，$b>0$ 时，$\int\frac{\mathrm{d}x}{ax^2+b}=\frac{1}{\sqrt{ab}}\int\frac{\mathrm{d}\left(\sqrt{\frac{a}{b}}x\right)}{\left(\sqrt{\frac{a}{b}}x\right)^2+1}=\frac{1}{\sqrt{ab}}\arctan\sqrt{\frac{a}{b}}x+C$；

当 $a>0$，$b<0$ 时，$\int\frac{\mathrm{d}x}{ax^2+b}=\int\frac{\mathrm{d}x}{(\sqrt{a}x+\sqrt{-b})(\sqrt{a}x-\sqrt{-b})}=\frac{1}{2\sqrt{-b}}\int\left(\frac{1}{\sqrt{a}x-\sqrt{-b}}-\frac{1}{\sqrt{a}x+\sqrt{-b}}\right)\mathrm{d}x$

$$=\frac{1}{2\sqrt{-b}}\cdot\frac{1}{\sqrt{a}}\left[\int\frac{1}{\sqrt{a}x-\sqrt{-b}}\mathrm{d}(\sqrt{a}x-\sqrt{-b})-\int\frac{1}{\sqrt{a}x+\sqrt{-b}}\mathrm{d}(\sqrt{a}x+\sqrt{-b})\right]$$

$$=\frac{1}{2\sqrt{-ab}}\ln\left|\frac{\sqrt{a}x-\sqrt{-b}}{\sqrt{a}x+\sqrt{-b}}\right|+C.$$

7. 解　$\int\frac{x}{ax^2+b}\mathrm{d}x=\frac{1}{2a}\int\frac{\mathrm{d}(ax^2+b)}{ax^2+b}=\frac{1}{2a}\ln|ax^2+b|+C.$

8. 解　$\int\frac{\mathrm{d}x}{1-x^2}=\int\frac{\mathrm{d}x}{(1+x)(1-x)}=\frac{1}{2}\int\left(\frac{1}{1+x}+\frac{1}{1-x}\right)\mathrm{d}x$

$$=\frac{1}{2}\int\frac{1}{1+x}\mathrm{d}(1+x)-\frac{1}{2}\int\frac{1}{1-x}\mathrm{d}(1-x)=\frac{1}{2}\ln\left|\frac{1+x}{1-x}\right|+C.$$

9. 解　$\int\frac{x}{1+x^2}\mathrm{d}x=\frac{1}{2}\int\frac{1}{1+x^2}\mathrm{d}(1+x^2)=\frac{1}{2}\ln(1+x^2)+C.$

10. 解　$\int\frac{\mathrm{d}x}{1+x^3}=\int\frac{\mathrm{d}x}{(1+x)(x^2-x+1)}=\frac{1}{3}\int\left(\frac{1}{1+x}+\frac{2-x}{x^2-x+1}\right)\mathrm{d}x=\frac{1}{3}\ln|1+x|+\frac{1}{3}\int\frac{-\frac{1}{2}(2x-1)+\frac{3}{2}}{x^2-x+1}\mathrm{d}x$

$$=\frac{1}{3}\ln|1+x|-\frac{1}{6}\int\frac{\mathrm{d}(x^2-x+1)}{x^2-x+1}+\frac{1}{2}\int\frac{\mathrm{d}x}{x^2-x+1}$$

$$=\frac{1}{3}\ln|1+x|-\frac{1}{6}\ln(x^2-x+1)+\frac{1}{2}\int\frac{\mathrm{d}\left(x-\frac{1}{2}\right)}{\left(x-\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$=\frac{1}{3}\ln|1+x|-\frac{1}{3}\ln\sqrt{x^2-x+1}+\frac{1}{2}\times\frac{2}{\sqrt{3}}\arctan\frac{x-\frac{1}{2}}{\frac{\sqrt{3}}{2}}+C$$

$$=\frac{1}{3}\ln\frac{|1+x|}{\sqrt{x^2-x+1}}+\frac{1}{\sqrt{3}}\arctan\frac{2x-1}{\sqrt{3}}+C.$$

11. 解　$\int\frac{\mathrm{d}x}{1-x^3}=\int\frac{\mathrm{d}x}{(1-x)(1+x+x^2)}=\frac{1}{3}\int\left(\frac{1}{1-x}+\frac{x+2}{1+x+x^2}\right)\mathrm{d}x=\frac{1}{3}\int\frac{1}{1-x}\mathrm{d}x+\frac{1}{3}\int\frac{x+2}{1+x+x^2}\mathrm{d}x$

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$$=-\frac{1}{3}\int\frac{\mathrm{d}(1-x)}{1-x}+\frac{1}{3}\int\frac{\frac{1}{2}(2x+1)+\frac{3}{2}}{1+x+x^2}\mathrm{d}x=-\frac{1}{3}\ln|1-x|+\frac{1}{6}\int\frac{\mathrm{d}(1+x+x^2)}{1+x+x^2}+\frac{1}{2}\int\frac{\mathrm{d}x}{1+x+x^2}$$

$$=-\frac{1}{3}\ln|1-x|+\frac{1}{3}\ln\sqrt{1+x+x^2}+\frac{1}{2}\int\frac{\mathrm{d}\left(x+\frac{1}{2}\right)}{\left(x+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$=\frac{1}{3}\ln\frac{\sqrt{1+x+x^2}}{|1-x|}+\frac{1}{\sqrt{3}}\arctan\frac{2x+1}{\sqrt{3}}+C.$$

12. 解 $$\int\frac{\mathrm{d}x}{1+x^4}=\frac{1}{2}\int\left(\frac{x^2+1}{1+x^4}-\frac{x^2-1}{1+x^4}\right)\mathrm{d}x=\frac{1}{2}\int\frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}\mathrm{d}x-\frac{1}{2}\int\frac{1-\frac{1}{x^2}}{x^2+\frac{1}{x^2}}\mathrm{d}x$$

$$=\frac{1}{2}\int\frac{\mathrm{d}\left(x-\frac{1}{x}\right)}{\left(x-\frac{1}{x}\right)^2+2}-\frac{1}{2}\int\frac{\mathrm{d}\left(x+\frac{1}{x}\right)}{\left(x+\frac{1}{x}\right)^2-2}=\frac{1}{2\sqrt{2}}\arctan\frac{x-\frac{1}{x}}{\sqrt{2}}-\frac{1}{4\sqrt{2}}\ln\left|\frac{x+\frac{1}{x}-\sqrt{2}}{x+\frac{1}{x}+\sqrt{2}}\right|+C$$

$$=\frac{\sqrt{2}}{4}\arctan\frac{\sqrt{2}(x^2-1)}{2x}-\frac{\sqrt{2}}{8}\ln\frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x+1}+C.$$

13. 解 $$\int\frac{\mathrm{d}x}{1-x^4}=\int\frac{\mathrm{d}x}{(1-x)(1+x)(1+x^2)}=\frac{1}{4}\int\frac{1}{1-x}\mathrm{d}x+\frac{1}{4}\int\frac{1}{1+x}\mathrm{d}x+\frac{1}{2}\int\frac{1}{1+x^2}\mathrm{d}x$$

$$=-\frac{1}{4}\int\frac{1}{1-x}\mathrm{d}(1-x)+\frac{1}{4}\int\frac{1}{1+x}\mathrm{d}(1+x)+\frac{1}{2}\int\frac{1}{1+x^2}\mathrm{d}x=\frac{1}{4}\ln\left|\frac{1+x}{1-x}\right|+\frac{1}{2}\arctan x+C.$$

14. 解 $$\int\frac{x}{1+x^3}\mathrm{d}x=\frac{1}{3}\int\left(-\frac{1}{1+x}+\frac{x+1}{x^2-x+1}\right)\mathrm{d}x=-\frac{1}{3}\ln|1+x|+\frac{1}{3}\int\frac{\frac{1}{2}(2x-1)+\frac{3}{2}}{x^2-x+1}\mathrm{d}x$$

$$=-\frac{1}{3}\ln|1+x|+\frac{1}{6}\ln(x^2-x+1)+\frac{1}{2}\int\frac{\mathrm{d}\left(x-\frac{1}{2}\right)}{\left(x-\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$=\frac{1}{6}\ln\frac{x^2-x+1}{(1+x)^2}+\frac{1}{\sqrt{3}}\arctan\frac{2x-1}{\sqrt{3}}+C.$$

15. 解 $$\int\frac{x}{1-x^3}\mathrm{d}x=\frac{1}{3}\int\left(\frac{1}{1-x}+\frac{x-1}{x^2+x+1}\right)\mathrm{d}x=-\frac{1}{3}\int\frac{1}{1-x}\mathrm{d}(1-x)+\frac{1}{3}\int\frac{\frac{1}{2}(2x+1)-\frac{3}{2}}{x^2+x+1}\mathrm{d}x$$

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$$=-\frac{1}{3}\ln|1-x|+\frac{1}{6}\int\frac{\mathrm{d}(x^2+x+1)}{x^2+x+1}-\frac{1}{2}\int\frac{\mathrm{d}\left(x+\frac{1}{2}\right)}{\left(x+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$=\frac{1}{6}\ln\frac{1+x+x^2}{(1-x)^2}-\frac{1}{2}\times\frac{2}{\sqrt{3}}\arctan\frac{x+\frac{1}{2}}{\frac{\sqrt{3}}{2}}+C$$

$$=\frac{1}{6}\ln\frac{1+x+x^2}{(1-x)^2}-\frac{1}{\sqrt{3}}\arctan\frac{2x+1}{\sqrt{3}}+C.$$

16. 解 $\int\frac{x}{1+x^4}\mathrm{d}x=\frac{1}{2}\int\frac{1}{1+(x^2)^2}\mathrm{d}(x^2)=\frac{1}{2}\arctan x^2+C.$

17. 解 $\int\frac{x}{1-x^4}\mathrm{d}x=\int\frac{x}{(1+x^2)(1-x^2)}\mathrm{d}x=\frac{1}{2}\int\frac{x}{1+x^2}\mathrm{d}x+\frac{1}{2}\int\frac{x}{1-x^2}\mathrm{d}x$

$$=\frac{1}{2}\times\frac{1}{2}\int\frac{1}{1+x^2}\mathrm{d}(1+x^2)+\frac{1}{2}\times\left(-\frac{1}{2}\right)\int\frac{1}{1-x^2}\mathrm{d}(1-x^2)$$

$$=\frac{1}{4}\ln(1+x^2)-\frac{1}{4}\ln|1-x^2|+C=\frac{1}{4}\ln\left|\frac{1+x^2}{1-x^2}\right|+C.$$

18. 解 $\int\frac{\mathrm{d}x}{c^2+x^2}=\frac{1}{c}\int\frac{\mathrm{d}\left(\frac{x}{c}\right)}{1+\left(\frac{x}{c}\right)^2}=\frac{1}{c}\arctan\frac{x}{c}+C.$

19. 解 $\int\frac{\mathrm{d}x}{(c^2+x^2)^2}=-\frac{1}{2}\int\frac{1}{x}\mathrm{d}\left(\frac{1}{c^2+x^2}\right)=-\frac{1}{2x(c^2+x^2)}-\frac{1}{2}\int\frac{1}{x^2(c^2+x^2)}\mathrm{d}x$

$$=-\frac{1}{2x(c^2+x^2)}-\frac{1}{2c^2}\int\left(\frac{1}{x^2}-\frac{1}{c^2+x^2}\right)\mathrm{d}x$$

$$=-\frac{1}{2x(c^2+x^2)}-\frac{1}{2c^2}\left(-\frac{1}{x}-\frac{1}{c}\arctan\frac{x}{c}\right)+C$$

$$=\frac{1}{2c^3}\left(\frac{cx}{c^2+x^2}+\arctan\frac{x}{c}\right)+C.$$

20. 解 $\int\frac{\mathrm{d}x}{(c^2+x^2)^3}=-\frac{1}{4}\int\frac{1}{x}\mathrm{d}\left[\frac{1}{(c^2+x^2)^2}\right]=-\frac{1}{4x(c^2+x^2)^2}-\frac{1}{4}\int\frac{1}{x^2(c^2+x^2)^2}\mathrm{d}x$

$$=-\frac{1}{4x(c^2+x^2)^2}-\frac{1}{4}\int\left[\frac{1}{c^4x^2}-\frac{1}{c^4(c^2+x^2)}-\frac{1}{c^2(c^2+x^2)^2}\right]\mathrm{d}x$$

$$=-\frac{1}{4x(c^2+x^2)^2}+\frac{1}{4c^4x}+\frac{1}{4c^5}\arctan\frac{x}{c}+\frac{1}{8c^5}\left(\frac{cx}{c^2+x^2}+\arctan\frac{x}{c}\right)+C$$

$$=\frac{1}{8c^5}\left(\frac{cx}{c^2+x^2}+3\arctan\frac{x}{c}\right)-\frac{1}{4x(c^2+x^2)^2}+\frac{1}{4c^4x}+C.$$

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21. 解 $\int \frac{x}{c^2+x^2}\mathrm{d}x=\frac{1}{2}\int \frac{\mathrm{d}(c^2+x^2)}{c^2+x^2}=\frac{1}{2}\ln(c^2+x^2)+C.$

22. 解 $\int \frac{x}{(c^2+x^2)^2}\mathrm{d}x=-\frac{1}{2}\int \mathrm{d}\left(\frac{1}{c^2+x^2}\right)=-\frac{1}{2(c^2+x^2)}+C.$

23. 解 $\int \frac{x^2}{c^2+x^2}\mathrm{d}x=\int \frac{c^2+x^2-c^2}{c^2+x^2}\mathrm{d}x=\int 1\mathrm{d}x-c^2\int \frac{1}{c^2+x^2}\mathrm{d}x=x-c\arctan\frac{x}{c}+C.$

24. 解 $\int \frac{x^2}{(c^2+x^2)^2}\mathrm{d}x=-\frac{1}{2}\int x\mathrm{d}\left(\frac{1}{c^2+x^2}\right)=-\frac{x}{2(c^2+x^2)}+\frac{1}{2}\int \frac{1}{c^2+x^2}\mathrm{d}x$

$$=-\frac{x}{2(c^2+x^2)}+\frac{1}{2c}\arctan\frac{x}{c}+C.$$

25. 解 $\int \frac{x^3}{c^2+x^2}\mathrm{d}x=\frac{1}{2}\int x^2\mathrm{d}[\ln(c^2+x^2)]=\frac{1}{2}x^2\ln(c^2+x^2)-\int x\ln(c^2+x^2)\,\mathrm{d}x$

$$=\frac{1}{2}x^2\ln(c^2+x^2)-\frac{1}{2}\int \ln(c^2+x^2)\mathrm{d}(c^2+x^2)$$

$$=\frac{1}{2}x^2\ln(c^2+x^2)-\frac{1}{2}(c^2+x^2)\ln(c^2+x^2)+\int x\mathrm{d}x=\frac{x^2}{2}-\frac{c^2}{2}\ln(c^2+x^2)+C.$$

26. 解 $\int \frac{x^3}{(c^2+x^2)^2}\mathrm{d}x=-\frac{1}{2}\int x^2\mathrm{d}\left(\frac{1}{c^2+x^2}\right)=-\frac{x^2}{2(c^2+x^2)}+\int \frac{x}{c^2+x^2}\mathrm{d}x$

$$=-\frac{x^2}{2(c^2+x^2)}+\frac{1}{2}\int \frac{1}{c^2+x^2}\mathrm{d}(c^2+x^2)=-\frac{x^2}{2(c^2+x^2)}+\frac{1}{2}\ln(c^2+x^2)+C_1$$

$$=\frac{c^2}{2(c^2+x^2)}+\frac{1}{2}\ln(c^2+x^2)+C.$$

27. 解 $\int \frac{\mathrm{d}x}{x(c^2+x^2)}=\frac{1}{c^2}\left(\int \frac{1}{x}\mathrm{d}x-\int \frac{x}{c^2+x^2}\mathrm{d}x\right)=\frac{1}{c^2}\left[\ln|x|-\frac{1}{2}\ln(c^2+x^2)\right]+C=\frac{1}{2c^2}\ln\frac{x^2}{c^2+x^2}+C.$

28. 解 $\int \frac{\mathrm{d}x}{x(c^2+x^2)^2}=\int\left[\frac{1}{c^4x}-\frac{x}{c^4(c^2+x^2)}-\frac{x}{c^2(c^2+x^2)^2}\right]\mathrm{d}x$

$$=\frac{1}{c^4}\ln|x|-\frac{1}{2c^4}\ln(c^2+x^2)+\frac{1}{2c^2(c^2+x^2)}+C$$

$$=\frac{1}{2c^2(c^2+x^2)}+\frac{1}{2c^4}\ln\frac{x^2}{c^2+x^2}+C.$$

29. 解 $\int \frac{\mathrm{d}x}{x^2(c^2+x^2)}=\frac{1}{c^2}\left(\int \frac{\mathrm{d}x}{x^2}-\int \frac{\mathrm{d}x}{c^2+x^2}\right)=\frac{1}{c^2}\left(-\frac{1}{x}-\frac{1}{c}\arctan\frac{x}{c}\right)+C$

$$=-\frac{1}{c^2x}-\frac{1}{c^3}\arctan\frac{x}{c}+C.$$

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30. 解 $\int \frac{\mathrm{d}x}{x^2(c^2+x^2)^2}=\int\left[\frac{1}{c^4x^2}-\frac{1}{c^4(c^2+x^2)}-\frac{1}{c^2(c^2+x^2)^2}\right]\mathrm{d}x$

$$=-\frac{1}{c^4x}-\frac{x}{2c^4(c^2+x^2)}-\frac{3}{2c^5}\arctan\frac{x}{c}+C.$$

31. 解 $\int \frac{\mathrm{d}x}{c^2-x^2}=\frac{1}{2c}\int\left(\frac{1}{c-x}+\frac{1}{c+x}\right)\mathrm{d}x=\frac{1}{2c}\left(\ln|c+x|-\ln|c-x|\right)+C=\frac{1}{2c}\ln\left|\frac{c+x}{c-x}\right|+C.$

32. 解 $\int \frac{\mathrm{d}x}{(c^2-x^2)^2}=\frac{1}{2}\int\frac{1}{x}\mathrm{d}\left(\frac{1}{c^2-x^2}\right)=\frac{1}{2x(c^2-x^2)}+\frac{1}{2}\int\frac{\mathrm{d}x}{x^2(c^2-x^2)}$

$$=\frac{1}{2x(c^2-x^2)}+\frac{1}{2c^2}\int\left(\frac{1}{x^2}+\frac{1}{c^2-x^2}\right)\mathrm{d}x=\frac{1}{2x(c^2-x^2)}-\frac{1}{2c^2x}+\frac{1}{4c^3}\ln\left|\frac{c+x}{c-x}\right|+C$$

$$=\frac{x}{2c^2(c^2-x^2)}+\frac{1}{4c^3}\ln\left|\frac{c+x}{c-x}\right|+C.$$

33. 解 $\int \frac{x^2}{c^2-x^2}\mathrm{d}x=-\int\frac{c^2-x^2-c^2}{c^2-x^2}\mathrm{d}x=-\int\left(1-\frac{c^2}{c^2-x^2}\right)\mathrm{d}x=-\int\left[1-\frac{c}{2}\left(\frac{1}{c+x}+\frac{1}{c-x}\right)\right]\mathrm{d}x$

$$=-x+\frac{c}{2}\ln\left|\frac{c+x}{c-x}\right|+C.$$

34. 解 $\int \frac{x^2}{(c^2-x^2)^2}\mathrm{d}x=\frac{1}{2}\int x\mathrm{d}\left(\frac{1}{c^2-x^2}\right)=\frac{x}{2(c^2-x^2)}-\frac{1}{2}\int\frac{1}{c^2-x^2}\mathrm{d}x=\frac{x}{2(c^2-x^2)}-\frac{1}{4c}\ln\left|\frac{c+x}{c-x}\right|+C.$

35. 解 $\int \frac{\mathrm{d}x}{x^2(c^2-x^2)}=\frac{1}{c^2}\left(\int\frac{\mathrm{d}x}{x^2}+\int\frac{\mathrm{d}x}{c^2-x^2}\right)=\frac{1}{c^2}\left(-\frac{1}{x}+\frac{1}{2c}\ln\left|\frac{c+x}{c-x}\right|\right)+C=-\frac{1}{c^2x}+\frac{1}{2c^3}\ln\left|\frac{c+x}{c-x}\right|+C.$

36. 解 $\int \frac{\mathrm{d}x}{x^2-c^2}=\frac{1}{2c}\int\left(\frac{1}{x-c}-\frac{1}{x+c}\right)\mathrm{d}x=\frac{1}{2c}(\ln|x-c|-\ln|x+c|)+C=\frac{1}{2c}\ln\left|\frac{x-c}{x+c}\right|+C.$

37. 解 $\int \frac{x}{x^2-c^2}\mathrm{d}x=\frac{1}{2}\int\frac{\mathrm{d}(x^2-c^2)}{x^2-c^2}=\frac{1}{2}\ln|x^2-c^2|+C.$

38. 解 $\int \frac{a+x}{a^3+x^3}\mathrm{d}x=\int\frac{a+x}{(a+x)(x^2-ax+a^2)}\mathrm{d}x=\int\frac{1}{x^2-ax+a^2}\mathrm{d}x=\int\frac{1}{\left(x-\frac{a}{2}\right)^2+\frac{3}{4}a^2}\mathrm{d}x$

$$=\frac{2}{\sqrt{3}a}\int\frac{\mathrm{d}\left(\frac{x-\frac{a}{2}}{\frac{\sqrt{3}}{2}a}\right)}{\left(\frac{x-\frac{a}{2}}{\frac{\sqrt{3}}{2}a}\right)^2+1}=\frac{2}{\sqrt{3}a}\arctan\frac{2x-a}{\sqrt{3}a}+C.$$

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39. 解 $$\int\frac{a+x}{a^3-x^3}\mathrm{d}x=\int\frac{a+x}{(a-x)(x^2+ax+a^2)}\mathrm{d}x=-\int\left[\frac{2}{3a}\cdot\frac{1}{x-a}-\frac{2x+a}{3a(x^2+ax+a^2)}\right]\mathrm{d}x$$

$$=-\frac{2}{3a}\ln|x-a|+\frac{1}{3a}\int\frac{1}{x^2+ax+a^2}\mathrm{d}(x^2+ax+a^2)$$

$$=-\frac{1}{3a}\ln\frac{(a-x)^2}{x^2+ax+a^2}+C.$$

40. 解 $$\int\frac{a-x}{a^3+x^3}\mathrm{d}x=\int\frac{a-x}{(a+x)(x^2-ax+a^2)}\mathrm{d}x=\int\left[\frac{2}{3a}\cdot\frac{1}{x+a}-\frac{2x-a}{3a(x^2-ax+a^2)}\right]\mathrm{d}x$$

$$=\frac{2}{3a}\ln|x+a|-\frac{1}{3a}\int\frac{1}{x^2-ax+a^2}\mathrm{d}(x^2-ax+a^2)$$

$$=\frac{1}{3a}\ln\frac{(a+x)^2}{x^2-ax+a^2}+C.$$

41. 解 $$\int\frac{a-x}{a^3-x^3}\mathrm{d}x=\int\frac{a-x}{(a-x)(x^2+ax+a^2)}\mathrm{d}x=\int\frac{\mathrm{d}x}{x^2+ax+a^2}=\int\frac{\mathrm{d}x}{\left(x+\frac{a}{2}\right)^2+\frac{3a^2}{4}}$$

$$=\frac{2}{\sqrt{3}a}\int\frac{\mathrm{d}\left(\frac{x+\frac{a}{2}}{\frac{\sqrt{3}}{2}a}\right)}{\left(\frac{x+\frac{a}{2}}{\frac{\sqrt{3}}{2}a}\right)^2+1}=\frac{2}{\sqrt{3}a}\arctan\frac{2x+a}{\sqrt{3}a}+C.$$

42. 解 令被积函数分母 $ax^2+bx+c=0$，则 $\Delta=b^2-4ac$.

①若 $\Delta>0$，则 $x_1=\dfrac{-b+\sqrt{b^2-4ac}}{2a}$，$x_2=\dfrac{-b-\sqrt{b^2-4ac}}{2a}$，

$$ax^2+bx+c=a\left(x-\frac{-b+\sqrt{b^2-4ac}}{2a}\right)\left(x-\frac{-b-\sqrt{b^2-4ac}}{2a}\right)$$

$$=a\left(x+\frac{b-\sqrt{b^2-4ac}}{2a}\right)\left(x+\frac{b+\sqrt{b^2-4ac}}{2a}\right),$$

$$\int\frac{\mathrm{d}x}{ax^2+bx+c}=\frac{1}{a}\int\frac{\mathrm{d}x}{\left(x+\frac{b-\sqrt{b^2-4ac}}{2a}\right)\left(x+\frac{b+\sqrt{b^2-4ac}}{2a}\right)}$$

$$=\frac{1}{\sqrt{b^2-4ac}}\int\left(\frac{1}{x+\frac{b-\sqrt{b^2-4ac}}{2a}}-\frac{1}{x+\frac{b+\sqrt{b^2-4ac}}{2a}}\right)\mathrm{d}x$$

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$$=\frac{1}{\sqrt{b^2-4ac}}\ln\left|\frac{2ax+b-\sqrt{b^2-4ac}}{2ax+b+\sqrt{b^2-4ac}}\right|+C;$$

②若 $\Delta<0$，则

$$\int\frac{\mathrm{d}x}{ax^2+bx+c}=\int\frac{\mathrm{d}x}{a\left(x+\frac{b}{2a}\right)^2+c-\frac{b^2}{4a}}=\frac{2}{\sqrt{4ac-b^2}}\int\frac{\mathrm{d}\left(\frac{2ax+b}{\sqrt{4ac-b^2}}\right)}{\left(\frac{2ax+b}{\sqrt{4ac-b^2}}\right)^2+1}$$

$$=\frac{2}{\sqrt{4ac-b^2}}\arctan\frac{2ax+b}{\sqrt{4ac-b^2}}+C;$$

③若 $\Delta=0$，则 $x_1=x_2=-\frac{b}{2a}$，

$$ax^2+bx+c=a\left(x+\frac{b}{2a}\right)^2,$$

$$\int\frac{\mathrm{d}x}{ax^2+bx+c}=\frac{1}{a}\int\frac{\mathrm{d}x}{\left(x+\frac{b}{2a}\right)^2}=-\frac{1}{a}\cdot\frac{1}{x+\frac{b}{2a}}+C=-\frac{2}{2ax+b}+C.$$

第二组

1. 解　$\int\sqrt{ax+b}\,\mathrm{d}x=\frac{1}{a}\int\sqrt{ax+b}\mathrm{d}(ax+b)=\frac{1}{a}\cdot\frac{2}{3}(ax+b)^{\frac{3}{2}}+C=\frac{2}{3a}\sqrt{(ax+b)^3}+C.$

2. 解　令 $\sqrt{ax+b}=t(t\geqslant 0)$，则 $x=\frac{t^2-b}{a}$，$\mathrm{d}x=\frac{2t}{a}\mathrm{d}t$，$x\sqrt{ax+b}=\frac{t^2-b}{a}\cdot t$.

所以　$$\int x\sqrt{ax+b}\mathrm{d}x=\int\frac{t^2-b}{a}\cdot t\cdot\frac{2t}{a}\mathrm{d}t=\frac{2}{a^2}\int(t^4-bt^2)\mathrm{d}t$$

$$=\frac{2}{5a^2}\cdot t^5-\frac{2b}{3a^2}\cdot t^3+C=\frac{2t^3}{15a^2}(3t^2-5b)+C.$$

将 $t=\sqrt{ax+b}$ 代入上式，得

$$\int x\sqrt{ax+b}\mathrm{d}x=\frac{2}{15a^2}\left[3(ax+b)-5b\right]\cdot\sqrt{(ax+b)^3}+C$$

$$=\frac{2}{15a^2}(3ax-2b)\cdot\sqrt{(ax+b)^3}+C.$$

3. 解　令 $\sqrt{ax+b}=t(t\geqslant 0)$，则 $x=\frac{t^2-b}{a}$，$\mathrm{d}x=\frac{2t}{a}\mathrm{d}t$.

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故 $$\int\frac{\mathrm{d}x}{x\sqrt{ax+b}}=\int\frac{1}{\frac{t^2-b}{a}\cdot t}\cdot\frac{2t}{a}\mathrm{d}t=\int\frac{2}{t^2-b}\mathrm{d}t.$$

①当 $b>0$ 时，$\int\frac{2}{t^2-b}\mathrm{d}t=2\int\frac{1}{t^2-(\sqrt{b})^2}\mathrm{d}t=\frac{1}{\sqrt{b}}\ln\left|\frac{t-\sqrt{b}}{t+\sqrt{b}}\right|+C$.

将 $t=\sqrt{ax+b}$ 代入上式，得 $\int\frac{\mathrm{d}x}{x\sqrt{ax+b}}=\frac{1}{\sqrt{b}}\ln\left|\frac{\sqrt{ax+b}-\sqrt{b}}{\sqrt{ax+b}+\sqrt{b}}\right|+C$.

②当 $b<0$ 时，$\int\frac{2}{t^2-b}\mathrm{d}t=2\int\frac{1}{t^2+(\sqrt{-b})^2}\mathrm{d}t=\frac{2}{\sqrt{-b}}\arctan\frac{t}{\sqrt{-b}}+C$.

将 $t=\sqrt{ax+b}$ 代入上式，得 $\int\frac{\mathrm{d}x}{x\sqrt{ax+b}}=\frac{2}{\sqrt{-b}}\arctan\sqrt{\frac{ax+b}{-b}}+C$.

综上，得

$$\int\frac{\mathrm{d}x}{x\sqrt{ax+b}}=\begin{cases}\frac{1}{\sqrt{b}}\ln\left|\frac{\sqrt{ax+b}-\sqrt{b}}{\sqrt{ax+b}+\sqrt{b}}\right|+C, & b>0,\\ \frac{2}{\sqrt{-b}}\arctan\sqrt{\frac{ax+b}{-b}}+C, & b<0.\end{cases}$$

4. 解 $$\int\frac{\sqrt{ax+b}}{x^2}\mathrm{d}x=-\int\sqrt{ax+b}\,\mathrm{d}\left(\frac{1}{x}\right)=-\frac{\sqrt{ax+b}}{x}+\int\frac{1}{x}\mathrm{d}(\sqrt{ax+b})$$

$$=-\frac{\sqrt{ax+b}}{x}+\int\frac{1}{x}\cdot(ax+b)^{-\frac{1}{2}}\cdot\frac{a}{2}\mathrm{d}x=-\frac{\sqrt{ax+b}}{x}+\frac{a}{2}\int\frac{\mathrm{d}x}{x\sqrt{ax+b}}.$$

5. 解 $$\int\frac{x}{\sqrt{x^2-a^2}}\mathrm{d}x=\frac{1}{2}\int\frac{1}{\sqrt{x^2-a^2}}\mathrm{d}(x^2-a^2)=\frac{1}{2}\cdot2(x^2-a^2)^{\frac{1}{2}}+C=\sqrt{x^2-a^2}+C.$$

6. 解 $$\int\frac{x}{\sqrt{(x^2-a^2)^3}}\mathrm{d}x=\frac{1}{2}\int\frac{1}{\sqrt{(x^2-a^2)^3}}\mathrm{d}(x^2-a^2)=\frac{1}{2}\cdot(-2)\cdot(x^2-a^2)^{-\frac{1}{2}}+C=-\frac{1}{\sqrt{x^2-a^2}}+C.$$

7. 解 $$\int\frac{x^2}{\sqrt{x^2-a^2}}\mathrm{d}x=\int\frac{x^2-a^2+a^2}{\sqrt{x^2-a^2}}\mathrm{d}x=\int\sqrt{x^2-a^2}\,\mathrm{d}x+a^2\int\frac{1}{\sqrt{x^2-a^2}}\mathrm{d}x,$$

$$\int\sqrt{x^2-a^2}\mathrm{d}x=\frac{x}{2}\cdot\sqrt{x^2-a^2}-\frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right|+C_1, \quad ①$$

$$a^2\int\frac{\mathrm{d}x}{\sqrt{x^2-a^2}}=a^2\ln\left|x+\sqrt{x^2-a^2}\right|+C_2, \quad ②$$

由①+②得

$$\int\frac{x^2}{\sqrt{x^2-a^2}}\mathrm{d}x=\frac{x}{2}\sqrt{x^2-a^2}+\frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right|+C.$$

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8. 解　当$x>a$时，令$x=a\sec t\left(0<t<\frac{\pi}{2}\right)$，$\mathrm{d}x=a\sec t\tan t\mathrm{d}t$，

$$原式=\int\frac{a\sec t\cdot\tan t}{a\sec t\cdot a\tan t}\mathrm{d}t=\frac{1}{a}\int 1\mathrm{d}t=\frac{t}{a}+C=\frac{1}{a}\arccos\frac{a}{x}+C.$$

同理可得，当$x<-a$时，

$$原式=\frac{1}{a}\arccos\left(-\frac{a}{x}\right)+C.$$

综上，原式$=\frac{1}{a}\arccos\frac{a}{|x|}+C.$

9. 解　令$x=a\sec t\left(0<t<\frac{\pi}{2}\right)$，$\mathrm{d}x=a\sec t\tan t\mathrm{d}t$，

$$原式=\int\frac{a\sec t\tan t}{a^2\sec^2 t\cdot a\tan t}\mathrm{d}t=\frac{1}{a^2}\int\cos t\,\mathrm{d}t=\frac{1}{a^2}\cdot\sin t+C=\frac{\sqrt{x^2-a^2}}{a^2x}+C.$$

10. 解　$\int x\sqrt{x^2-a^2}\mathrm{d}x=\frac{1}{2}\int(x^2-a^2)^{\frac{1}{2}}\mathrm{d}(x^2-a^2)=\frac{1}{3}\sqrt{(x^2-a^2)^3}+C.$

11. 解　当$x>a$时，令$x=a\sec t\left(0<t<\frac{\pi}{2}\right)$，则$\frac{\sqrt{x^2-a^2}}{x}=\frac{a\tan t}{a\sec t}$，$\mathrm{d}x=a\sec t\tan t\mathrm{d}t$，

$$\int\frac{\sqrt{x^2-a^2}}{x}\mathrm{d}x=\int\frac{a\tan t}{a\sec t}a\sec t\tan t\mathrm{d}t=\int a\tan^2 t\mathrm{d}t=a\int\frac{\sin^2 t}{\cos^2 t}\mathrm{d}t=a\int\frac{1-\cos^2 t}{\cos^2 t}\mathrm{d}t$$
$$=a\int\frac{1}{\cos^2 t}\mathrm{d}t-a\int\mathrm{d}t=a\tan t-at+C=\sqrt{x^2-a^2}-a\arccos\frac{a}{x}+C,$$

同理可得当$x<-a$时，

$$\int\frac{\sqrt{x^2-a^2}}{x}\mathrm{d}x=\sqrt{x^2-a^2}-a\arccos\frac{a}{-x}+C.$$

综上，
$$\int\frac{\sqrt{x^2-a^2}}{x}\mathrm{d}x=\sqrt{x^2-a^2}-a\arccos\frac{a}{|x|}+C.$$

12. 解　$\int\frac{\sqrt{x^2-a^2}}{x^2}\mathrm{d}x=-\int\sqrt{x^2-a^2}\mathrm{d}\left(\frac{1}{x}\right)=-\frac{\sqrt{x^2-a^2}}{x}+\int\frac{1}{x}\mathrm{d}(\sqrt{x^2-a^2})$

$$=-\frac{\sqrt{x^2-a^2}}{x}+\int\frac{1}{x}\frac{1}{2}\cdot 2x(x^2-a^2)^{-\frac{1}{2}}\mathrm{d}x=-\frac{\sqrt{x^2-a^2}}{x}+\int\frac{1}{\sqrt{x^2-a^2}}\mathrm{d}x,$$

又
$$\int\frac{1}{\sqrt{x^2-a^2}}\mathrm{d}x=\ln\left|x+\sqrt{x^2-a^2}\right|+C,$$

所以原式$=-\frac{\sqrt{x^2-a^2}}{x}+\ln\left|x+\sqrt{x^2-a^2}\right|+C.$

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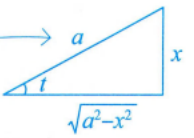
13. 解　令$x=a\sin t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，则$\mathrm{d}x=a\cos t\mathrm{d}t$.



又$-\frac{\pi}{2}<t<\frac{\pi}{2}$，$\cos t>0$，则

$$\int\frac{\mathrm{d}x}{\sqrt{(a^2-x^2)^3}}=\int\frac{a\cos t}{a^3\cos^3 t}\mathrm{d}t=\frac{1}{a^2}\int\frac{1}{\cos^2 t}\mathrm{d}t=\frac{1}{a^2}\tan t+C=\frac{x}{a^2\sqrt{a^2-x^2}}+C.$$

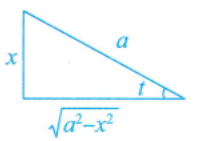
14. 解　$$\int\frac{x}{\sqrt{a^2-x^2}}\mathrm{d}x=-\frac{1}{2}\int(a^2-x^2)^{-\frac{1}{2}}\mathrm{d}(a^2-x^2)$$

$$=-\frac{1}{2}\cdot\frac{1}{1-\frac{1}{2}}(a^2-x^2)^{1-\frac{1}{2}}+C=-\sqrt{a^2-x^2}+C.$$

15. 解　$$\int\frac{x}{\sqrt{(a^2-x^2)^3}}\mathrm{d}x=-\frac{1}{2}\int(a^2-x^2)^{-\frac{3}{2}}\mathrm{d}(a^2-x^2)$$

$$=-\frac{1}{2}\cdot\frac{1}{-\frac{1}{2}}(a^2-x^2)^{-\frac{1}{2}}+C=\frac{1}{\sqrt{a^2-x^2}}+C.$$

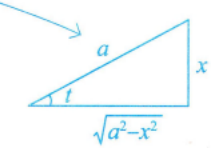
16. 解　令$x=a\sin t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，则$\mathrm{d}x=a\cos t\mathrm{d}t$，故



$$\int\frac{x^2}{\sqrt{a^2-x^2}}\mathrm{d}x=\int\frac{a^2\sin^2 t}{a\cos t}a\cos t\mathrm{d}t=a^2\int\sin^2 t\mathrm{d}t=a^2\int\frac{1-\cos 2t}{2}\mathrm{d}t$$

$$=\frac{1}{2}a^2t-\frac{a^2}{4}\sin 2t+C=\frac{a^2}{2}\arcsin\frac{x}{a}-\frac{x}{2}\sqrt{a^2-x^2}+C.$$

17. 解　令$x=a\sin t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，则$\mathrm{d}x=a\cos t\mathrm{d}t$. 故

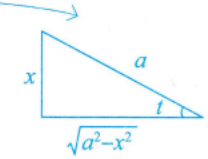


$$\int\frac{x^2}{\sqrt{(a^2-x^2)^3}}\mathrm{d}x=\int\frac{\sin^2 t}{a\cos^3 t}\cdot a\cos t\mathrm{d}t=\int\frac{\sin^2 t}{\cos^2 t}\mathrm{d}t=\int\frac{1-\cos^2 t}{\cos^2 t}\mathrm{d}t$$

$$=\int\frac{1}{\cos^2 t}\mathrm{d}t-\int 1\mathrm{d}t=\tan t-t+C$$

$$=\frac{x}{\sqrt{a^2-x^2}}-\arcsin\frac{x}{a}+C.$$

18. 解　令$x=a\sin t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，则$\mathrm{d}x=a\cos t\mathrm{d}t$. 故



$$\int\frac{1}{x^2\sqrt{a^2-x^2}}\mathrm{d}x=\int\frac{1}{a^3\sin^2 t\cos t}\cdot a\cos t\mathrm{d}t=\frac{1}{a^2}\int\frac{1}{\sin^2 t}\mathrm{d}t$$

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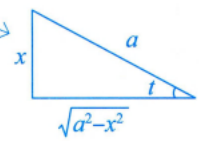
$$=-\frac{1}{a^2}\cot t+C=-\frac{\sqrt{a^2-x^2}}{a^2x}+C.$$

19. 解 $\int x\sqrt{a^2-x^2}\,\mathrm{d}x=-\frac{1}{2}\int(a^2-x^2)^{\frac{1}{2}}\mathrm{d}(a^2-x^2)$

$$=-\frac{1}{2}\cdot\frac{2}{3}(a^2-x^2)^{\frac{3}{2}}+C=-\frac{1}{3}\sqrt{(a^2-x^2)^3}+C.$$

20. 解 令 $x=a\sin t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，$\mathrm{d}x=a\cos t\mathrm{d}t$，



$$\frac{\sqrt{a^2-x^2}}{x^2}=\frac{a\cos t}{a^2\sin^2 t}=\frac{\cos t}{a\sin^2 t},$$

所以 $$\int\frac{\sqrt{a^2-x^2}}{x^2}\mathrm{d}x=\int\frac{\cos t}{a\sin^2 t}a\cos t\mathrm{d}t=\int\frac{\cos^2 t}{\sin^2 t}\mathrm{d}t=\int\frac{1-\sin^2 t}{\sin^2 t}\mathrm{d}t=\int\csc^2 t\mathrm{d}t-\int 1\mathrm{d}t$$

$$=-\cot t-t+C=-\frac{\sqrt{a^2-x^2}}{x}-\arcsin\frac{x}{a}+C.$$

21. 解 $\int\sqrt{(a^2-x^2)^3}\,\mathrm{d}x=x(a^2-x^2)^{\frac{3}{2}}-\int x\mathrm{d}\left[(a^2-x^2)^{\frac{3}{2}}\right]=x(a^2-x^2)^{\frac{3}{2}}-\int x\cdot\frac{3}{2}(a^2-x^2)^{\frac{1}{2}}\cdot(-2x)\mathrm{d}x$

$$=x(a^2-x^2)^{\frac{3}{2}}+3\int x^2(a^2-x^2)^{\frac{1}{2}}\mathrm{d}x=x(a^2-x^2)^{\frac{3}{2}}+3\int(x^2-a^2+a^2)(a^2-x^2)^{\frac{1}{2}}\mathrm{d}x$$

$$=x(a^2-x^2)^{\frac{3}{2}}-3\int(a^2-x^2)^{\frac{3}{2}}\mathrm{d}x+3a^2\int(a^2-x^2)^{\frac{1}{2}}\mathrm{d}x,$$

移项整理得

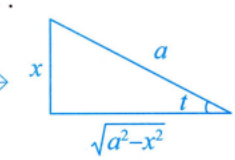
$$\int\sqrt{(a^2-x^2)^3}\,\mathrm{d}x=\frac{x}{4}(a^2-x^2)^{\frac{3}{2}}+\frac{3a^2}{4}\int(a^2-x^2)^{\frac{1}{2}}\mathrm{d}x.$$

又 $$\int(a^2-x^2)^{\frac{1}{2}}\mathrm{d}x=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\arcsin\frac{x}{a}+C_1,$$

代入得 $$\int\sqrt{(a^2-x^2)^3}\,\mathrm{d}x=\frac{x}{8}(5a^2-2x^2)\sqrt{a^2-x^2}+\frac{3}{8}a^4\arcsin\frac{x}{a}+C$$

$$=\frac{1}{4}\left[x\sqrt{(a^2-x^2)^3}+\frac{3a^2x}{2}\sqrt{a^2-x^2}+\frac{3a^4}{2}\arcsin\frac{x}{a}\right]+C.$$

22. 解 令 $x=a\sin t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，$\mathrm{d}x=a\cos t\mathrm{d}t$，$x^2\sqrt{a^2-x^2}=a^2\sin^2 t\cdot a\cos t$.



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$$\int x^2\sqrt{a^2-x^2}\,\mathrm{d}x=\int a^2\sin^2 t\cdot a\cos t\cdot a\cos t\mathrm{d}t=\int a^4\sin^2 t\cos^2 t\mathrm{d}t$$
$$=\int a^4\sin^2 t(1-\sin^2 t)\mathrm{d}t=\underbrace{a^4\int\sin^2 t\mathrm{d}t}_{①}-\underbrace{a^4\int\sin^4 t\mathrm{d}t}_{②}.$$

$$①=a^4\int\frac{1-\cos 2t}{2}\mathrm{d}t=\frac{a^4}{2}t-\frac{a^4}{4}\sin 2t+C_1,$$

$$②=a^4\int\left(\frac{1-\cos 2t}{2}\right)^2\mathrm{d}t=a^4\int\frac{1+\cos^2 2t-2\cos 2t}{4}\mathrm{d}t=a^4\int\left(\frac{1}{4}-\frac{1}{2}\cos 2t+\frac{\cos^2 2t}{4}\right)\mathrm{d}t$$
$$=a^4\int\left(\frac{1}{4}-\frac{1}{2}\cos 2t+\frac{1+\cos 4t}{8}\right)\mathrm{d}t=a^4\int\left(\frac{3}{8}-\frac{1}{2}\cos 2t+\frac{1}{8}\cos 4t\right)\mathrm{d}t=\frac{3}{8}a^4t-\frac{a^4}{4}\sin 2t+\frac{a^4}{32}\sin 4t+C_2.$$

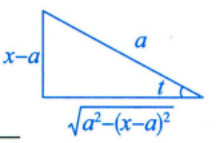
①−②得，原式$=\frac{1}{8}a^4t-\frac{a^4}{32}\sin 4t+C=-\frac{x}{4}\sqrt{(a^2-x^2)^3}+\frac{a^2}{8}\left(x\sqrt{a^2-x^2}+a^2\arcsin\frac{x}{a}\right)+C.$

23. 解　$\int\sqrt{2ax-x^2}\mathrm{d}x=\int\sqrt{-(x^2-2ax)}\mathrm{d}x=\int\sqrt{a^2-(x-a)^2}\mathrm{d}x.$

令$x-a=a\sin t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，$\mathrm{d}x=a\cos t\mathrm{d}t$，则

$$\text{原式}=\int a\cos t\cdot a\cos t\mathrm{d}t=\int a^2\cos^2 t\mathrm{d}t=a^2\int\frac{1+\cos 2t}{2}\mathrm{d}t=\frac{a^2}{2}t+\frac{a^2}{4}\sin 2t+C,$$

由$t=\arcsin\frac{x-a}{a}$，$\sin t=\frac{x-a}{a}$，$\cos t=\frac{\sqrt{a^2-(x-a)^2}}{a}$，有



$$\text{原式}=\frac{a^2}{2}\arcsin\frac{x-a}{a}+\frac{a^2}{4}\cdot 2\cdot\frac{x-a}{a}\cdot\frac{\sqrt{a^2-(x-a)^2}}{a}+C$$
$$=\frac{a^2}{2}\arcsin\frac{x-a}{a}+\frac{x-a}{2}\sqrt{a^2-(x-a)^2}+C.$$

24. 解　$\int\frac{1}{\sqrt{2ax-x^2}}\mathrm{d}x=\int\frac{1}{\sqrt{a^2-(x-a)^2}}\mathrm{d}x.$

令$x-a=a\cos t\ (0<t<\pi)$，$\mathrm{d}x=-a\sin t\mathrm{d}t$，$\sqrt{a^2-(x-a)^2}=a\sin t$，

$$\text{原式}=\int\frac{1}{a\sin t}\cdot(-a\sin t)\mathrm{d}t=-\int 1\mathrm{d}t=-t+C=-\arccos\frac{x-a}{a}+C.$$

同理，令$x-a=a\sin t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，$\mathrm{d}x=a\cos t\mathrm{d}t$，$\sqrt{a^2-(x-a)^2}=a\cos t$，

$$\text{原式}=\int\frac{1}{a\cos t}\cdot a\cos t\mathrm{d}t=\int 1\mathrm{d}t=t+C=\arcsin\frac{x-a}{a}+C.$$

25. 解　$\int\frac{1}{\sqrt{2ax+x^2}}\mathrm{d}x=\int\frac{1}{\sqrt{(x+a)^2-a^2}}\mathrm{d}x.$

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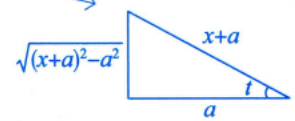
令 $x+a=a\sec t\left(0<t<\frac{\pi}{2}\right)$，$\mathrm{d}x=a\sec t\tan t\mathrm{d}t$.



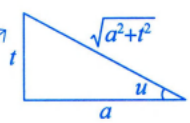
$$\text{原式}=\int\frac{a\sec t\tan t}{a\tan t}\mathrm{d}t=\int\sec t\mathrm{d}t=\ln|\sec t+\tan t|+C_1$$

$$=\ln\left|\frac{x+a}{a}+\frac{\sqrt{2ax+x^2}}{a}\right|+C_1=\ln\left|x+a+\sqrt{2ax+x^2}\right|+C.$$

26. 解　令 $t=x^{\frac{n}{2}}$，$\ln t=\frac{n}{2}\ln x$，$\frac{1}{t}\mathrm{d}t=\frac{n}{2}\frac{1}{x}\mathrm{d}x$，则

$$\int\frac{1}{x\sqrt{x^n+a^2}}\mathrm{d}x=\frac{2}{n}\int\frac{1}{t\sqrt{t^2+a^2}}\mathrm{d}t.$$

令 $t=a\tan u\left(-\frac{\pi}{2}<u<\frac{\pi}{2}\right)$，$\mathrm{d}t=a\sec^2 u\mathrm{d}u$，则



一手+V：kaoyan33311

$$\text{原式}=\frac{2}{n}\int\frac{1}{a\tan u\cdot a\sec u}a\sec^2 u\mathrm{d}u=\frac{2}{an}\int\frac{1}{\sin u}\mathrm{d}u$$

$$=\frac{2}{an}\ln|\csc u-\cot u|+C=\frac{2}{an}\ln\left|\frac{\sqrt{a^2+t^2}}{t}-\frac{a}{t}\right|+C$$

$$=\frac{2}{an}\ln\left|\frac{\sqrt{a^2+x^n}}{x^{\frac{n}{2}}}-\frac{a}{x^{\frac{n}{2}}}\right|+C=\frac{2}{an}\ln\frac{\sqrt{a^2+x^n}-a}{\sqrt{x^n}}+C.$$

27. 解　令 $t=x^{\frac{n}{2}}$，$\ln t=\frac{n}{2}\ln x$，$\frac{1}{t}\mathrm{d}t=\frac{n}{2}\frac{1}{x}\mathrm{d}x$，则

$$\int\frac{1}{x\sqrt{x^n-a^2}}\mathrm{d}x=\frac{2}{n}\int\frac{1}{t}\frac{1}{\sqrt{t^2-a^2}}\mathrm{d}t.$$

令 $t=a\csc u\left(0<u<\frac{\pi}{2}\right)$，$\mathrm{d}t=-a\csc u\cot u\mathrm{d}u$，则

$$\text{原式}=-\frac{2}{n}\int\frac{1}{a\csc u\cdot a\cot u}a\csc u\cot u\mathrm{d}u=-\frac{2}{n}\int\frac{1}{a}\mathrm{d}u=-\frac{2}{an}u+C=-\frac{2}{an}\arcsin\frac{a}{t}+C$$

$$=-\frac{2}{an}\arcsin\frac{a}{\sqrt{x^n}}+C.$$

28. 解　令 $t=\sqrt{x}$，$x=t^2$，$\mathrm{d}x=2t\mathrm{d}t$.

$$\int\sqrt{\frac{x}{a^3-x^3}}\mathrm{d}x=\int\frac{t}{\sqrt{a^3-t^6}}2t\mathrm{d}t=\frac{2}{3}\int\frac{1}{\sqrt{a^3-t^6}}\mathrm{d}(t^3),$$

令 $t^3=u$，则

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$$原式=\frac{2}{3}\int\frac{1}{\sqrt{a^3-u^2}}\mathrm{d}u,$$

令 $u=a^{\frac{3}{2}}\sin z\left(-\frac{\pi}{2}<z<\frac{\pi}{2}\right)$，则 $\mathrm{d}u=a^{\frac{3}{2}}\cos z\mathrm{d}z$.

$$原式=\frac{2}{3}\int\frac{1}{a^{\frac{3}{2}}\cos z}a^{\frac{3}{2}}\cos z\mathrm{d}z=\frac{2}{3}\int 1\mathrm{d}z=\frac{2}{3}z+C$$

$$=\frac{2}{3}\arcsin\frac{u}{a^{\frac{3}{2}}}+C=\frac{2}{3}\arcsin\frac{t^3}{a^{\frac{3}{2}}}+C=\frac{2}{3}\arcsin\left(\frac{x}{a}\right)^{\frac{3}{2}}+C.$$

29. 解 $\int\sqrt{\frac{a+bx}{a-bx}}\mathrm{d}x=\int\sqrt{\frac{(a+bx)^2}{(a-bx)(a+bx)}}\mathrm{d}x=\int\sqrt{\frac{(a+bx)^2}{a^2-b^2x^2}}\mathrm{d}x.$

令 $x=\frac{a}{b}\sin t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，$\mathrm{d}x=\frac{a}{b}\cos t\mathrm{d}t$，$t=\arcsin\frac{bx}{a}$，$\cos t=\frac{\sqrt{a^2-b^2x^2}}{a}$，则

$$\int\sqrt{\frac{(a+bx)^2}{a^2-b^2x^2}}\mathrm{d}x=\int\sqrt{\frac{\left(a+b\cdot\frac{a}{b}\sin t\right)^2}{a^2(1-\sin^2 t)}}\cdot\frac{a}{b}\cos t\mathrm{d}t=\int\sqrt{\frac{a^2(1+\sin t)^2}{a^2\cos^2 t}}\cdot\frac{a}{b}\cos t\mathrm{d}t$$

$$=\frac{a}{b}\int(1+\sin t)\,\mathrm{d}t=\frac{a}{b}(t-\cos t)+C.$$

将 t 反代回去，得

$$原式=\frac{a}{b}\left(\arcsin\frac{bx}{a}-\frac{\sqrt{a^2-b^2x^2}}{a}\right)+C$$

$$=-\frac{1}{b}\sqrt{(a+bx)(a-bx)}+\frac{a}{b}\arcsin\frac{bx}{a}+C.$$

30. 解 $\int\sqrt{\frac{a-bx}{a+bx}}\mathrm{d}x=\int\sqrt{\frac{(a-bx)^2}{(a+bx)(a-bx)}}\mathrm{d}x=\int\sqrt{\frac{(a-bx)^2}{a^2-b^2x^2}}\mathrm{d}x.$

令 $x=\frac{a}{b}\sin t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，$\mathrm{d}x=\frac{a}{b}\cos t\mathrm{d}t$，$t=\arcsin\frac{bx}{a}$，$\cos t=\frac{\sqrt{a^2-b^2x^2}}{a}$，则

$$\int\sqrt{\frac{(a-bx)^2}{a^2-b^2x^2}}\mathrm{d}x=\int\sqrt{\frac{\left(a-b\cdot\frac{a}{b}\sin t\right)^2}{a^2(1-\sin^2 t)}}\cdot\frac{a}{b}\cos t\mathrm{d}t=\int\sqrt{\frac{a^2(1-\sin t)^2}{a^2\cos^2 t}}\cdot\frac{a}{b}\cos t\mathrm{d}t=\frac{a}{b}\int(1-\sin t)\,\mathrm{d}t$$

$$=\frac{a}{b}(t+\cos t)+C=\frac{a}{b}\left(\arcsin\frac{bx}{a}+\frac{\sqrt{a^2-b^2x^2}}{a}\right)+C$$

$$=\frac{1}{b}\sqrt{(a+bx)(a-bx)}+\frac{a}{b}\arcsin\frac{bx}{a}+C.$$

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31. 解 $\int\sqrt{\frac{1+x}{1-x}}\mathrm{d}x=\int\sqrt{\frac{(1+x)^2}{(1-x)(1+x)}}\mathrm{d}x=\int\sqrt{\frac{(1+x)^2}{1-x^2}}\mathrm{d}x$.

令 $x=\sin t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，$\mathrm{d}x=\cos t\mathrm{d}t$，$t=\arcsin x$，$\cos t=\sqrt{1-x^2}$，则

$$\int\sqrt{\frac{(1+x)^2}{1-x^2}}\mathrm{d}x=\int\sqrt{\frac{(1+\sin t)^2}{\cos^2 t}}\cdot\cos t\mathrm{d}t=\int(1+\sin t)\,\mathrm{d}t=t-\cos t+C=\arcsin x-\sqrt{1-x^2}+C.$$

32. 解 令 $x=a\tan t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，$\mathrm{d}x=a\sec^2 t\mathrm{d}t$，$\sec t=\frac{\sqrt{x^2+a^2}}{a}$.

$$\int\frac{\mathrm{d}x}{\sqrt{x^2+a^2}}=\int\frac{a\sec^2 t}{\sqrt{a^2(1+\tan^2 t)}}\mathrm{d}t=\int\sec t\,\mathrm{d}t=\ln|\sec t+\tan t|+C_1$$

$$=\ln\left|\frac{\sqrt{x^2+a^2}}{a}+\frac{x}{a}\right|+C_1=\ln(x+\sqrt{x^2+a^2})+C.$$

33. 解 令 $x=a\tan t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，$\mathrm{d}x=a\sec^2 t\mathrm{d}t$.

$$\int\frac{\mathrm{d}x}{\sqrt{(x^2+a^2)^3}}=\int\frac{1}{a^3\sec^3 t}\cdot a\sec^2 t\mathrm{d}t=\frac{1}{a^2}\int\frac{1}{\sec t}\mathrm{d}t=\frac{1}{a^2}\int\cos t\mathrm{d}t=\frac{1}{a^2}\sin t+C=\frac{x}{a^2\sqrt{x^2+a^2}}+C.$$

34. 解 $\int\frac{x}{\sqrt{x^2+a^2}}\mathrm{d}x=\frac{1}{2}\int\frac{1}{\sqrt{x^2+a^2}}\mathrm{d}(x^2+a^2)=\frac{1}{2}\times2(x^2+a^2)^{\frac{1}{2}}+C=\sqrt{x^2+a^2}+C$.

35. 解 $\int\frac{x}{\sqrt{(x^2+a^2)^3}}\mathrm{d}x=\frac{1}{2}\int\frac{1}{\sqrt{(x^2+a^2)^3}}\mathrm{d}(x^2+a^2)=\frac{1}{2}\cdot(-2)\cdot(x^2+a^2)^{-\frac{1}{2}}+C=-\frac{1}{\sqrt{x^2+a^2}}+C$.

36. 解 $$\int\frac{x^2}{\sqrt{x^2+a^2}}\mathrm{d}x=\int\frac{x^2+a^2-a^2}{\sqrt{x^2+a^2}}\mathrm{d}x=\int\sqrt{x^2+a^2}\,\mathrm{d}x-a^2\int\frac{1}{\sqrt{x^2+a^2}}\mathrm{d}x,$$

其中 $$\int\sqrt{x^2+a^2}\,\mathrm{d}x=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\ln(x+\sqrt{x^2+a^2})+C_1,$$

$$\int\frac{1}{\sqrt{x^2+a^2}}\mathrm{d}x=\ln(x+\sqrt{x^2+a^2})+C_2,$$

所以 $$\int\frac{x^2}{\sqrt{x^2+a^2}}\mathrm{d}x=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\ln(x+\sqrt{x^2+a^2})-a^2\ln(x+\sqrt{x^2+a^2})+C$$

$$=\frac{x}{2}\sqrt{x^2+a^2}-\frac{a^2}{2}\ln(x+\sqrt{x^2+a^2})+C.$$

37. 解 $\int\frac{\mathrm{d}x}{x^2\sqrt{x^2+a^2}}=-\int\frac{1}{\sqrt{x^2+a^2}}\mathrm{d}\left(\frac{1}{x}\right)$.

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令 $\frac{1}{x}=t$，则 $x=\frac{1}{t}(t\neq 0)$，

$$-\int\frac{1}{\sqrt{x^2+a^2}}\mathrm{d}\left(\frac{1}{x}\right)=-\int\frac{1}{\sqrt{\frac{1}{t^2}+a^2}}\mathrm{d}t=-\int\frac{t}{\sqrt{1+a^2t^2}}\mathrm{d}t$$

$$=-\frac{1}{2a^2}\int\frac{1}{\sqrt{1+a^2t^2}}\mathrm{d}(1+a^2t^2)=-\frac{1}{a^2}\sqrt{1+a^2t^2}+C,$$

将 $t=\frac{1}{x}$ 代回上式得

$$\int\frac{\mathrm{d}x}{x^2\sqrt{x^2+a^2}}=-\frac{\sqrt{x^2+a^2}}{a^2x}+C.$$

38. 解 $\int\sqrt{x^2+a^2}\mathrm{d}x=x\sqrt{x^2+a^2}-\int x\mathrm{d}(\sqrt{x^2+a^2})$

$$=x\sqrt{x^2+a^2}-\int\frac{x^2}{\sqrt{x^2+a^2}}\mathrm{d}x=x\sqrt{x^2+a^2}-\int\sqrt{x^2+a^2}\mathrm{d}x+a^2\int\frac{\mathrm{d}x}{\sqrt{x^2+a^2}},$$

移项即得 $\int\sqrt{x^2+a^2}\mathrm{d}x=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\int\frac{\mathrm{d}x}{\sqrt{x^2+a^2}}=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\ln(x+\sqrt{x^2+a^2})+C$.

39. 解 $\int x\sqrt{x^2+a^2}\mathrm{d}x=\frac{1}{2}\int\sqrt{x^2+a^2}\mathrm{d}(x^2+a^2)=\frac{1}{2}\times\frac{2}{3}(x^2+a^2)^{\frac{3}{2}}+C=\frac{1}{3}\sqrt{(x^2+a^2)^3}+C$.

40. 解 令 $x=a\tan t\left(-\frac{\pi}{2}<t<\frac{\pi}{2}\right)$，$\mathrm{d}x=a\sec^2t\mathrm{d}t$，则

$$\int\frac{\sqrt{x^2+a^2}}{x^2}\mathrm{d}x=\int\frac{\sec t}{a\tan^2t}\cdot a\sec^2t\mathrm{d}t=\int\frac{\sec t}{\tan^2t}(1+\tan^2t)\mathrm{d}t=\int\frac{\sec t}{\tan^2t}\mathrm{d}t+\int\sec t\mathrm{d}t$$

$$=\int\frac{\cos t}{\sin^2t}\mathrm{d}t+\int\sec t\mathrm{d}t=\int\frac{1}{\sin^2t}\mathrm{d}(\sin t)+\int\sec t\mathrm{d}t=\ln|\sec t+\tan t|-\frac{1}{\sin t}+C_1$$

$$=-\frac{\sqrt{x^2+a^2}}{x}+\ln(x+\sqrt{x^2+a^2})+C.$$

41. 解 当 $b^2-4ac>0$ 时，

$$\int\frac{\mathrm{d}x}{\sqrt{ax^2+bx+c}}=2\sqrt{a}\int\frac{\mathrm{d}x}{\sqrt{(2ax+b)^2-(\sqrt{b^2-4ac})^2}}=\frac{2\sqrt{a}}{2a}\int\frac{\mathrm{d}(2ax+b)}{\sqrt{(2ax+b)^2-(\sqrt{b^2-4ac})^2}}$$

$$=\frac{1}{\sqrt{a}}\ln\left|2ax+b+\sqrt{(2ax+b)^2-(\sqrt{b^2-4ac})^2}\right|+C$$

$$=\frac{1}{\sqrt{a}}\ln\left|2ax+b+2\sqrt{a}\sqrt{ax^2+bx+c}\right|+C;$$

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当 $b^2-4ac=0$ 时，

$$\int\frac{\mathrm{d}x}{\sqrt{ax^2+bx+c}}=\int\frac{\mathrm{d}x}{\sqrt{a}\left(x+\frac{b}{2a}\right)}=\frac{1}{\sqrt{a}}\ln\left|x+\frac{b}{2a}\right|+C;$$

当 $b^2-4ac<0$ 时，

$$\int\frac{\mathrm{d}x}{\sqrt{ax^2+bx+c}}=2\sqrt{a}\int\frac{\mathrm{d}x}{\sqrt{(2ax+b)^2+(\sqrt{4ac-b^2})^2}}=\frac{2\sqrt{a}}{2a}\int\frac{\mathrm{d}(2ax+b)}{\sqrt{(2ax+b)^2+(\sqrt{4ac-b^2})^2}}$$

$$=\frac{1}{\sqrt{a}}\ln\left|2ax+b+2\sqrt{a}\sqrt{ax^2+bx+c}\right|+C.$$

42. 解 $$\int\frac{\mathrm{d}x}{\sqrt{c+bx-ax^2}}=2\sqrt{a}\int\frac{1}{\sqrt{b^2+4ac-(2ax-b)^2}}\mathrm{d}x=\frac{2\sqrt{a}}{2a}\int\frac{\mathrm{d}(2ax-b)}{\sqrt{\left(\sqrt{b^2+4ac}\right)^2-(2ax-b)^2}}$$

$$=\frac{1}{\sqrt{a}}\arcsin\frac{2ax-b}{\sqrt{b^2+4ac}}+C.$$

43. 解 令 $t=\sqrt{\frac{x-a}{x-b}}$，则 $x=\frac{a-bt^2}{1-t^2}$，$\mathrm{d}x=\frac{2t(a-b)}{(1-t^2)^2}\mathrm{d}t$，故

$$\int\sqrt{\frac{x-a}{x-b}}\mathrm{d}x=\int t\cdot\frac{2t(a-b)}{(1-t^2)^2}\mathrm{d}t=2(a-b)\int\frac{t^2}{(1-t^2)^2}\mathrm{d}t=2(b-a)\int\frac{1-t^2-1}{(1-t^2)^2}\mathrm{d}t$$

$$=2(b-a)\int\left[\frac{1}{1-t^2}-\frac{1}{(1-t^2)^2}\right]\mathrm{d}t=2(a-b)\int\frac{1}{t^2-1}\mathrm{d}t+2(a-b)\int\frac{1}{(1-t^2)^2}\mathrm{d}t$$

$$=2(a-b)\cdot\frac{1}{2}\ln\left|\frac{t-1}{t+1}\right|+2(a-b)\int\frac{1}{(1-t^2)^2}\mathrm{d}t=(a-b)\ln\left|\frac{t-1}{t+1}\right|+2(a-b)\int\frac{1}{(1-t^2)^2}\mathrm{d}t,$$

其中，$\int\frac{\mathrm{d}t}{(1-t^2)^2}=\frac{t}{2(1-t^2)}+\frac{1}{4}\ln\left|\frac{1+t}{1-t}\right|+C_1$，则 （由第一组32式）

$$\int\sqrt{\frac{x-a}{x-b}}\mathrm{d}x=(a-b)\ln\left|\frac{t-1}{t+1}\right|+2(a-b)\left[\frac{t}{2(1-t^2)}+\frac{1}{4}\ln\left|\frac{1+t}{1-t}\right|\right]+C_2$$

$$=(a-b)\ln\left|\frac{t-1}{t+1}\right|+\frac{(a-b)t}{1-t^2}+\frac{a-b}{2}\ln\left|\frac{1+t}{1-t}\right|+C_2$$

$$=\frac{(a-b)t}{1-t^2}+\frac{b-a}{2}\ln\left|\frac{1+t}{1-t}\right|+C_2.$$

将 $t=\sqrt{\frac{x-a}{x-b}}$ 代入上式得

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$$\int\sqrt{\frac{x-a}{x-b}}\mathrm{d}x=(x-b)\sqrt{\frac{x-a}{x-b}}+\frac{b-a}{2}\ln\left|\frac{1+\sqrt{\frac{x-a}{x-b}}}{1-\sqrt{\frac{x-a}{x-b}}}\right|+C_2$$

$$=(x-b)\sqrt{\frac{x-a}{x-b}}+(b-a)\ln\left(\sqrt{|x-a|}+\sqrt{|x-b|}\right)+C.$$

44. 解　令 $t=\sqrt{\frac{x-a}{b-x}}$，则 $x=\frac{a+bt^2}{1+t^2}$，$\mathrm{d}x=\frac{2t(b-a)}{(1+t^2)^2}\mathrm{d}t$，则

$$\int\sqrt{\frac{x-a}{b-x}}\mathrm{d}x=\int t\cdot\frac{2t(b-a)}{(1+t^2)^2}\mathrm{d}t=2(b-a)\int\frac{t^2}{(1+t^2)^2}\mathrm{d}t=2(b-a)\int\left[\frac{1}{1+t^2}-\frac{1}{(1+t^2)^2}\right]\mathrm{d}t$$

$$=2(b-a)\arctan t-2(b-a)\int\frac{1}{(1+t^2)^2}\mathrm{d}t,$$

令 $t=\tan k$，$\mathrm{d}t=\sec^2 k\mathrm{d}k$，则

$$\int\frac{1}{(1+t^2)^2}\mathrm{d}t=\int\frac{1}{\sec^4 k}\sec^2 k\mathrm{d}k=\int\cos^2 k\mathrm{d}k=\frac{1}{2}\int(1+\cos 2k)\,\mathrm{d}k=\frac{k}{2}+\frac{1}{4}\sin 2k+C_1$$

$$=\frac{k}{2}+\frac{1}{4}\cdot 2\sin k\cos k+C_1=\frac{1}{2}\arctan t+\frac{1}{2}\cdot\frac{1}{\sqrt{t^2+1}}\cdot\frac{t}{\sqrt{t^2+1}}+C_1$$

$$=\frac{1}{2}\arctan t+\frac{t}{2(t^2+1)}+C_1.$$

所以 $$\int\sqrt{\frac{x-a}{b-x}}\mathrm{d}x=2(b-a)\arctan t-2(b-a)\left[\frac{1}{2}\arctan t+\frac{t}{2(t^2+1)}+C_1\right].$$

将 $t=\sqrt{\frac{x-a}{b-x}}$ 反代回去，得

$$\int\sqrt{\frac{x-a}{b-x}}\mathrm{d}x=(b-a)\arctan\sqrt{\frac{x-a}{b-x}}+(x-b)\sqrt{\frac{x-a}{b-x}}+C.$$

45. 解　$$\int\frac{\mathrm{d}x}{\sqrt{(x-a)(b-x)}}=\int\frac{1}{|x-a|}\cdot\sqrt{\frac{x-a}{b-x}}\mathrm{d}x.$$

令 $t=\sqrt{\frac{x-a}{b-x}}$，则 $x=\frac{a+bt^2}{1+t^2}$，$\mathrm{d}x=\frac{2t(b-a)}{(1+t^2)^2}\mathrm{d}t$，因为 $b>a$，所以 $|x-a|=(b-a)\cdot\frac{t^2}{1+t^2}$，于是

$$\int\frac{1}{|x-a|}\cdot\sqrt{\frac{x-a}{b-x}}\mathrm{d}x=\int\frac{1}{b-a}\cdot\frac{1+t^2}{t^2}\cdot t\cdot\frac{2t(b-a)}{(1+t^2)^2}\mathrm{d}t=2\int\frac{1}{1+t^2}\mathrm{d}t$$

$$=2\arctan t+C=2\arctan\sqrt{\frac{x-a}{b-x}}+C.$$

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令 $\tan u=\sqrt{\dfrac{x-a}{b-x}}$，则 $u=\arctan\sqrt{\dfrac{x-a}{b-x}}$，$\sin u=\sqrt{\dfrac{x-a}{b-a}}$，$u=\arcsin\sqrt{\dfrac{x-a}{b-a}}$，所以

$$\int\frac{\mathrm{d}x}{\sqrt{(x-a)(b-x)}}=2\arcsin\sqrt{\frac{x-a}{b-a}}+C,\quad a<b.$$

46. 解　因 $(x-a)(b-x)>0$ 且 $a<b$，故 $a<x<b$，则有 $\int\sqrt{(x-a)(b-x)}\mathrm{d}x=\int(x-a)\sqrt{\dfrac{b-x}{x-a}}\,\mathrm{d}x$.

令 $t=\sqrt{\dfrac{b-x}{x-a}}$，则 $x=\dfrac{b+at^2}{1+t^2}$，$\mathrm{d}x=\dfrac{2t(a-b)}{(1+t^2)^2}\mathrm{d}t$，所以

$$\int\sqrt{(x-a)(b-x)}\mathrm{d}x=\int\frac{b-a}{1+t^2}\cdot t\cdot\frac{2t(a-b)}{(1+t^2)^2}\mathrm{d}t=-2(a-b)^2\int\frac{t^2}{(1+t^2)^3}\mathrm{d}t.$$

对于 $\int\dfrac{t^2}{(1+t^2)^3}\mathrm{d}t$，令 $t=\tan k$，$(t^2+1)^3=\sec^6 k$，$\mathrm{d}t=\sec^2 k\mathrm{d}k$，则

$$\int\frac{t^2}{(1+t^2)^3}\mathrm{d}t=\int\frac{\tan^2 k}{\sec^6 k}\cdot\sec^2 k\mathrm{d}k=\int\sin^2 k\cdot\cos^2 k\mathrm{d}k=\frac{1}{4}\int\sin^2 2k\mathrm{d}k=\frac{1}{8}\left(k-\frac{1}{4}\sin 4k\right)+C_1$$

$$=\frac{k}{8}-\frac{1}{32}\sin 4k+C_1=\frac{k}{8}-\frac{1}{32}(4\sin k\cos^3 k-4\sin^3 k\cos k)+C_1$$

$$=\frac{k}{8}-\frac{1}{8}\sin k\cos^3 k+\frac{1}{8}\sin^3 k\cos k+C_1,$$

则
$$\int\sqrt{(x-a)(b-x)}\mathrm{d}x=-\frac{(b-a)^2}{4}\left[\arcsin\frac{t}{\sqrt{t^2+1}}+\frac{t(t^2-1)}{(t^2+1)^2}\right]+C.$$

将 $t=\sqrt{\dfrac{b-x}{x-a}}$ 代入上式得 $\int\sqrt{(x-a)(b-x)}\mathrm{d}x=\dfrac{2x-a-b}{4}\sqrt{(x-a)(b-x)}+\dfrac{(b-a)^2}{4}\arcsin\sqrt{\dfrac{x-a}{b-a}}+C$,

其中 $\arcsin\sqrt{\dfrac{x-a}{b-a}}+\arcsin\sqrt{\dfrac{b-x}{b-a}}=\dfrac{\pi}{2}$.

第三组

1. 解　$\int\cos^3 ax\mathrm{d}x=\dfrac{1}{a}\int\cos^3 ax\mathrm{d}(ax)=\dfrac{1}{a}\int(1-\sin^2 ax)\mathrm{d}(\sin ax)=\dfrac{1}{a}\sin ax-\dfrac{1}{3a}\sin^3 ax+C$.

2. 解　$\int\tan^2 ax\mathrm{d}x=\dfrac{1}{a}\int(\sec^2 ax-1)\,\mathrm{d}(ax)=\dfrac{1}{a}\tan ax-x+C$.

3. 解　$\int\tan^3 ax\mathrm{d}x=\dfrac{1}{a}\int\tan^3 ax\mathrm{d}(ax)=\dfrac{1}{a}\int\tan ax(\sec^2 ax-1)\,\mathrm{d}(ax)$

$$=\frac{1}{a}\int\tan ax\sec^2 ax\mathrm{d}(ax)-\frac{1}{a}\int\tan ax\mathrm{d}(ax)=\frac{1}{2a}\tan^2 ax+\frac{1}{a}\ln|\cos ax|+C.$$

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4. 解 $\int \cot^2 ax\mathrm{d}x=\frac{1}{a}\int(\csc^2 ax-1)\,\mathrm{d}(ax)=-\frac{1}{a}\cot ax-x+C$.

5. 解 $\int \cot^3 ax\mathrm{d}x=\frac{1}{a}\int \cot^3 ax\mathrm{d}(ax)=\frac{1}{a}\int \cot ax(\csc^2 ax-1)\,\mathrm{d}(ax)$

$$=\frac{1}{a}\int \cot ax\csc^2 ax\mathrm{d}(ax)-\frac{1}{a}\int \cot ax\mathrm{d}(ax)=-\frac{1}{2a}\cot^2 ax-\frac{1}{a}\ln|\sin ax|+C.$$

6. 解 $\int\frac{\mathrm{d}x}{\sin^3 ax}=\frac{1}{a}\int\frac{1}{\sin^3 ax}\mathrm{d}(ax)=-\frac{1}{a}\int\frac{1}{\sin ax}\cdot\left(-\frac{1}{\sin^2 ax}\right)\mathrm{d}(ax)$

$$=-\frac{1}{a}\int\frac{1}{\sin ax}\mathrm{d}(\cot ax)=-\frac{1}{a}\cdot\frac{\cot ax}{\sin ax}-\frac{1}{a}\int \cot ax\cdot\csc ax\cdot\cot ax\mathrm{d}(ax)$$

$$=-\frac{1}{a}\cdot\frac{\cot ax}{\sin ax}-\frac{1}{a}\int(\csc^2 ax-1)\csc ax\mathrm{d}(ax)$$

$$=-\frac{1}{a}\cdot\frac{\cot ax}{\sin ax}-\frac{1}{a}\int\csc^3 ax\mathrm{d}(ax)+\frac{1}{a}\int\csc ax\mathrm{d}(ax),$$

即 $\int\frac{\mathrm{d}x}{\sin^3 ax}=-\frac{1}{a}\cot ax\csc ax-\int\frac{\mathrm{d}x}{\sin^3 ax}+\frac{1}{a}\int\csc ax\mathrm{d}(ax)$，故

$$\int\frac{\mathrm{d}x}{\sin^3 ax}=\frac{-1}{2a}\cot ax\csc ax+\frac{1}{2a}\ln|\csc ax-\cot ax|+C$$

$$=-\frac{\cos ax}{2a\sin^2 ax}+\frac{1}{2a}\ln\left|\frac{1-\cos ax}{\sin ax}\right|+C$$

$$=-\frac{\cos ax}{2a\sin^2 ax}+\frac{1}{2a}\ln\left|\tan\frac{ax}{2}\right|+C.$$

7. 解 $\int\frac{\mathrm{d}x}{\cos^3 ax}=\frac{1}{a}\int\frac{1}{\cos^3 ax}\mathrm{d}(ax)=\frac{1}{a}\int\frac{1}{\cos ax}\cdot\frac{1}{\cos^2 ax}\mathrm{d}(ax)$

$$=\frac{1}{a}\int\frac{1}{\cos ax}\mathrm{d}(\tan ax)=\frac{1}{a}\cdot\frac{\tan ax}{\cos ax}-\frac{1}{a}\int\tan ax\sec ax\tan ax\mathrm{d}(ax)$$

$$=\frac{1}{a}\cdot\frac{\tan ax}{\cos ax}-\frac{1}{a}\int(\sec^2 ax-1)\sec ax\mathrm{d}(ax)$$

$$=\frac{1}{a}\cdot\frac{\tan ax}{\cos ax}-\frac{1}{a}\int\sec^3 ax\mathrm{d}(ax)+\frac{1}{a}\int\sec ax\mathrm{d}(ax)$$

$$=\frac{1}{a}\cdot\frac{\tan ax}{\cos ax}-\int\frac{1}{\cos^3 ax}\mathrm{d}x+\frac{1}{a}\ln|\sec ax+\tan ax|,$$

故 $$\int\frac{\mathrm{d}x}{\cos^3 ax}=\frac{1}{2a}\cdot\frac{\tan ax}{\cos ax}+\frac{1}{2a}\ln|\sec ax+\tan ax|+C$$

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$$=\frac{\sin ax}{2a\cos^2 ax}+\frac{1}{2a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right|+C.$$

8. 解 $\int \sin mx\sin nx\mathrm{d}x=\int\frac{1}{2}[\cos(m-n)x-\cos(m+n)x]\mathrm{d}x$

$$=\frac{1}{2(m-n)}\sin(m-n)x-\frac{1}{2(m+n)}\sin(m+n)x+C,\quad m^2\neq n^2.$$

9. 解 $\int \sin mx\cos nx\mathrm{d}x=\frac{1}{2}\int[\sin(m+n)x+\sin(m-n)x]\mathrm{d}x$

$$=-\frac{1}{2(m+n)}\cos(m+n)x-\frac{1}{2(m-n)}\cos(m-n)x+C,\quad m^2\neq n^2.$$

10. 解 $\int \cos mx\cos nx\mathrm{d}x=\frac{1}{2}\int[\cos(m+n)x+\cos(m-n)x]\mathrm{d}x$

$$=\frac{1}{2(m+n)}\sin(m+n)x+\frac{1}{2(m-n)}\sin(m-n)x+C,\quad m^2\neq n^2.$$

11. 解 $\int \sin ax\cos ax\mathrm{d}x=\frac{1}{a}\int\sin ax\mathrm{d}(\sin ax)=\frac{1}{2a}\sin^2 ax+C.$

12. 解 $\int \sin^2 ax\cos ax\mathrm{d}x=\frac{1}{a}\int\sin^2 ax\mathrm{d}(\sin ax)=\frac{1}{3a}\sin^3 ax+C.$

13. 解 $\int \sin^2 ax\cos^2 ax\mathrm{d}x=\frac{1}{8a}\int\sin^2 2ax\mathrm{d}(2ax)=\frac{1}{8a}\int\frac{1-\cos 4ax}{2}\mathrm{d}(2ax)=\frac{x}{8}-\frac{\sin 4ax}{32a}+C.$

14. 解 $\int \sin^2 ax\cos^3 ax\mathrm{d}x=\frac{1}{a}\int\sin^2 ax\cos^2 ax\mathrm{d}(\sin ax)$

$$=\frac{1}{a}\int\sin^2 ax(1-\sin^2 ax)\mathrm{d}(\sin ax)=\frac{1}{3a}\sin^3 ax-\frac{1}{5a}\sin^5 ax+C.$$

15. 解 $\int \sin^3 ax\cos ax\mathrm{d}x=\frac{1}{a}\int\sin^3 ax\mathrm{d}(\sin ax)=\frac{1}{4a}\sin^4 ax+C.$

16. 解 $\int \sin^3 ax\cos^2 ax\mathrm{d}x=-\frac{1}{a}\int\sin^2 ax\cos^2 ax\mathrm{d}(\cos ax)=-\frac{1}{a}\int(1-\cos^2 ax)\cos^2 ax\mathrm{d}(\cos ax)$

$$=-\frac{1}{a}\left(\frac{1}{3}\cos^3 ax-\frac{1}{5}\cos^5 ax\right)+C.$$

17. 解 $\int \sin^3 ax\cos^3 ax\mathrm{d}x=\int\frac{1}{8}\sin^3 2ax\mathrm{d}x=-\frac{1}{2a}\cdot\frac{1}{8}\int\sin^2 2ax\mathrm{d}(\cos 2ax)$

$$=-\frac{1}{16a}\int(1-\cos^2 2ax)\,\mathrm{d}(\cos 2ax)=-\frac{1}{16a}\left(\cos 2ax-\frac{1}{3}\cos^3 2ax\right)+C$$

$$=-\frac{1}{16a}\cos 2ax+\frac{1}{48a}\cos^3 2ax+C.$$

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18. 解 $\int \frac{\sin ax}{\cos^2 ax}\mathrm{d}x = -\frac{1}{a}\int \frac{1}{\cos^2 ax}\mathrm{d}(\cos ax) = -\frac{1}{a}\left(-\frac{1}{\cos ax}\right)+C = \frac{1}{a\cos ax}+C = \frac{1}{a}\sec ax + C$.

19. 解 令 $I = \int \frac{\sin^2 ax}{\cos^3 ax}\mathrm{d}x$，则

$$I = \int \tan^2 ax \cdot \sec ax\mathrm{d}x = \frac{1}{a}\int \tan ax\mathrm{d}(\sec ax) = \frac{1}{a}\left(\tan ax\cdot\sec ax - a\int \sec^3 ax\mathrm{d}x\right)$$

$$= \frac{1}{a}\tan ax\cdot\sec ax - \int \sec^3 ax\mathrm{d}x = \frac{1}{a}\tan ax\cdot\sec ax - \int (\tan^2 ax+1)\sec ax\mathrm{d}x$$

$$= \frac{1}{a}\tan ax\cdot\sec ax - I - \int \sec ax\mathrm{d}x,$$

故

$$I = \frac{1}{2a}\tan ax\sec ax - \frac{1}{2}\int \sec ax\mathrm{d}x = \frac{1}{2a}\cdot\frac{\sin ax}{\cos^2 ax} - \frac{1}{2a}\ln|\sec ax+\tan ax| + C$$

$$= \frac{1}{2a}\cdot\frac{\sin ax}{\cos^2 ax} - \frac{1}{2a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right| + C.$$

20. 解 $\int \frac{\sin^2 ax}{\cos^4 ax}\mathrm{d}x = \int \frac{1-\cos^2 ax}{\cos^4 ax}\mathrm{d}x = \int \sec^4 ax\mathrm{d}x - \int \sec^2 ax\mathrm{d}x = \frac{1}{a}\int \sec^2 ax\mathrm{d}(\tan ax) - \frac{1}{a}\tan ax$

$$= \frac{1}{a}\int (1+\tan^2 ax)\mathrm{d}(\tan ax) - \frac{1}{a}\tan ax = \frac{1}{a}\tan ax + \frac{1}{3a}\tan^3 ax - \frac{1}{a}\tan ax + C$$

$$= \frac{1}{3a}\tan^3 ax + C.$$

21. 解 $\int \frac{\sin^3 ax}{\cos ax}\mathrm{d}x = -\frac{1}{a}\int \frac{(1-\cos^2 ax)}{\cos ax}\mathrm{d}(\cos ax) = -\frac{1}{a}\left[\int \frac{1}{\cos ax}\mathrm{d}(\cos ax) - \int \cos ax\mathrm{d}(\cos ax)\right]$

$$= -\frac{1}{a}\left(\ln|\cos ax| - \frac{1}{2}\cos^2 ax\right) + C_1 = -\frac{1}{2a}\sin^2 ax - \frac{1}{a}\ln|\cos ax| + C.$$

22. 解 $\int \frac{\sin^3 ax}{\cos^2 ax}\mathrm{d}x = \int \frac{(1-\cos^2 ax)\sin ax}{\cos^2 ax}\mathrm{d}x = -\frac{1}{a}\int \frac{1-\cos^2 ax}{\cos^2 ax}\mathrm{d}(\cos ax)$

$$= -\frac{1}{a}\left[\int \frac{1}{\cos^2 ax}\mathrm{d}(\cos ax) - \int \mathrm{d}(\cos ax)\right] = -\frac{1}{a}\left(-\frac{1}{\cos ax} - \cos ax\right) + C$$

$$= \frac{1}{a\cos ax} + \frac{\cos ax}{a} + C.$$

23. 解 $\int \frac{\sin^3 ax}{\cos^6 ax}\mathrm{d}x = \int \frac{(1-\cos^2 ax)\sin ax}{\cos^6 ax}\mathrm{d}x = -\frac{1}{a}\int \frac{1-\cos^2 ax}{\cos^6 ax}\mathrm{d}(\cos ax)$

$$= -\frac{1}{a}\int \left(\frac{1}{\cos^6 ax} - \frac{1}{\cos^4 ax}\right)\mathrm{d}(\cos ax) = -\frac{1}{a}\left(-\frac{1}{5}\frac{1}{\cos^5 ax} + \frac{1}{3}\frac{1}{\cos^3 ax}\right) + C$$

$$= \frac{1}{5a\cos^5 ax} - \frac{1}{3a\cos^3 ax} + C.$$

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24. 解 $\int\frac{\cos^2 ax}{\sin ax}\mathrm{d}x=\int\frac{1-\sin^2 ax}{\sin ax}\mathrm{d}x=\int\frac{1}{\sin ax}\mathrm{d}x-\int\sin ax\mathrm{d}x=\int\csc ax\mathrm{d}x+\frac{1}{a}\cos ax$

$=\frac{1}{a}\ln|\csc ax-\cot ax|+\frac{1}{a}\cos ax+C=\frac{1}{a}\ln\left|\frac{1-\cos ax}{\sin ax}\right|+\frac{1}{a}\cos ax+C$

$=\frac{1}{a}\ln\left|\tan\frac{ax}{2}\right|+\frac{1}{a}\cos ax+C$.

25. 解 令 $I=\int\frac{\cos^2 ax}{\sin^3 ax}\mathrm{d}x=\int\frac{1-\sin^2 ax}{\sin^3 ax}\mathrm{d}x=\int\csc^3 ax\mathrm{d}x-\int\csc ax\mathrm{d}x$，再令 $I_1=\int\csc^3 ax\mathrm{d}x$，则

$I_1=\int\csc^3 ax\mathrm{d}x=-\frac{1}{a}\int\csc ax\mathrm{d}(\cot ax)=-\frac{1}{a}\left(\csc ax\cdot\cot ax+a\int\cot ax\cdot\csc ax\cdot\cot ax\mathrm{d}x\right)$

$=-\frac{1}{a}\left[\csc ax\cdot\cot ax+a\int(\csc^2 ax-1)\csc ax\mathrm{d}x\right]=-\frac{1}{a}\left(\csc ax\cdot\cot ax+a\int\csc^3 ax\mathrm{d}x\right)+\int\csc ax\mathrm{d}x$，

故 $I_1=-\frac{1}{2a}\csc ax\cot ax+\frac{1}{2}\int\csc ax\mathrm{d}x$.

于是 $I=I_1-\int\csc ax\mathrm{d}x=-\frac{1}{2a}\csc ax\cot ax+\frac{1}{2}\int\csc ax\mathrm{d}x-\int\csc ax\mathrm{d}x$

$=-\frac{1}{2a}\frac{\cos ax}{\sin^2 ax}-\frac{1}{2a}\ln\left|\tan\frac{ax}{2}\right|+C$.

26. 解 $\int\frac{\cos^2 ax}{\sin^4 ax}\mathrm{d}x=\int\cot^2 ax\cdot\csc^2 ax\mathrm{d}x=-\frac{1}{a}\int\cot^2 ax\mathrm{d}(\cot ax)=-\frac{1}{a}\cdot\frac{1}{3}\cot^3 ax+C$

$=-\frac{1}{3a}\cot^3 ax+C$.

27. 解 $\int\frac{\cos^3 ax}{\sin ax}\mathrm{d}x=\int\frac{(1-\sin^2 ax)\cos ax}{\sin ax}\mathrm{d}x=\frac{1}{a}\int\frac{1-\sin^2 ax}{\sin ax}\mathrm{d}(\sin ax)$

$=\frac{1}{a}\left[\int\frac{1}{\sin ax}\mathrm{d}(\sin ax)-\int\sin ax\mathrm{d}(\sin ax)\right]=\frac{1}{a}\left(\ln|\sin ax|-\frac{1}{2}\sin^2 ax\right)+C_1$

$=\frac{\cos^2 ax}{2a}+\frac{1}{a}\ln|\sin ax|+C$.

28. 解 $\int\frac{\cos^3 ax}{\sin^2 ax}\mathrm{d}x=\int\frac{(1-\sin^2 ax)}{\sin^2 ax}\cdot\cos ax\mathrm{d}x=\frac{1}{a}\int\frac{1-\sin^2 ax}{\sin^2 ax}\mathrm{d}(\sin ax)$

$=\frac{1}{a}\left[\int\frac{1}{\sin^2 ax}\mathrm{d}(\sin ax)-\int\mathrm{d}(\sin ax)\right]=\frac{1}{a}\left(-\frac{1}{\sin ax}-\sin ax\right)+C$.

29. 解 $\int\frac{\cos^3 ax}{\sin^4 ax}\mathrm{d}x=\int\frac{(1-\sin^2 ax)\cos ax}{\sin^4 ax}\mathrm{d}x=\frac{1}{a}\int\frac{1-\sin^2 ax}{\sin^4 ax}\mathrm{d}(\sin ax)$

$=\frac{1}{a}\left[\int\frac{1}{\sin^4 ax}\mathrm{d}(\sin ax)-\int\frac{1}{\sin^2 ax}\mathrm{d}(\sin ax)\right]=\frac{1}{a}\left(-\frac{1}{3}\frac{1}{\sin^3 ax}+\frac{1}{\sin ax}\right)+C$.

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30. 解 $$\int\frac{1}{\sin ax\cos ax}\mathrm{d}x=\int\frac{1}{\frac{1}{2}\sin 2ax}\mathrm{d}x=\int 2\csc 2ax\mathrm{d}x=2\cdot\frac{1}{2a}\ln|\csc 2ax-\cot 2ax|+C$$

$$=\frac{1}{a}\ln\left|\frac{1-\cos 2ax}{\sin 2ax}\right|+C=\frac{1}{a}\ln|\tan ax|+C.$$

31. 解 $$\int\frac{1}{\sin ax\cos^2 ax}\mathrm{d}x=\int\frac{\sin^2 ax+\cos^2 ax}{\sin ax\cos^2 ax}\mathrm{d}x=\int\frac{\sin ax}{\cos^2 ax}\mathrm{d}x+\int\frac{1}{\sin ax}\mathrm{d}x$$

$$=-\frac{1}{a}\int\frac{1}{\cos^2 ax}\mathrm{d}(\cos ax)+\int\csc ax\mathrm{d}x$$

$$=-\frac{1}{a}\left(-\frac{1}{\cos ax}\right)+\frac{1}{a}\ln|\csc ax-\cot ax|+C$$

$$=\frac{1}{a\cos ax}+\frac{1}{a}\ln\left|\frac{1-\cos ax}{\sin ax}\right|+C=\frac{1}{a\cos ax}+\frac{1}{a}\ln\left|\tan\frac{a}{2}x\right|+C$$

$$=\frac{1}{a}\left(\sec ax+\ln\left|\tan\frac{ax}{2}\right|\right)+C.$$

32. 解 $$\int\frac{1}{\sin^2 ax\cos ax}\mathrm{d}x=\int\frac{\sin^2 ax+\cos^2 ax}{\sin^2 ax\cos ax}\mathrm{d}x=\int\sec ax\mathrm{d}x+\int\frac{\cos ax}{\sin^2 ax}\mathrm{d}x$$

$$=\frac{1}{a}\ln|\sec ax+\tan ax|+\frac{1}{a}\int\frac{1}{\sin^2 ax}\mathrm{d}(\sin ax)=\frac{1}{a}\ln\left|\frac{1+\sin ax}{\cos ax}\right|-\frac{1}{a}\frac{1}{\sin ax}+C$$

$$=\frac{1}{a}\ln\left|\frac{1-\cos\left(\frac{\pi}{2}+ax\right)}{\sin\left(\frac{\pi}{2}+ax\right)}\right|-\frac{1}{a}\csc ax+C=\frac{1}{a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right|-\frac{1}{a}\csc ax+C.$$

33. 解 $$\int\frac{1}{\sin^2 ax\cos^2 ax}\mathrm{d}x=\int\frac{1}{\frac{1}{4}\sin^2 2ax}\mathrm{d}x=4\int\csc^2 2ax\mathrm{d}x=4\cot 2ax\cdot\left(-\frac{1}{2a}\right)+C=-\frac{2}{a}\cot 2ax+C.$$

34. 解 $$\int\frac{\mathrm{d}x}{\sin^2 ax\cos^3 ax}=\int\frac{\sin^2 ax+\cos^2 ax}{\sin^2 ax\cdot\cos^3 ax}\mathrm{d}x=\int\sec^3 ax\mathrm{d}x+\int\frac{1}{\sin^2 ax\cdot\cos ax}\mathrm{d}x$$

由7式 ← $$=\frac{\sin ax}{2a\cdot\cos^2 ax}+\frac{1}{2a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right|+\frac{1}{a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right|-\frac{1}{a}\cdot\csc ax+C$$ → 由32式

$$=\frac{1}{2a\sin ax}\left(\frac{1}{\cos^2 ax}-3\right)+\frac{3}{2a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right|+C.$$

35. 解 $$\int\frac{\mathrm{d}x}{\sin^2 ax\cos^4 ax}=\int\frac{\sin^2 ax+\cos^2 ax}{\sin^2 ax\cdot\cos^4 ax}\mathrm{d}x=\int\sec^4 ax\mathrm{d}x+\int\frac{1}{\sin^2 ax\cdot\cos^2 ax}\mathrm{d}x$$

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$$=\frac{1}{a}\int(\tan^2 ax+1)\mathrm{d}(\tan ax)-\frac{2}{a}\cot 2ax \quad \text{由33式}$$

$$=\frac{1}{a}\cdot\left(\frac{1}{3}\tan^3 ax+\tan ax\right)-\frac{2}{a}\cot 2ax+C.$$

36. 解 $\int\frac{1}{\sin^3 ax\cos ax}\mathrm{d}x=\int\frac{\sin^2 ax+\cos^2 ax}{\sin^3 ax\cdot\cos ax}\mathrm{d}x=\int\frac{1}{\sin ax\cdot\cos ax}\mathrm{d}x+\int\frac{\cos ax}{\sin^3 ax}\mathrm{d}x$

由30式 $=\frac{1}{a}\ln|\tan ax|-\frac{1}{a}\cdot\frac{1}{2\sin^2 ax}+C=\frac{1}{a}\ln|\tan ax|-\frac{1}{2a\sin^2 ax}+C.$

37. 解 $\int\frac{1}{\sin^3 ax\cos^2 ax}\mathrm{d}x=\int\frac{\sin^2 ax+\cos^2 ax}{\sin^3 ax\cdot\cos^2 ax}\mathrm{d}x=\int\frac{1}{\sin ax\cdot\cos^2 ax}\mathrm{d}x+\int\csc^3 ax\mathrm{d}x$

由31式 由6式 $=\frac{1}{a}\left(\sec ax+\ln\left|\tan\frac{ax}{2}\right|\right)-\frac{\cos ax}{2a\sin^2 ax}+\frac{1}{2a}\ln\left|\tan\frac{ax}{2}\right|+C$

$$=\frac{1}{a\cos ax}-\frac{\cos ax}{2a\sin^2 ax}+\frac{3}{2a}\ln\left|\tan\frac{ax}{2}\right|+C.$$

38. 解 $\int\frac{1}{\sin^3 ax\cos^3 ax}\mathrm{d}x=\int\frac{\sin^2 ax+\cos^2 ax}{\sin^3 ax\cdot\cos^3 ax}\mathrm{d}x=\int\frac{1}{\sin ax\cdot\cos^3 ax}\mathrm{d}x+\int\frac{1}{\sin^3 ax\cdot\cos ax}\mathrm{d}x$

由36式 $=\int\frac{\sin^2 ax+\cos^2 ax}{\sin ax\cdot\cos^3 ax}\mathrm{d}x-\frac{1}{2a\sin^2 ax}+\frac{1}{a}\ln|\tan ax|$

$$=\int\frac{\sin ax}{\cos^3 ax}\mathrm{d}x+\int\frac{1}{\sin ax\cdot\cos ax}\mathrm{d}x-\frac{1}{2a\sin^2 ax}+\frac{1}{a}\ln|\tan ax|$$

由30式 $=\frac{1}{a}\cdot\frac{1}{2\cos^2 ax}+\frac{1}{a}\ln|\tan ax|-\frac{1}{2a\sin^2 ax}+\frac{1}{a}\ln|\tan ax|+C$

$$=-\frac{2\cos 2ax}{a\sin^2 2ax}+\frac{2}{a}\ln|\tan ax|+C.$$

39. 解 $\int\frac{1}{\sin^3 ax\cdot\cos^4 ax}\mathrm{d}x$

$$=\int\frac{\sin^2 ax+\cos^2 ax}{\sin^3 ax\cdot\cos^4 ax}\mathrm{d}x=\int\frac{1}{\sin ax\cdot\cos^4 ax}\mathrm{d}x+\int\frac{1}{\sin^3 ax\cdot\cos^2 ax}\mathrm{d}x$$

由37式 $=\int\frac{\sin^2 ax+\cos^2 ax}{\sin ax\cdot\cos^4 ax}\mathrm{d}x+\frac{1}{a\cos ax}-\frac{\cos ax}{2a\sin^2 ax}+\frac{3}{2a}\ln\left|\tan\frac{ax}{2}\right|$

$$=\int\frac{\sin ax}{\cos^4 ax}\mathrm{d}x+\int\frac{1}{\sin ax\cdot\cos^2 ax}\mathrm{d}x+\frac{1}{a\cos ax}-\frac{\cos ax}{2a\sin^2 ax}+\frac{3}{2a}\ln\left|\tan\frac{ax}{2}\right|$$

由31式 $=\frac{1}{a}\cdot\frac{1}{3\cos^3 ax}+\frac{1}{a}\left(\sec ax+\ln\left|\tan\frac{ax}{2}\right|\right)+\frac{1}{a\cos ax}-\frac{\cos ax}{2a\sin^2 ax}+\frac{3}{2a}\ln\left|\tan\frac{ax}{2}\right|+C$

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$$=\frac{2}{a\cos ax}+\frac{1}{3a\cos^3 ax}-\frac{\cos ax}{2a\sin^2 ax}+\frac{5}{2a}\ln\left|\tan\frac{ax}{2}\right|+C.$$

40. 解 $$\int\frac{1}{1\pm\sin ax}\mathrm{d}x=\int\frac{1\mp\sin ax}{(1\pm\sin ax)(1\mp\sin ax)}\mathrm{d}x=\int\frac{1}{1-\sin^2 ax}\mathrm{d}x\mp\int\frac{\sin ax}{1-\sin^2 ax}\mathrm{d}x$$

$$=\int\sec^2 ax\mathrm{d}x\mp\int\frac{\sin ax}{\cos^2 ax}\mathrm{d}x=\frac{1}{a}\left(\tan ax\mp\frac{1}{\cos ax}\right)+C=\mp\frac{1}{a}\tan\left(\frac{\pi}{4}\mp\frac{ax}{2}\right)+C.$$

41. 解 $$\int\frac{x}{1+\sin ax}\mathrm{d}x=\frac{1}{a}\int x\mathrm{d}(\tan ax-\sec ax)=\frac{1}{a}x(\tan ax-\sec ax)-\frac{1}{a}\int(\tan ax-\sec ax)\mathrm{d}x$$

$$=-\frac{x}{a}\tan\left(\frac{\pi}{4}-\frac{ax}{2}\right)-\frac{1}{a^2}\left(-\ln|\cos ax|-\ln|\tan ax+\sec ax|\right)+C$$

$$=-\frac{x}{a}\tan\left(\frac{\pi}{4}-\frac{ax}{2}\right)+\frac{1}{a^2}\ln|1+\sin ax|+C.$$

42. 解 $$\int\frac{x}{1-\sin ax}\mathrm{d}x=\frac{1}{a}\int x\mathrm{d}(\tan ax+\sec ax)=\frac{1}{a}x(\tan ax+\sec ax)-\frac{1}{a}\int(\tan ax+\sec ax)\mathrm{d}x$$

$$=\frac{x}{a}\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)-\frac{1}{a^2}\left(-\ln|\cos ax|+\ln|\tan ax+\sec ax|\right)+C$$

$$=\frac{x}{a}\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)+\frac{1}{a^2}\ln|1-\sin ax|+C.$$

43. 解 ①$$\int\frac{\sin ax}{1+\sin ax}\mathrm{d}x=\int\left(1-\frac{1}{1+\sin ax}\right)\mathrm{d}x=x-\int\frac{1}{1+\sin ax}\mathrm{d}x=x+\frac{1}{a}\tan\left(\frac{\pi}{4}-\frac{ax}{2}\right)+C.$$ 由40式

②$$\int\frac{\sin ax}{1-\sin ax}\mathrm{d}x=\int\left(-1+\frac{1}{1-\sin ax}\right)\mathrm{d}x=-x+\int\frac{1}{1-\sin ax}\mathrm{d}x=-x+\frac{1}{a}\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)+C.$$

故原式$=\pm x+\frac{1}{a}\tan\left(\frac{\pi}{4}\mp\frac{ax}{2}\right)+C.$

44. 解 $$\int\frac{\cos ax}{1\pm\sin ax}\mathrm{d}x=\frac{1}{a}\int\frac{1}{1\pm\sin ax}\mathrm{d}(\sin ax)=\pm\frac{1}{a}\int\frac{1}{1\pm\sin ax}\mathrm{d}(1\pm\sin ax)=\pm\frac{1}{a}\ln|1\pm\sin ax|+C.$$

45. 解 ①$$\int\frac{1}{\sin ax(1+\sin ax)}\mathrm{d}x=\int\left(\frac{1}{\sin ax}-\frac{1}{1+\sin ax}\right)\mathrm{d}x=\frac{1}{a}\ln\left|\tan\frac{ax}{2}\right|+\frac{1}{a}\tan\left(\frac{\pi}{4}-\frac{ax}{2}\right)+C.$$ 由40式

②$$\int\frac{1}{\sin ax(1-\sin ax)}\mathrm{d}x=\int\left(\frac{1}{\sin ax}+\frac{1}{1-\sin ax}\right)\mathrm{d}x=\frac{1}{a}\ln\left|\tan\frac{ax}{2}\right|+\frac{1}{a}\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)+C.$$

故原式$=\frac{1}{a}\ln\left|\tan\frac{ax}{2}\right|+\frac{1}{a}\tan\left(\frac{\pi}{4}\mp\frac{ax}{2}\right)+C.$

46. 解 $$\int\frac{1}{\sin ax(1\pm\cos ax)}\mathrm{d}x=\int\frac{1\mp\cos ax}{\sin ax(1\pm\cos ax)(1\mp\cos ax)}\mathrm{d}x=\int\frac{1\mp\cos ax}{\sin^3 ax}\mathrm{d}x$$

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$$=\int\frac{1}{\sin^3 ax}\mathrm{d}x\mp\int\frac{\cos ax}{\sin^3 ax}\mathrm{d}x$$

$$=-\frac{\cos ax}{2a\sin^2 ax}+\frac{1}{2a}\ln\left|\tan\frac{ax}{2}\right|\pm\frac{1}{2a\sin^2 ax}+C$$

由6式

$$=\pm\frac{1}{2a(1\pm\cos ax)}+\frac{1}{2a}\ln\left|\tan\frac{ax}{2}\right|+C.$$

47. 解 $\displaystyle\int\frac{1}{\cos ax(1\pm\sin ax)}\mathrm{d}x=\int\frac{1\mp\sin ax}{\cos ax(1\pm\sin ax)(1\mp\sin ax)}\mathrm{d}x$

$$=\int\frac{1\mp\sin ax}{\cos^3 ax}\mathrm{d}x=\int\frac{1}{\cos^3 ax}\mathrm{d}x\mp\int\frac{\sin ax}{\cos^3 ax}\mathrm{d}x$$

$$=\frac{\sin ax}{2a\cos^2 ax}+\frac{1}{2a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right|\mp\frac{1}{2a\cos^2 ax}+C$$

由7式

$$=\mp\frac{1}{2a(1\pm\sin ax)}+\frac{1}{2a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right|+C.$$

48. 解 $\displaystyle\int\frac{\sin ax}{\cos ax(1\pm\sin ax)}\mathrm{d}x=\pm\int\frac{\pm\sin ax+1-1}{\cos ax(1\pm\sin ax)}\mathrm{d}x=\pm\left[\int\frac{1}{\cos ax}\mathrm{d}x-\int\frac{1}{\cos ax(1\pm\sin ax)}\mathrm{d}x\right]$

$$=\pm\frac{1}{a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right|+\frac{1}{2a(1\pm\sin ax)}\mp\frac{1}{2a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right|+C$$

由47式

$$=\frac{1}{2a(1\pm\sin ax)}\pm\frac{1}{2a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right|+C.$$

49. 解 $\displaystyle\int\frac{\sin ax}{\cos ax(1\pm\cos ax)}\mathrm{d}x=-\frac{1}{a}\int\frac{1}{\cos ax(1\pm\cos ax)}\mathrm{d}(\cos ax)=-\frac{1}{a}\int\left(\frac{1}{\cos ax}\mp\frac{1}{1\pm\cos ax}\right)\mathrm{d}(\cos ax)$

$$=\frac{1}{a}\ln\left|\frac{1\pm\cos ax}{\cos ax}\right|+C=\frac{1}{a}\ln|\sec ax\pm1|+C.$$

50. 解 $\displaystyle\int\frac{\cos ax}{\sin ax(1\pm\sin ax)}\mathrm{d}x=\frac{1}{a}\int\frac{1}{\sin ax(1\pm\sin ax)}\mathrm{d}(\sin ax)=\frac{1}{a}\int\left(\frac{1}{\sin ax}\mp\frac{1}{1\pm\sin ax}\right)\mathrm{d}(\sin ax)$

$$=-\frac{1}{a}\ln\left|\frac{1\pm\sin ax}{\sin ax}\right|+C=-\frac{1}{a}\ln|\csc ax\pm1|+C.$$

51. 解 $\displaystyle\int\frac{\cos ax}{\sin ax(1\pm\cos ax)}\mathrm{d}x=\pm\int\frac{\pm\cos ax+1-1}{\sin ax(1\pm\cos ax)}\mathrm{d}x=\pm\left[\int\frac{1}{\sin ax}\mathrm{d}x-\int\frac{1}{\sin ax(1\pm\cos ax)}\mathrm{d}x\right]$

$$=\pm\left[\frac{1}{a}\ln\left|\tan\frac{ax}{2}\right|\mp\frac{1}{2a(1\pm\cos ax)}-\frac{1}{2a}\ln\left|\tan\frac{ax}{2}\right|\right]+C$$

由46式

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$$=-\frac{1}{2a(1\pm\cos ax)}\pm\frac{1}{2a}\ln\left|\tan\frac{ax}{2}\right|+C.$$

52. 解　原式 $=\int\frac{\mathrm{d}x}{\left(\sin^2\frac{ax}{2}+\cos^2\frac{ax}{2}+2\sin\frac{ax}{2}\cos\frac{ax}{2}\right)^2}=\int\frac{\mathrm{d}x}{\left(\sin\frac{ax}{2}+\cos\frac{ax}{2}\right)^4}$

$$=\int\frac{\mathrm{d}x}{\left[\sqrt{2}\sin\left(\frac{ax}{2}+\frac{\pi}{4}\right)\right]^4}=\frac{1}{4}\int\csc^4\left(\frac{ax}{2}+\frac{\pi}{4}\right)\mathrm{d}x,$$

由于　$\int\csc^4x\mathrm{d}x=-\int\csc^2x\mathrm{d}(\cot x)=-\cot x\cdot\csc^2x-2\int\cot^2x\csc^2x\mathrm{d}x$

$$=-\cot x\cdot\csc^2x-2\int(\csc^2x-1)\csc^2x\mathrm{d}x=-\cot x\cdot\csc^2x-2\int\csc^4x\mathrm{d}x-2\cot x,$$

则　$\int\csc^4x\mathrm{d}x=-\frac{1}{3}\cot x(\csc^2x+2)+C_0=-\frac{1}{3}\cot x(\cot^2x+3)+C_0=-\frac{1}{3}\cot^3x-\cot x+C_0$,

故　原式 $=\frac{1}{2a}\int\csc^4\left(\frac{ax}{2}+\frac{\pi}{4}\right)\mathrm{d}\left(\frac{ax}{2}+\frac{\pi}{4}\right)=-\frac{1}{6a}\cot^3\left(\frac{ax}{2}+\frac{\pi}{4}\right)-\frac{1}{2a}\cot\left(\frac{ax}{2}+\frac{\pi}{4}\right)+C.$

53. 解　原式 $=\int\frac{\mathrm{d}x}{\left(\sin^2\frac{ax}{2}+\cos^2\frac{ax}{2}-2\sin\frac{ax}{2}\cos\frac{ax}{2}\right)^2}=\int\frac{\mathrm{d}x}{\left(\sin\frac{ax}{2}-\cos\frac{ax}{2}\right)^4}$

$$=\int\frac{\mathrm{d}x}{\left[\sqrt{2}\sin\left(\frac{ax}{2}-\frac{\pi}{4}\right)\right]^4}=\frac{1}{4}\int\csc^4\left(\frac{ax}{2}-\frac{\pi}{4}\right)\mathrm{d}x,$$

由上题可知，$\int\csc^4x\mathrm{d}x=-\frac{1}{3}\cot^3x-\cot x+C_0$，故

$$原式=\frac{1}{2a}\int\csc^4\left(\frac{ax}{2}-\frac{\pi}{4}\right)\mathrm{d}\left(\frac{ax}{2}-\frac{\pi}{4}\right)=-\frac{1}{6a}\cot^3\left(\frac{ax}{2}-\frac{\pi}{4}\right)-\frac{1}{2a}\cot\left(\frac{ax}{2}-\frac{\pi}{4}\right)+C.$$

54. 解　借助万能公式，令 $t=\tan\frac{ax}{2}$，则 $x=\frac{2}{a}\arctan t$，$\sin ax=\frac{2t}{1+t^2}$，故

$$原式=\int\frac{\frac{2t}{1+t^2}}{\left(1+\frac{2t}{1+t^2}\right)^2}\cdot\frac{2}{a}\cdot\frac{1}{1+t^2}\mathrm{d}t=\frac{4}{a}\int\frac{t}{(1+t^2+2t)^2}\mathrm{d}t=\frac{4}{a}\int\frac{t+1-1}{(t+1)^4}\mathrm{d}t$$

$$=\frac{4}{a}\int\frac{1}{(t+1)^3}\mathrm{d}t-\frac{4}{a}\int\frac{1}{(t+1)^4}\mathrm{d}t=-\frac{2}{a}\cdot\frac{1}{(t+1)^2}+\frac{4}{3a}\frac{1}{(t+1)^3}+C$$

$$=-\frac{2}{a}\frac{1}{\left(\tan\frac{ax}{2}+1\right)^2}+\frac{4}{3a}\frac{1}{\left(\tan\frac{ax}{2}+1\right)^3}+C.$$

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55. 解　借助万能公式，令 $t=\tan\frac{ax}{2}$，则 $x=\frac{2}{a}\arctan t$，$\sin ax=\frac{2t}{1+t^2}$，故

$$原式=\int\frac{\frac{2t}{1+t^2}}{\left(1-\frac{2t}{1+t^2}\right)^2}\cdot\frac{2}{a}\cdot\frac{1}{1+t^2}\mathrm{d}t=\frac{4}{a}\int\frac{t}{(1+t^2-2t)^2}\mathrm{d}t=\frac{4}{a}\int\frac{t-1+1}{(t-1)^4}\mathrm{d}t$$

$$=\frac{4}{a}\int\frac{1}{(t-1)^3}\mathrm{d}t+\frac{4}{a}\int\frac{1}{(t-1)^4}\mathrm{d}t=-\frac{2}{a}\cdot\frac{1}{(t-1)^2}-\frac{4}{3a}\frac{1}{(t-1)^3}+C$$

$$=-\frac{2}{a}\frac{1}{\left(\tan\frac{ax}{2}-1\right)^2}-\frac{4}{3a}\frac{1}{\left(\tan\frac{ax}{2}-1\right)^3}+C.$$

56. 解　$原式=\int\frac{1}{2\cos^2\frac{ax}{2}}\mathrm{d}x=\frac{1}{2}\cdot\frac{2}{a}\int\sec^2\frac{ax}{2}\mathrm{d}\left(\frac{ax}{2}\right)=\frac{1}{a}\tan\frac{ax}{2}+C.$

57. 解　$原式=\int\frac{1}{2\sin^2\frac{ax}{2}}\mathrm{d}x=\frac{1}{2}\cdot\frac{2}{a}\int\csc^2\frac{ax}{2}\mathrm{d}\left(\frac{ax}{2}\right)=-\frac{1}{a}\cot\frac{ax}{2}+C.$

58. 解　$原式=\frac{1}{2}\int\frac{x}{\cos^2\frac{ax}{2}}\mathrm{d}x=\frac{1}{a}\int x\sec^2\frac{ax}{2}\mathrm{d}\left(\frac{ax}{2}\right)=\frac{1}{a}\int x\mathrm{d}\left(\tan\frac{ax}{2}\right)$

$$=\frac{x}{a}\tan\frac{ax}{2}-\frac{1}{a}\int\tan\frac{ax}{2}\mathrm{d}x=\frac{x}{a}\tan\frac{ax}{2}-\frac{2}{a^2}\int\tan\frac{ax}{2}\mathrm{d}\left(\frac{ax}{2}\right)$$

$$=\frac{x}{a}\tan\frac{ax}{2}+\frac{2}{a^2}\ln\left|\cos\frac{ax}{2}\right|+C.$$

59. 解　$原式=\frac{1}{2}\int\frac{x}{\sin^2\frac{ax}{2}}\mathrm{d}x=\frac{1}{a}\int x\csc^2\frac{ax}{2}\mathrm{d}\left(\frac{ax}{2}\right)=-\frac{1}{a}\int x\mathrm{d}\left(\cot\frac{ax}{2}\right)$

$$=-\frac{x}{a}\cot\frac{ax}{2}+\frac{1}{a}\int\cot\frac{ax}{2}\mathrm{d}x=-\frac{x}{a}\cot\frac{ax}{2}+\frac{2}{a^2}\int\cot\frac{ax}{2}\mathrm{d}\left(\frac{ax}{2}\right)$$

$$=-\frac{x}{a}\cot\frac{ax}{2}+\frac{2}{a^2}\ln\left|\sin\frac{ax}{2}\right|+C.$$

60. 解　$原式=-\frac{1}{a}\int\frac{\mathrm{d}(\cos ax)}{1\pm\cos ax}=\mp\frac{1}{a}\int\frac{\mathrm{d}(1\pm\cos ax)}{1\pm\cos ax}=\mp\frac{1}{a}\ln(1\pm\cos ax)+C.$

61. 解　$原式=\int\frac{2\cos^2\frac{ax}{2}-1}{2\cos^2\frac{ax}{2}}\mathrm{d}x=x-\frac{1}{2}\int\sec^2\frac{ax}{2}\mathrm{d}x=x-\frac{1}{a}\int\sec^2\frac{ax}{2}\mathrm{d}\left(\frac{ax}{2}\right)=x-\frac{1}{a}\tan\frac{ax}{2}+C.$

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62. 解　原式$=\int\frac{1-2\sin^2\frac{ax}{2}}{2\sin^2\frac{ax}{2}}\mathrm{d}x=\frac{1}{2}\int\csc^2\frac{ax}{2}\mathrm{d}x-x=\frac{1}{a}\int\csc^2\frac{ax}{2}\mathrm{d}\left(\frac{ax}{2}\right)-x=-\frac{1}{a}\cot\frac{ax}{2}-x+C.$

63. 解　借助万能公式，令$t=\tan\frac{ax}{2}$，则$x=\frac{2}{a}\arctan t$，$\cos ax=\frac{1-t^2}{1+t^2}$，故

$$原式=\int\frac{1}{\frac{1-t^2}{1+t^2}\cdot\frac{2}{1+t^2}}\cdot\frac{2}{a}\cdot\frac{1}{1+t^2}\mathrm{d}t=\frac{1}{a}\int\frac{1+t^2}{1-t^2}\mathrm{d}t=\frac{1}{a}\int\frac{t^2-1+2}{1-t^2}\mathrm{d}t=-\frac{1}{a}t-\frac{2}{a}\int\frac{1}{t^2-1}\mathrm{d}t$$

$$=-\frac{1}{a}t-\frac{1}{a}\ln\left|\frac{t-1}{t+1}\right|+C=-\frac{1}{a}\tan\frac{ax}{2}-\frac{1}{a}\ln\left|\frac{\tan\frac{ax}{2}-1}{\tan\frac{ax}{2}+1}\right|+C.$$

64. 解　借助万能公式，令$t=\tan\frac{ax}{2}$，则$x=\frac{2}{a}\arctan t$，$\cos ax=\frac{1-t^2}{1+t^2}$，故

$$原式=\int\frac{1}{\frac{1-t^2}{1+t^2}\cdot\frac{2t^2}{1+t^2}}\cdot\frac{2}{a}\cdot\frac{1}{1+t^2}\mathrm{d}t=\frac{1}{a}\int\frac{1+t^2}{t^2(1-t^2)}\mathrm{d}t=\frac{1}{a}\left(\int\frac{1}{t^2}\mathrm{d}t+\int\frac{2}{1-t^2}\mathrm{d}t\right)$$

$$=-\frac{1}{at}-\frac{1}{a}\ln\left|\frac{t-1}{t+1}\right|+C=-\frac{1}{a}\cot\frac{ax}{2}-\frac{1}{a}\ln\left|\frac{\tan\frac{ax}{2}-1}{\tan\frac{ax}{2}+1}\right|+C.$$

65. 解　借助万能公式，令$t=\tan\frac{ax}{2}$，则$x=\frac{2}{a}\arctan t$，$\cos ax=\frac{1-t^2}{1+t^2}$，故

$$原式=\int\frac{1}{\left(1+\frac{1-t^2}{1+t^2}\right)^2}\cdot\frac{2}{a}\cdot\frac{1}{1+t^2}\mathrm{d}t=\frac{1}{2a}\int(1+t^2)\mathrm{d}t=\frac{1}{2a}\left(t+\frac{1}{3}t^3\right)+C=\frac{1}{2a}\tan\frac{ax}{2}+\frac{1}{6a}\tan^3\frac{ax}{2}+C.$$

66. 解　借助万能公式，令$t=\tan\frac{ax}{2}$，则$x=\frac{2}{a}\arctan t$，$\cos ax=\frac{1-t^2}{1+t^2}$，故

$$原式=\int\frac{1}{\left(1-\frac{1-t^2}{1+t^2}\right)^2}\cdot\frac{2}{a}\cdot\frac{1}{1+t^2}\mathrm{d}t=\frac{1}{2a}\int\frac{1+t^2}{t^4}\mathrm{d}t=\frac{1}{2a}\left(-\frac{1}{3t^3}-\frac{1}{t}\right)+C=-\frac{1}{6a}\cot^3\frac{ax}{2}-\frac{1}{2a}\cot\frac{ax}{2}+C.$$

67. 解　借助万能公式，令$t=\tan\frac{ax}{2}$，则$x=\frac{2}{a}\arctan t$，$\cos ax=\frac{1-t^2}{1+t^2}$，故

$$原式=\int\frac{\frac{1-t^2}{1+t^2}}{\left(1+\frac{1-t^2}{1+t^2}\right)^2}\cdot\frac{2}{a}\cdot\frac{1}{1+t^2}\mathrm{d}t=\frac{1}{2a}\int(1-t^2)\mathrm{d}t=\frac{1}{2a}t-\frac{1}{6a}t^3+C=\frac{1}{2a}\tan\frac{ax}{2}-\frac{1}{6a}\tan^3\frac{ax}{2}+C.$$

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68. 解　借助万能公式，令 $t=\tan\frac{ax}{2}$，则 $x=\frac{2}{a}\arctan t$，$\cos ax=\frac{1-t^2}{1+t^2}$，故

$$\text{原式}=\int\frac{\frac{1-t^2}{1+t^2}}{\left(1-\frac{1-t^2}{1+t^2}\right)^2}\cdot\frac{2}{a}\cdot\frac{1}{1+t^2}\mathrm{d}t=\frac{1}{2a}\int\frac{1-t^2}{t^4}\mathrm{d}t=-\frac{1}{6at^3}+\frac{1}{2at}+C=-\frac{1}{6a}\cot^3\frac{ax}{2}+\frac{1}{2a}\cot\frac{ax}{2}+C.$$

69. 解　令 $ax=t$，则

$$\text{原式}=\frac{1}{a}\int\frac{1}{1+\cos t+\sin t}\mathrm{d}t\xlongequal{令u=\tan\frac{t}{2}}\frac{1}{a}\int\frac{1}{1+\frac{1-u^2}{1+u^2}+\frac{2u}{1+u^2}}\cdot\frac{2\mathrm{d}u}{1+u^2}$$

$$=\frac{1}{a}\int\frac{1}{1+u}\mathrm{d}u=\frac{1}{a}\ln|1+u|+C=\frac{1}{a}\ln\left|1+\tan\frac{ax}{2}\right|+C.$$

70. 解　$\int\frac{\mathrm{d}x}{1+\sin^2 ax}=\frac{1}{a}\int\frac{\mathrm{d}(ax)}{1+\sin^2 ax}\xlongequal{令ax=t}\frac{1}{a}\int\frac{\mathrm{d}t}{1+\sin^2 t}$，其中

$$\int\frac{\mathrm{d}t}{1+\sin^2 t}=\int\frac{\mathrm{d}t}{2\sin^2 t+\cos^2 t}=\int\frac{\mathrm{d}(\tan t)}{2\tan^2 t+1}=\frac{1}{\sqrt{2}}\arctan(\sqrt{2}\tan t)+C_1,$$

故
$$\text{原式}=\frac{1}{\sqrt{2}a}\arctan(\sqrt{2}\tan ax)+C.$$

71. 解　$$\int\frac{\mathrm{d}x}{1+b\sin^2 ax}\xlongequal{令ax=t}\frac{1}{a}\int\frac{\mathrm{d}t}{1+b\sin^2 t}=\frac{1}{a}\int\frac{\mathrm{d}t}{(b+1)\sin^2 t+\cos^2 t}=\frac{1}{a}\int\frac{\mathrm{d}(\tan t)}{(b+1)\tan^2 t+1}$$

$$=\frac{1}{a\sqrt{1+b}}\arctan(\sqrt{1+b}\tan ax)+C.$$

72. 解　$$\int\frac{\mathrm{d}x}{1-\sin^2 ax}\xlongequal{令ax=t}\frac{1}{a}\int\frac{\mathrm{d}t}{1-\sin^2 t}=\frac{1}{a}\int\sec^2 t\mathrm{d}t=\frac{1}{a}\tan ax+C.$$

73. 解　令 $t=ax$，原式$=\frac{1}{a}\int\frac{\mathrm{d}t}{1-b\sin^2 t}$.

当 $0<b<1$ 时，原式$=\frac{1}{a}\int\frac{\mathrm{d}t}{(1-b)\sin^2 t+\cos^2 t}=\frac{1}{a}\int\frac{\mathrm{d}(\tan t)}{(1-b)\tan^2 t+1}=\frac{1}{a\sqrt{1-b}}\arctan(\sqrt{1-b}\tan ax)+C$;

当 $b>1$ 时，原式$=-\frac{1}{a}\int\frac{\mathrm{d}t}{-\cos^2 t+(b-1)\sin^2 t}=\frac{-1}{a}\int\frac{\mathrm{d}(\tan t)}{(-1)+(b-1)\tan^2 t}=\frac{1}{2a\sqrt{b-1}}\ln\left|\frac{\sqrt{b-1}\tan t+1}{\sqrt{b-1}\tan t-1}\right|+C$

$$=\frac{1}{2a\sqrt{b-1}}\ln\left|\frac{\sqrt{b-1}\tan ax+1}{\sqrt{b-1}\tan ax-1}\right|+C.$$

74. 解　$\int\frac{\mathrm{d}x}{(1-b\sin^2 ax)^2}\xlongequal{令t=ax}\frac{1}{a}\int\frac{\mathrm{d}t}{(1-b\sin^2 t)^2}$，其中

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$$\int\frac{\mathrm{d}t}{(1-b\sin^2 t)^2}=\int\frac{\mathrm{d}t}{[\cos^2 t+(1-b)\sin^2 t]^2}=\int\frac{\sec^2 t\mathrm{d}(\tan t)}{[1+(1-b)\tan^2 t]^2}$$

$$\overset{令u=\tan t}{=\!=\!=}\int\frac{(u^2+1)\mathrm{d}u}{[1+(1-b)u^2]^2}=\frac{1}{(1-b)^2}\int\frac{u^2+1}{\left(u^2+\dfrac{1}{1-b}\right)^2}\mathrm{d}u$$

$$=-\frac{b}{2(1-b)^2}\frac{\tan ax}{\tan^2 ax+\dfrac{1}{1-b}}+\frac{2-b}{2\sqrt{1-b}(1-b)}\arctan\frac{\tan ax}{\sqrt{\dfrac{1}{1-b}}}+C.$$

75. 解 $$\int\frac{\sin ax\cos ax}{1\pm b\sin^2 ax}\mathrm{d}x\overset{令t=ax}{=\!=\!=}\frac{1}{a}\int\frac{\sin t\cos t}{1\pm b\sin^2 t}\mathrm{d}t=\frac{1}{a}\int\frac{\sin t}{1\pm b\sin^2 t}\mathrm{d}(\sin t)\overset{令u=\sin t}{=\!=\!=}\frac{1}{2a}\int\frac{\mathrm{d}(u^2)}{1\pm bu^2}$$

$$=\pm\frac{1}{2ab}\ln\left|1\pm bu^2\right|+C=\pm\frac{1}{2ab}\ln\left|1\pm b\sin^2 ax\right|+C.$$

76. 解 $$\int\frac{\mathrm{d}x}{1+b\cos^2 ax}\overset{令t=ax}{=\!=\!=}\frac{1}{a}\int\frac{\mathrm{d}t}{1+b\cos^2 t}=\frac{1}{a}\int\frac{\mathrm{d}t}{\sin^2 t+(b+1)\cos^2 t}=\frac{1}{a}\int\frac{\mathrm{d}(\tan t)}{\tan^2 t+(b+1)}$$

$$=\frac{1}{a\sqrt{1+b}}\arctan\frac{\tan t}{\sqrt{1+b}}+C=\frac{1}{a\sqrt{1+b}}\arctan\frac{\tan ax}{\sqrt{1+b}}+C.$$

77. 解 $$\int\frac{\mathrm{d}x}{1-\cos^2 ax}\overset{令ax=t}{=\!=\!=}\frac{1}{a}\int\frac{\mathrm{d}t}{1-\cos^2 t}=\frac{1}{a}\int\csc^2 t\mathrm{d}t=-\frac{1}{a}\cot t+C=-\frac{1}{a}\cot ax+C.$$

78. 解 当$0<b<1$时，

$$\text{原式}\overset{令t=ax}{=\!=\!=}\frac{1}{a}\int\frac{\mathrm{d}t}{1-b\cos^2 t}=\frac{1}{a}\int\frac{\mathrm{d}t}{\sin^2 t+(1-b)\cos^2 t}=\frac{1}{a}\int\frac{\mathrm{d}(\tan t)}{\tan^2 t+(1-b)}=\frac{1}{a\sqrt{1-b}}\arctan\frac{\tan ax}{\sqrt{1-b}}+C;$$

当$b>1$时，

$$\text{原式}\overset{令t=ax}{=\!=\!=}\frac{1}{a}\int\frac{\mathrm{d}(\tan t)}{\tan^2 t-(b-1)}=\frac{1}{2a\sqrt{b-1}}\ln\left|\frac{\tan ax-\sqrt{b-1}}{\tan ax+\sqrt{b-1}}\right|+C.$$

79. 解 $$\int\frac{\mathrm{d}x}{(1-b\cos^2 ax)^2}\overset{令t=ax}{=\!=\!=}\frac{1}{a}\int\frac{\mathrm{d}t}{(1-b\cos^2 t)^2}=\frac{1}{a}\int\frac{\mathrm{d}t}{\left[\sin^2 t+(1-b)\cos^2 t\right]^2}=\frac{1}{a}\int\frac{\sec^2 t\mathrm{d}(\tan t)}{\left[\tan^2 t+(1-b)\right]^2}$$

$$\overset{令u=\tan t}{=\!=\!=}\frac{1}{a}\int\frac{u^2+1}{[u^2+(1-b)]^2}\mathrm{d}u.$$

当$0<b<1$时，

$$\text{原式}=\frac{1}{a}\int\frac{u^2}{\left[u^2+(1-b)\right]^2}\mathrm{d}u+\frac{1}{a}\int\frac{1}{\left[u^2+(1-b)\right]^2}\mathrm{d}u$$

由第一组24，19式

$$=\frac{b\tan ax}{2a(1-b)(1-b+\tan^2 ax)}+\frac{2-b}{2a(1-b)^{\frac{3}{2}}}\arctan\frac{\tan ax}{\sqrt{1-b}}+C$$

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$$=\frac{b\sin 2ax}{4a(1-b)(1-b\cos^2 ax)}+\frac{2-b}{2a(1-b)}\cdot\frac{1}{\sqrt{1-b}}\arctan\frac{\tan ax}{\sqrt{1-b}}+C;$$

当$b>1$时，

$$\text{原式}=\frac{1}{a}\int\frac{u^2}{\left[u^2-(b-1)\right]^2}\mathrm{d}u+\frac{1}{a}\int\frac{1}{\left[u^2-(b-1)\right]^2}\mathrm{d}u$$

由第一组34，32式

$$=\frac{b\tan ax}{2a(b-1)(b-1-\tan^2 ax)}+\frac{2-b}{4a(b-1)^{\frac{3}{2}}}\ln\left|\frac{\tan ax+\sqrt{b-1}}{\tan ax-\sqrt{b-1}}\right|+C$$

$$=\frac{b\sin 2ax}{4a(1-b)(1-b\cos^2 ax)}+\frac{2-b}{2a(1-b)}\cdot\frac{1}{2\sqrt{b-1}}\ln\left|\frac{\tan ax-\sqrt{b-1}}{\tan ax+\sqrt{b-1}}\right|+C.$$

80. 解 $\displaystyle\int\frac{\sin^2 ax}{1+b\cos^2 ax}\mathrm{d}x\overset{令t=ax}{=\!=\!=}\frac{1}{a}\int\frac{\sin^2 t}{1+b\cos^2 t}\mathrm{d}t=\frac{1}{a}\int\frac{\sin^2 t}{\sin^2 t+(1+b)\cos^2 t}\mathrm{d}t$

$$=\frac{1}{a}\int\frac{\tan^2 t}{\tan^2 t+(1+b)}\mathrm{d}t=\frac{1}{a}\int\frac{\sec^2 t-1}{\tan^2 t+(1+b)}\mathrm{d}t,$$

其中 $$\int\frac{\sec^2 t}{\tan^2 t+(1+b)}\mathrm{d}t=\int\frac{\mathrm{d}(\tan t)}{\tan^2 t+(1+b)}=\frac{1}{\sqrt{1+b}}\arctan\frac{\tan t}{\sqrt{1+b}}+C_1,$$

$$\int\frac{1}{\tan^2 t+(1+b)}\mathrm{d}t\overset{令u=\tan t}{=\!=\!=}\int\frac{1}{u^2+(1+b)}\cdot\frac{1}{u^2+1}\mathrm{d}u=\frac{-1}{b}\int\left[\frac{1}{u^2+(1+b)}-\frac{1}{u^2+1}\right]\mathrm{d}u$$

$$=\frac{-1}{b}\left(\frac{1}{\sqrt{1+b}}\arctan\frac{u}{\sqrt{1+b}}-\arctan u\right)+C_2$$

$$=-\frac{1}{b}\cdot\frac{1}{\sqrt{1+b}}\arctan\frac{\tan t}{\sqrt{1+b}}+\frac{1}{b}\arctan\tan t+C_2$$

$$=\frac{t}{b}-\frac{1}{b\sqrt{1+b}}\arctan\frac{\tan t}{\sqrt{1+b}}+C_2,$$

故原式$=-\dfrac{x}{b}+\dfrac{\sqrt{1+b}}{ab}\arctan\dfrac{\tan ax}{\sqrt{1+b}}+C$.

81. 解 $\displaystyle\int\frac{\cos^2 ax}{1+b\cos^2 ax}\mathrm{d}x\overset{令t=ax}{=\!=\!=}\frac{1}{a}\int\frac{\cos^2 t}{1+b\cos^2 t}\mathrm{d}t=\frac{1}{a}\int\frac{\cos^2 t}{\sin^2 t+(1+b)\cos^2 t}\mathrm{d}t$

$$=\frac{1}{a(b+1)}\int\frac{\sin^2 t+(b+1)\cos^2 t-\sin^2 t}{\sin^2 t+(b+1)\cos^2 t}\mathrm{d}t$$

$$=\frac{1}{a(b+1)}\int\left[1-\frac{\sin^2 t}{\sin^2 t+(b+1)\cos^2 t}\right]\mathrm{d}t$$

$$=\frac{t}{a(b+1)}-\frac{1}{a(b+1)}\int\frac{\sin^2 t}{\sin^2 t+(b+1)\cos^2 t}\mathrm{d}t,$$

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$$\frac{1}{a}\int\frac{\sin^2 t}{\sin^2 t+(b+1)\cos^2 t}\mathrm{d}t=\int\frac{\sin^2 ax}{1+b\cos^2 ax}\mathrm{d}x=-\frac{x}{b}+\frac{\sqrt{1+b}}{ab}\arctan\frac{\tan ax}{\sqrt{1+b}}+C_0$$，故

（由80式）

$$\text{原式}=\frac{x}{b+1}-\frac{1}{b+1}\left(-\frac{x}{b}+\frac{\sqrt{1+b}}{ab}\arctan\frac{\tan ax}{\sqrt{1+b}}\right)+C=\frac{x}{b}-\frac{1}{ab\sqrt{1+b}}\arctan\frac{\tan ax}{\sqrt{1+b}}+C.$$

82. 解

$$\int\frac{\sin ax\cos ax}{1\pm b\cos^2 ax}\mathrm{d}x\overset{令t=ax}{=\!=\!=}\frac{1}{a}\int\frac{\sin t\cos t}{1\pm b\cos^2 t}\mathrm{d}t=-\frac{1}{a}\int\frac{\cos t\mathrm{d}(\cos t)}{1\pm b\cos^2 t}=\mp\frac{1}{2ab}\ln\left|1\pm b\cos^2 t\right|+C$$

$$=\mp\frac{1}{2ab}\ln\left|1\pm b\cos^2 ax\right|+C=\mp\frac{1}{ab}\ln\sqrt{\left|1\pm b\cos^2 ax\right|}+C.$$

83. 解

$$\int\frac{\sin^2 ax}{1-b\cos^2 ax}\mathrm{d}x\overset{令t=ax}{=\!=\!=}\frac{1}{a}\int\frac{\sin^2 t}{1-b\cos^2 t}\mathrm{d}t=\frac{1}{a}\int\frac{\sin^2 t}{\sin^2 t+(1-b)\cos^2 t}\mathrm{d}t$$

$$=\frac{1}{a}\int\frac{\tan^2 t}{\tan^2 t+(1-b)}\mathrm{d}t=\frac{1}{a}\left[\int\frac{\sec^2 t-1}{\tan^2 t+(1-b)}\mathrm{d}t\right],$$

当$0<b<1$时，

$$\int\frac{\sec^2 t}{\tan^2 t+(1-b)}\mathrm{d}t=\int\frac{\mathrm{d}(\tan t)}{\tan^2 t+(1-b)}=\frac{1}{\sqrt{1-b}}\arctan\frac{\tan t}{\sqrt{1-b}}+C_1,$$

$$\int\frac{1}{\tan^2 t+(1-b)}\mathrm{d}t\overset{令u=\tan t}{=\!=\!=}\int\frac{1}{u^2+(1-b)}\cdot\frac{1}{u^2+1}\mathrm{d}u=-\frac{t}{b}+\frac{1}{b\sqrt{1-b}}\arctan\frac{\tan t}{\sqrt{1-b}}+C_2$$（与80题类似），

故原式$=-\dfrac{\sqrt{1-b}}{ab}\arctan\dfrac{\tan ax}{\sqrt{1-b}}+\dfrac{x}{b}+C$.

当$b>1$时，

$$\int\frac{\sec^2 t}{\tan^2 t-(b-1)}\mathrm{d}t=\int\frac{\mathrm{d}(\tan t)}{\tan^2 t-(b-1)}=\frac{1}{2\sqrt{b-1}}\ln\left|\frac{\tan t-\sqrt{b-1}}{\tan t+\sqrt{b-1}}\right|+C_1,$$

$$\int\frac{1}{\tan^2 t-(b-1)}\mathrm{d}t\overset{令u=\tan t}{=\!=\!=}\int\frac{1}{u^2-(b-1)}\cdot\frac{1}{u^2+1}\mathrm{d}u=\frac{1}{b}\int\left[\frac{1}{u^2-(b-1)}-\frac{1}{u^2+1}\right]\mathrm{d}u$$

$$=\frac{1}{b}\left(\frac{1}{2\sqrt{b-1}}\ln\left|\frac{\tan t-\sqrt{b-1}}{\tan t+\sqrt{b-1}}\right|-t\right)+C_2,$$

故原式$=\dfrac{\sqrt{b-1}}{2ab}\ln\left|\dfrac{\tan ax-\sqrt{b-1}}{\tan ax+\sqrt{b-1}}\right|+\dfrac{x}{b}+C$.

84. 解

$$\text{原式}=\int\frac{1-\sin^2 ax}{1-b\cos^2 ax}\mathrm{d}x=\int\frac{\mathrm{d}x}{1-b\cos^2 ax}-\int\frac{\sin^2 ax}{1-b\cos^2 ax}\mathrm{d}x$$

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$$=\begin{cases}\dfrac{1}{ab\sqrt{1-b}}\arctan\dfrac{\tan ax}{\sqrt{1-b}}-\dfrac{x}{b}+C, & 0<b<1,\\ \dfrac{1}{2ab\sqrt{b-1}}\ln\left|\dfrac{\tan ax-\sqrt{b-1}}{\tan ax+\sqrt{b-1}}\right|-\dfrac{x}{b}+C, & b>1.\end{cases}$$

→ 由78式和83式

85. 解　令 $t=\tan\dfrac{x}{2}$，则 $\sin x=\dfrac{2t}{1+t^2}$，$\mathrm{d}x=\dfrac{2}{1+t^2}\mathrm{d}t$，故

$$\text{原式}=\int\frac{1}{a+\dfrac{2bt}{1+t^2}}\cdot\frac{2}{1+t^2}\mathrm{d}t=2\int\frac{1}{a(1+t^2)+2bt}\mathrm{d}t=2\int\frac{1}{at^2+2bt+a}\mathrm{d}t$$

$$=2\int\frac{1}{a\left(t^2+2\dfrac{b}{a}t+\dfrac{b^2}{a^2}\right)-\dfrac{b^2}{a}+a}\mathrm{d}t=\frac{2}{a}\int\frac{1}{\left(t+\dfrac{b}{a}\right)^2+\dfrac{a^2-b^2}{a^2}}\mathrm{d}t.$$

当 $a^2>b^2$ 时，原式 $=\dfrac{2}{a}\cdot\dfrac{a}{\sqrt{a^2-b^2}}\arctan\dfrac{\left(t+\dfrac{b}{a}\right)a}{\sqrt{a^2-b^2}}+C=\dfrac{2}{\sqrt{a^2-b^2}}\arctan\dfrac{a\tan\dfrac{x}{2}+b}{\sqrt{a^2-b^2}}+C$；

当 $a^2<b^2$ 时，原式 $=\dfrac{2}{a}\displaystyle\int\frac{1}{\left(t+\dfrac{b}{a}\right)^2-\dfrac{b^2-a^2}{a^2}}\mathrm{d}t=\frac{2}{a}\cdot\frac{a}{2\sqrt{b^2-a^2}}\cdot\ln\left|\frac{t+\dfrac{b}{a}-\dfrac{\sqrt{b^2-a^2}}{a}}{t+\dfrac{b}{a}+\dfrac{\sqrt{b^2-a^2}}{a}}\right|+C$

$$=\frac{1}{\sqrt{b^2-a^2}}\ln\left|\frac{a\tan\dfrac{x}{2}+b-\sqrt{b^2-a^2}}{a\tan\dfrac{x}{2}+b+\sqrt{b^2-a^2}}\right|+C.$$

86. 解　令 $t=\tan\dfrac{x}{2}$，则 $\cos x=\dfrac{1-t^2}{1+t^2}$，$\mathrm{d}x=\dfrac{2\mathrm{d}t}{1+t^2}$，故

$$\int\frac{\mathrm{d}x}{a+b\cos x}=\int\frac{\dfrac{2\mathrm{d}t}{1+t^2}}{a+b\dfrac{1-t^2}{1+t^2}}=2\int\frac{\mathrm{d}t}{a(1+t^2)+b(1-t^2)}=2\int\frac{\mathrm{d}t}{a+b+at^2-bt^2}.$$

当 $a^2<b^2$ 时，

$$\text{原式}=\frac{2}{a-b}\int\frac{\mathrm{d}t}{\dfrac{a+b}{a-b}+t^2}=\frac{2}{a-b}\int\frac{\mathrm{d}t}{\left(t-\sqrt{\dfrac{a+b}{b-a}}\right)\left(t+\sqrt{\dfrac{a+b}{b-a}}\right)}$$

$$=\frac{2}{2(a-b)\sqrt{\dfrac{a+b}{b-a}}}\left[\int\frac{\mathrm{d}\left(t-\sqrt{\dfrac{a+b}{b-a}}\right)}{t-\sqrt{\dfrac{a+b}{b-a}}}-\int\frac{\mathrm{d}\left(t+\sqrt{\dfrac{a+b}{b-a}}\right)}{t+\sqrt{\dfrac{a+b}{b-a}}}\right]$$

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$$=-\frac{1}{\sqrt{b^2-a^2}}\ln\left|\frac{t-\sqrt{\frac{a+b}{b-a}}}{t+\sqrt{\frac{a+b}{b-a}}}\right|+C=-\frac{1}{\sqrt{b^2-a^2}}\ln\left|\frac{\tan\frac{x}{2}-\sqrt{\frac{a+b}{b-a}}}{\tan\frac{x}{2}+\sqrt{\frac{a+b}{b-a}}}\right|+C$$

$$=\frac{1}{\sqrt{b^2-a^2}}\ln\left|\frac{\tan\frac{x}{2}+\sqrt{\frac{a+b}{b-a}}}{\tan\frac{x}{2}-\sqrt{\frac{a+b}{b-a}}}\right|+C=\frac{1}{\sqrt{b^2-a^2}}\ln\left|\frac{\sqrt{b^2-a^2}\tan\frac{x}{2}+|a+b|}{\sqrt{b^2-a^2}\tan\frac{x}{2}-|a+b|}\right|+C;$$

当 $a=b$ 时，

$$\text{原式}=2\int\frac{\mathrm{d}t}{a+a}=\frac{1}{a}\tan\frac{x}{2}+C;$$

当 $a=-b$ 时，

$$\text{原式}=\frac{1}{a}\int\frac{\mathrm{d}t}{t^2}=-\frac{1}{a}\frac{1}{t}+C=-\frac{1}{a}\cot\frac{x}{2}+C;$$

当 $a^2>b^2$ 时，

$$\text{原式}=2\int\frac{\mathrm{d}t}{a+b+(a-b)t^2}=\frac{2}{a+b}\int\frac{\mathrm{d}t}{1+\frac{a-b}{a+b}t^2}=\frac{2}{a+b}\sqrt{\frac{a+b}{a-b}}\int\frac{\mathrm{d}\left(\sqrt{\frac{a-b}{a+b}}t\right)}{1+\left(\sqrt{\frac{a-b}{a+b}}t\right)^2}$$

$$=\frac{2\operatorname{sgn}(a+b)}{\sqrt{a^2-b^2}}\arctan\left(\frac{\sqrt{a^2-b^2}}{|a+b|}\tan\frac{x}{2}\right)+C.$$

87. 解

$$\underbrace{\int\frac{\mathrm{d}x}{(a+b\sin x)^2}}_{I}=\underbrace{\int\frac{\cos^2x\mathrm{d}x}{(a+b\sin x)^2}}_{I_1}+\underbrace{\int\frac{\sin^2x\mathrm{d}x}{(a+b\sin x)^2}}_{I_2},$$

$$I_1=\int\frac{\cos x}{(a+b\sin x)^2}\cos x\mathrm{d}x=\int\frac{\cos x}{(a+b\sin x)^2}\mathrm{d}(\sin x)=\int\frac{\frac{1}{b}\cos x}{(a+b\sin x)^2}\mathrm{d}(a+b\sin x)$$

$$=-\frac{1}{b}\int\cos x\mathrm{d}\left(\frac{1}{a+b\sin x}\right)=-\frac{1}{b}\frac{\cos x}{a+b\sin x}+\frac{1}{b}\int\frac{-\sin x}{a+b\sin x}\mathrm{d}x=-\frac{1}{b}\frac{\cos x}{a+b\sin x}-\frac{1}{b}\int\frac{\sin x}{a+b\sin x}\mathrm{d}x$$

$$=-\frac{1}{b}\frac{\cos x}{a+b\sin x}-\frac{1}{b^2}\int\frac{b\sin x+a-a}{a+b\sin x}\mathrm{d}x=-\frac{1}{b}\frac{\cos x}{a+b\sin x}-\frac{x}{b^2}+\frac{a}{b^2}\int\frac{1}{a+b\sin x}\mathrm{d}x,$$

$$I_2=\int\frac{\sin^2x\mathrm{d}x}{(a+b\sin x)^2}=\frac{1}{b^2}\int\frac{(a+b\sin x)^2-a^2-2ab\sin x}{(a+b\sin x)^2}\mathrm{d}x$$

$$=\frac{x}{b^2}-\frac{a^2}{b^2}\int\frac{1}{(a+b\sin x)^2}\mathrm{d}x-\int\frac{2a(a+b\sin x)-2a^2}{b^2(a+b\sin x)^2}\mathrm{d}x$$

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$$=\frac{x}{b^2}+\frac{a^2}{b^2}\int\frac{1}{(a+b\sin x)^2}\mathrm{d}x-\frac{2a}{b^2}\int\frac{1}{a+b\sin x}\mathrm{d}x,$$

$$I=I_1+I_2=-\frac{1}{b}\frac{\cos x}{a+b\sin x}-\cancel{\frac{x}{b^2}}+\frac{a}{b^2}\int\frac{1}{a+b\sin x}\mathrm{d}x+\cancel{\frac{x}{b^2}}+\frac{a^2}{b^2}\underbrace{\int\frac{1}{(a+b\sin x)^2}\mathrm{d}x}_{I}-\frac{2a}{b^2}\int\frac{1}{a+b\sin x}\mathrm{d}x,$$

$$\left(1-\frac{a^2}{b^2}\right)I=\frac{\cos x}{-b(a+b\sin x)}-\frac{a}{b^2}\int\frac{1}{a+b\sin x}\mathrm{d}x,$$

$$I=\left[\frac{\cos x}{-b(a+b\sin x)}-\frac{a}{b^2}\int\frac{\mathrm{d}x}{a+b\sin x}\right]\cdot\frac{b^2}{b^2-a^2}$$

$$=\begin{cases}\dfrac{b\cos x}{(a^2-b^2)(a+b\sin x)}+\dfrac{2a}{(a^2-b^2)^{\frac{3}{2}}}\arctan\dfrac{a\tan\frac{x}{2}+b}{\sqrt{a^2-b^2}}+C, & a^2>b^2,\\ \dfrac{b\cos x}{(a^2-b^2)(a+b\sin x)}-\dfrac{a}{(b^2-a^2)^{\frac{3}{2}}}\ln\left|\dfrac{a\tan\frac{x}{2}+b-\sqrt{b^2-a^2}}{a\tan\frac{x}{2}+b+\sqrt{b^2-a^2}}\right|+C, & a^2<b^2.\end{cases}$$

→ 由85式

88. 解 $\underbrace{\int\frac{\mathrm{d}x}{(a+b\cos x)^2}}_{I}=\int\frac{\sin^2x+\cos^2x}{(a+b\cos x)^2}\mathrm{d}x=\underbrace{\int\frac{\sin^2x\mathrm{d}x}{(a+b\cos x)^2}}_{I_1}+\underbrace{\int\frac{\cos^2x\mathrm{d}x}{(a+b\cos x)^2}}_{I_2}$,

$$I_1=\int\frac{\sin x}{(a+b\cos x)^2}\sin x\mathrm{d}x=\int\frac{-\sin x\mathrm{d}(\cos x)}{(a+b\cos x)^2}=\int\frac{-\sin x\mathrm{d}(a+b\cos x)}{b(a+b\cos x)^2}=\frac{1}{b}\int\sin x\mathrm{d}\left(\frac{1}{a+b\cos x}\right)$$

$$=\frac{\sin x}{b(a+b\cos x)}-\frac{1}{b}\int\frac{\cos x}{a+b\cos x}\mathrm{d}x=\frac{\sin x}{b(a+b\cos x)}-\frac{1}{b^2}\int\frac{b\cos x+a-a}{a+b\cos x}\mathrm{d}x$$

$$=\frac{\sin x}{b(a+b\cos x)}-\frac{x}{b^2}+\frac{a}{b^2}\int\frac{\mathrm{d}x}{a+b\cos x},$$

$$I_2=\int\frac{\cos^2x\mathrm{d}x}{(a+b\cos x)^2}\mathrm{d}x=\int\frac{(a+b\cos x)^2-a^2-2ab\cos x}{b^2(a+b\cos x)^2}\mathrm{d}x=\int\left[\frac{1}{b^2}-\frac{a^2}{b^2(a+b\cos x)^2}-\frac{2ab\cos x}{b^2(a+b\cos x)^2}\right]\mathrm{d}x$$

$$=\int\left[\frac{1}{b^2}-\frac{a^2}{b^2(a+b\cos x)^2}-\frac{2a(a+b\cos x)-2a^2}{b^2(a+b\cos x)^2}\right]\mathrm{d}x$$

$$=\frac{x}{b^2}-\frac{2a}{b^2}\int\frac{\mathrm{d}x}{a+b\cos x}+\frac{a^2}{b^2}\int\frac{\mathrm{d}x}{(a+b\cos x)^2}.$$

故 $I=I_1+I_2=\dfrac{\sin x}{b(a+b\cos x)}-\dfrac{x}{b^2}+\dfrac{a}{b^2}\int\dfrac{\mathrm{d}x}{a+b\cos x}+\dfrac{x}{b^2}-\dfrac{2a}{b^2}\int\dfrac{\mathrm{d}x}{a+b\cos x}+\dfrac{a^2}{b^2}\underbrace{\int\dfrac{\mathrm{d}x}{(a+b\cos x)^2}}_{I}$,

则 $$\left(1-\frac{a^2}{b^2}\right)I=\frac{\sin x}{b(a+b\cos x)}-\frac{a}{b^2}\int\frac{\mathrm{d}x}{a+b\cos x},$$

故 $$I=\left[\frac{\sin x}{b(a+b\cos x)}-\frac{a}{b^2}\int\frac{\mathrm{d}x}{a+b\cos x}\right]\cdot\frac{b^2}{b^2-a^2}$$

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$$=\frac{b\sin x}{(b^2-a^2)(a+b\cos x)}-\frac{a}{b^2-a^2}\int\frac{\mathrm{d}x}{a+b\cos x}.$$

89. 解 $\int\frac{\mathrm{d}x}{\sin x(a+b\sin x)}=\int\frac{A}{\sin x}\mathrm{d}x+\int\frac{B}{a+b\sin x}\mathrm{d}x=\int\frac{A(a+b\sin x)+B\sin x}{\sin x(a+b\sin x)}\mathrm{d}x$,

则有 $Aa+Ab\sin x+B\sin x=1$，即 $\begin{cases}Aa=1,\\ Ab+B=0,\end{cases}$ 解得 $\begin{cases}A=\frac{1}{a},\\ B=-\frac{b}{a},\end{cases}$ 故

$$\text{原式}=\frac{1}{a}\int\frac{1}{\sin x}\mathrm{d}x-\frac{b}{a}\int\frac{1}{a+b\sin x}\mathrm{d}x=\frac{1}{a}\ln\left|\tan\frac{x}{2}\right|-\frac{b}{a}\int\frac{1}{a+b\sin x}\mathrm{d}x.$$

90. 解 $\int\frac{\mathrm{d}x}{\cos x(a+b\cos x)}=\frac{1}{a}\int\frac{1}{\cos x}\mathrm{d}x-\frac{b}{a}\int\frac{1}{a+b\cos x}\mathrm{d}x=\frac{1}{a}\ln\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right|-\frac{b}{a}\int\frac{1}{a+b\cos x}\mathrm{d}x.$

91. 解 $\int\frac{\sin x}{a+b\sin x}\mathrm{d}x=\frac{1}{b}\int\frac{a+b\sin x-a}{a+b\sin x}\mathrm{d}x=\frac{x}{b}-\frac{a}{b}\int\frac{1}{a+b\sin x}\mathrm{d}x.$

92. 解 $\int\frac{\cos x}{a+b\cos x}\mathrm{d}x=\frac{1}{b}\int\frac{a+b\cos x-a}{a+b\cos x}\mathrm{d}x=\frac{x}{b}-\frac{a}{b}\int\frac{\mathrm{d}x}{a+b\cos x}.$

93. 解 $\int\frac{\cos x}{(a+b\cos x)^2}\mathrm{d}x=\frac{1}{b}\int\frac{a+b\cos x-a}{(a+b\cos x)^2}\mathrm{d}x$

$$=\frac{1}{b}\int\frac{\mathrm{d}x}{a+b\cos x}-\frac{a}{b}\int\frac{1}{(a+b\cos x)^2}\mathrm{d}x \quad \text{由88式}$$

$$=\frac{1}{b}\int\frac{\mathrm{d}x}{a+b\cos x}-\frac{a}{b}\cdot\frac{b\sin x}{(b^2-a^2)(a+b\cos x)}+\frac{a}{b}\cdot\frac{a}{b^2-a^2}\int\frac{\mathrm{d}x}{a+b\cos x}$$

$$=\left[\frac{1}{b}+\frac{a^2}{b(b^2-a^2)}\right]\int\frac{\mathrm{d}x}{a+b\cos x}-\frac{a\sin x}{(b^2-a^2)(a+b\cos x)}$$

$$=\frac{b}{b^2-a^2}\int\frac{\mathrm{d}x}{a+b\cos x}-\frac{a\sin x}{(b^2-a^2)(a+b\cos x)}$$

$$=\frac{a\sin x}{(a^2-b^2)(a+b\cos x)}-\frac{b}{a^2-b^2}\int\frac{\mathrm{d}x}{a+b\cos x}.$$

94. 解 $\int\frac{\mathrm{d}x}{a^2\cos^2x+b^2\sin^2x}=\int\frac{\sec^2x\mathrm{d}x}{a^2+b^2\tan^2x}=\int\frac{\mathrm{d}(\tan x)}{a^2+b^2\tan^2x}=\int\frac{\mathrm{d}\left(\frac{1}{a^2}\tan x\right)}{1+\left(\frac{b}{a}\tan x\right)^2}$

$$=\int\frac{\frac{1}{ab}\mathrm{d}\left(\frac{b}{a}\tan x\right)}{1+\left(\frac{b}{a}\tan x\right)^2}=\frac{1}{ab}\arctan\frac{b\tan x}{a}+C.$$

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95. 解 $$\int\frac{\sin cx}{a\cos cx+b\sin cx}\mathrm{d}x=\int\frac{A(a\cos cx+b\sin cx)'+B(a\cos cx+b\sin cx)}{a\cos cx+b\sin cx}\mathrm{d}x,$$

其中 $A(a\cos cx+b\sin cx)'+B(a\cos cx+b\sin cx)=-Aac\sin cx+Abc\cos cx+Ba\cos cx+Bb\sin cx=\sin cx.$

由比较系数法，得 $\begin{cases}Bb-Aac=1,\\Abc+Ba=0,\end{cases}$ 整理得 $\begin{cases}A=-\dfrac{a}{c(a^2+b^2)},\\B=\dfrac{b}{a^2+b^2},\end{cases}$

$$I=-\frac{a}{c(a^2+b^2)}\ln|a\cos cx+b\sin cx|+\frac{bx}{a^2+b^2}+C=\frac{1}{c(a^2+b^2)}(bcx-a\ln|a\cos cx+b\sin cx|)+C.$$

96. 解 $$\int\frac{\cos cx}{a\cos cx+b\sin cx}\mathrm{d}x=\int\frac{A(a\cos cx+b\sin cx)'+B(a\cos cx+b\sin cx)}{a\cos cx+b\sin cx}\mathrm{d}x,$$

其中 $A(a\cos cx+b\sin cx)'+B(a\cos cx+b\sin cx)=-Aac\sin cx+Abc\cos cx+Ba\cos cx+Bb\sin cx=\cos cx.$

由比较系数法，得 $\begin{cases}Bb-Aac=0,\\Abc+Ba=1,\end{cases}$ 整理得 $\begin{cases}A=\dfrac{b}{c(a^2+b^2)},\\B=\dfrac{a}{a^2+b^2},\end{cases}$

$$I=\frac{b}{c(a^2+b^2)}\ln|a\cos cx+b\sin cx|+\frac{ax}{a^2+b^2}+C=\frac{1}{c(a^2+b^2)}(acx+b\ln|a\cos cx+b\sin cx|)+C.$$

97. 解 $$\int\frac{\sin cx\cos cx}{a\cos^2cx+b\sin^2cx}\mathrm{d}x=\frac{1}{2bc-2ac}\int\frac{\mathrm{d}(a\cos^2cx+b\sin^2cx)}{a\cos^2cx+b\sin^2cx}$$

$$=\frac{1}{2c(b-a)}\ln|a\cos^2cx+b\sin^2cx|+C.$$

98. 解 $$\int\frac{\mathrm{d}x}{a^2+b^2\sin^2cx}=\int\frac{\sec^2cx\mathrm{d}x}{a^2\sec^2cx+b^2\tan^2cx}=\frac{1}{c}\int\frac{\mathrm{d}(\tan cx)}{a^2(\tan^2cx+1)+b^2\tan^2cx}$$

$$=\frac{1}{c}\int\frac{\mathrm{d}(\tan cx)}{(a^2+b^2)\tan^2cx+a^2}=\frac{1}{a^2c}\int\frac{\mathrm{d}(\tan cx)}{1+\dfrac{a^2+b^2}{a^2}\tan^2cx}$$

$$=\frac{1}{ac}\int\frac{\dfrac{1}{\sqrt{a^2+b^2}}\mathrm{d}\left(\dfrac{\sqrt{a^2+b^2}}{a}\tan cx\right)}{1+\left(\dfrac{\sqrt{a^2+b^2}}{a}\tan cx\right)^2}$$

$$=\frac{1}{ac\sqrt{a^2+b^2}}\arctan\frac{\sqrt{a^2+b^2}\tan cx}{a}+C.$$

99. 解 $$I=\int\frac{1-\sin^2cx}{a^2+b^2\sin^2cx}\mathrm{d}x=\underbrace{\int\frac{1}{a^2+b^2\sin^2cx}\mathrm{d}x}_{I_1}-\underbrace{\int\frac{\sin^2cx}{a^2+b^2\sin^2cx}\mathrm{d}x}_{I_2},$$

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其中 $$I_2=\frac{1}{b^2}\int\frac{b^2\sin^2cx+a^2-a^2}{a^2+b^2\sin^2cx}\mathrm{d}x=\frac{x}{b^2}-\frac{a^2}{b^2}\int\frac{1}{a^2+b^2\sin^2cx}\mathrm{d}x\ .$$

故 $$I=\left(1+\frac{a^2}{b^2}\right)\int\frac{1}{a^2+b^2\sin^2cx}\mathrm{d}x-\frac{x}{b^2}$$ （由98式）

$$=\frac{b^2+a^2}{b^2}\cdot\frac{1}{ac\sqrt{a^2+b^2}}\arctan\frac{\sqrt{a^2+b^2}\tan cx}{a}-\frac{x}{b^2}+C$$

$$=\frac{\sqrt{a^2+b^2}}{ab^2c}\arctan\frac{\sqrt{a^2+b^2}\tan cx}{a}-\frac{x}{b^2}+C\ .$$

100. 解 当 $a^2>b^2$ 时，

$$\int\frac{1}{a^2-b^2\sin^2cx}\mathrm{d}x=\int\frac{\sec^2cx\mathrm{d}x}{a^2\sec^2cx-b^2\tan^2cx}=\frac{1}{c}\int\frac{\mathrm{d}(\tan cx)}{a^2(\tan^2cx+1)-b^2\tan^2cx}$$

$$=\frac{1}{c}\int\frac{\mathrm{d}(\tan cx)}{(a^2-b^2)\tan^2cx+a^2}=\frac{1}{a^2c}\int\frac{\mathrm{d}(\tan cx)}{\frac{a^2-b^2}{a^2}\tan^2cx+1}$$

$$=\frac{1}{a^2c}\int\frac{\frac{a}{\sqrt{a^2-b^2}}\mathrm{d}\left(\frac{\sqrt{a^2-b^2}}{a}\tan cx\right)}{\left(\frac{\sqrt{a^2-b^2}}{a}\tan cx\right)^2+1}=\frac{1}{ac\sqrt{a^2-b^2}}\arctan\frac{\sqrt{a^2-b^2}\tan cx}{a}+C\ .$$

当 $a^2<b^2$ 时，

$$\int\frac{1}{a^2-b^2\sin^2cx}\mathrm{d}x=\int\frac{1}{(a-b\sin cx)(a+b\sin cx)}\mathrm{d}x=\frac{1}{2a}\left(\underbrace{\int\frac{1}{a-b\sin cx}\mathrm{d}x}_{I_1}+\underbrace{\int\frac{1}{a+b\sin cx}\mathrm{d}x}_{I_2}\right),$$

其中 $$I_2=\frac{1}{c}\int\frac{\mathrm{d}(cx)}{a+b\sin cx}$$ （由85式）

$$=\frac{1}{c\sqrt{b^2-a^2}}\ln\left|\frac{a\tan\frac{cx}{2}+b-\sqrt{b^2-a^2}}{a\tan\frac{cx}{2}+b+\sqrt{b^2-a^2}}\right|+C_1.$$

同理 $$I_1=\frac{1}{c}\int\frac{\mathrm{d}(cx)}{a-b\sin cx}=\frac{1}{c\sqrt{b^2-a^2}}\ln\left|\frac{a\tan\frac{cx}{2}-b-\sqrt{b^2-a^2}}{a\tan\frac{cx}{2}-b+\sqrt{b^2-a^2}}\right|+C_2.$$

故 $$\text{原式}=\frac{1}{2a}(I_1+I_2)=\frac{1}{2ac\sqrt{b^2-a^2}}\ln\left|\frac{\sqrt{b^2-a^2}\tan cx+a}{\sqrt{b^2-a^2}\tan cx-a}\right|+C.$$

101. 解 $$\int\frac{\mathrm{d}x}{a^2+b^2\cos^2cx}=\int\frac{\sec^2cx\mathrm{d}x}{a^2\sec^2cx+b^2}=\frac{1}{c}\int\frac{\mathrm{d}(\tan cx)}{a^2(\tan^2cx+1)+b^2}$$

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$$=\frac{1}{c}\int\frac{\mathrm{d}(\tan cx)}{a^2\tan^2 cx+(a^2+b^2)}=\frac{1}{(a^2+b^2)c}\int\frac{\mathrm{d}(\tan cx)}{\frac{a^2}{a^2+b^2}\tan^2 cx+1}$$

$$=\frac{1}{(a^2+b^2)c}\int\frac{\frac{\sqrt{a^2+b^2}}{a}\mathrm{d}\left(\frac{a}{\sqrt{a^2+b^2}}\tan cx\right)}{\left(\frac{a}{\sqrt{a^2+b^2}}\tan cx\right)^2+1}$$

$$=\frac{1}{ac\sqrt{a^2+b^2}}\arctan\frac{a\tan cx}{\sqrt{a^2+b^2}}+C.$$

102. 解　当$a^2>b^2$时，

$$\text{原式}=\int\frac{\sec^2 cx\mathrm{d}x}{a^2\sec^2 cx-b^2}=\int\frac{\sec^2 cx\mathrm{d}x}{a^2(\tan^2 cx+1)-b^2}=\frac{1}{c}\int\frac{\mathrm{d}(\tan cx)}{a^2\tan^2 cx+(a^2-b^2)}$$

$$=\frac{1}{c(a^2-b^2)}\int\frac{\mathrm{d}(\tan cx)}{\frac{a^2}{a^2-b^2}\tan^2 cx+1}=\frac{1}{c(a^2-b^2)}\int\frac{\frac{\sqrt{a^2-b^2}}{a}\mathrm{d}\left(\frac{a}{\sqrt{a^2-b^2}}\tan cx\right)}{\left(\frac{a}{\sqrt{a^2-b^2}}\tan cx\right)^2+1}$$

$$=\frac{1}{ac\sqrt{a^2-b^2}}\arctan\frac{a\tan cx}{\sqrt{a^2-b^2}}+C;$$

当$a^2<b^2$时，

$$\text{原式}=\int\frac{\mathrm{d}x}{(a-b\cos cx)(a+b\cos cx)}=\frac{1}{2a}\left(\underbrace{\int\frac{\mathrm{d}x}{a-b\cos cx}}_{I_1}+\underbrace{\int\frac{\mathrm{d}x}{a+b\cos cx}}_{I_2}\right),$$

其中

$$I_2=\frac{1}{c}\int\frac{\mathrm{d}(cx)}{a+b\cos cx}\xrightarrow{\text{由86式}}\frac{1}{c\sqrt{b^2-a^2}}\ln\left|\frac{\sqrt{b^2-a^2}\tan\frac{cx}{2}+a+b}{\sqrt{b^2-a^2}\tan\frac{cx}{2}-a-b}\right|+C_1,$$

$$I_1=\frac{1}{c}\int\frac{\mathrm{d}(cx)}{a+(-b)\cos cx}\xrightarrow{\text{由86式}}\frac{1}{c\sqrt{b^2-a^2}}\ln\left|\frac{\sqrt{b^2-a^2}\tan\frac{cx}{2}+b-a}{\sqrt{b^2-a^2}\tan\frac{cx}{2}-b+a}\right|+C_2.$$

故

$$\text{原式}=\frac{1}{2a}(I_1+I_2)=\frac{1}{2ac\sqrt{b^2-a^2}}\left(\ln\left|\frac{\sqrt{b^2-a^2}\tan\frac{cx}{2}+a+b}{\sqrt{b^2-a^2}\tan\frac{cx}{2}-a-b}\right|+\ln\left|\frac{\sqrt{b^2-a^2}\tan\frac{cx}{2}+b-a}{\sqrt{b^2-a^2}\tan\frac{cx}{2}-b+a}\right|\right)+C$$

$$=\frac{1}{2ac\sqrt{b^2-a^2}}\ln\left|\frac{a\tan cx-\sqrt{b^2-a^2}}{a\tan cx+\sqrt{b^2-a^2}}\right|+C.$$

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103. 解　令 $t=\tan\dfrac{cx}{2}$，则 $x=\dfrac{2}{c}\arctan t$，$\cos cx=\dfrac{1-t^2}{1+t^2}$.

$$\text{原式}=\int\frac{\mathrm{d}\left(\frac{2}{c}\arctan t\right)}{a^2+b^2-2ab\frac{1-t^2}{1+t^2}}=\frac{2}{c}\int\frac{1}{a^2+b^2-2ab\frac{1-t^2}{1+t^2}}\frac{1}{1+t^2}\mathrm{d}t=\frac{2}{c}\int\frac{1}{(a^2+b^2)(1+t^2)-2ab(1-t^2)}\mathrm{d}t$$

$$=\frac{2}{c}\int\frac{1}{(a-b)^2+(a+b)^2t^2}\mathrm{d}t=\frac{2}{c(a+b)}\int\frac{\mathrm{d}[(a+b)t]}{(a-b)^2+(a+b)^2t^2}$$

$$=\frac{2}{c(a+b)(a-b)}\int\frac{\mathrm{d}\left(\frac{a+b}{a-b}t\right)}{1+\left(\frac{a+b}{a-b}t\right)^2}=\frac{2}{c(a+b)(a-b)}\arctan\frac{(a+b)t}{a-b}+C$$

$$=\frac{2}{c(a^2-b^2)}\arctan\left(\frac{a+b}{a-b}\tan\frac{cx}{2}\right)+C.$$

104. 解　$$\text{原式}=\int\frac{x+2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}\mathrm{d}x=\int\left(\frac{1}{2}x\sec^2\frac{x}{2}+\tan\frac{x}{2}\right)\mathrm{d}x=\int x\mathrm{d}\left(\tan\frac{x}{2}\right)+\int\tan\frac{x}{2}\mathrm{d}x$$

$$=x\tan\frac{x}{2}-\int\tan\frac{x}{2}\mathrm{d}x+\int\tan\frac{x}{2}\mathrm{d}x=x\tan\frac{x}{2}+C.$$

105. 解　$$\int\frac{x-\sin x}{1-\cos x}\mathrm{d}x=\int\frac{x-2\sin\frac{x}{2}\cos\frac{x}{2}}{2\sin^2\frac{x}{2}}\mathrm{d}x=\int\left(\frac{1}{2}x\csc^2\frac{x}{2}-\cot\frac{x}{2}\right)\mathrm{d}x$$

$$=-\int x\mathrm{d}\left(\cot\frac{x}{2}\right)-\int\cot\frac{x}{2}\mathrm{d}x=-x\cot\frac{x}{2}+\int\cot\frac{x}{2}\mathrm{d}x-\int\cot\frac{x}{2}\mathrm{d}x$$

$$=-x\cot\frac{x}{2}+C.$$

106. 解　$$\text{原式}=\int\frac{1}{a}\cdot\frac{1}{\sqrt{2}\sin\left(ax\pm\frac{\pi}{4}\right)}\mathrm{d}\left(ax\pm\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}a}\int\csc\left(ax\pm\frac{\pi}{4}\right)\mathrm{d}\left(ax\pm\frac{\pi}{4}\right)$$

$$=\frac{1}{\sqrt{2}a}\ln\left|\tan\left(\frac{ax\pm\frac{\pi}{4}}{2}\right)\right|+C=\frac{1}{\sqrt{2}a}\ln\left|\tan\left(\frac{ax}{2}\pm\frac{\pi}{8}\right)\right|+C.$$

107. 解　$$\text{原式}=\int\frac{1}{1\pm\sin 2ax}\mathrm{d}x=\int\frac{1\mp\sin 2ax}{\cos^2 2ax}\mathrm{d}x=\frac{1}{2a}\int\sec^2 2ax\mathrm{d}(2ax)\pm\frac{1}{2a}\int\sec^2 2ax\mathrm{d}(\cos 2ax)$$

$$=\frac{1}{2a}\tan 2ax\mp\frac{1}{2a}\sec 2ax+C=\frac{1}{2a}\cdot\frac{\sin 2ax\mp 1}{\cos 2ax}+C=\frac{1}{2a}\cdot\frac{\sin ax\mp\cos ax}{\cos ax\pm\sin ax}+C$$

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$$=\frac{1}{2a}\cdot\frac{\tan ax\mp 1}{1\pm\tan ax}+C=\frac{1}{2a}\tan\left(ax\mp\frac{\pi}{4}\right)+C.$$

108. 解 原式$=\int\frac{\frac{\sin ax+\cos ax}{2}+\frac{\sin ax-\cos ax}{2}}{\sin ax\pm\cos ax}\mathrm{d}x=\int\left(\frac{1}{2}+\frac{1}{2}\frac{\sin ax\mp\cos ax}{\sin ax\pm\cos ax}\right)\mathrm{d}x$

$$=\frac{x}{2}\mp\frac{1}{2a}\int\frac{1}{\sin ax\pm\cos ax}\mathrm{d}(\sin ax\pm\cos ax)$$

$$=\frac{x}{2}\mp\frac{1}{2a}\ln|\sin ax\pm\cos ax|+C$$

$$=\frac{1}{2a}(ax\mp\ln|\sin ax\pm\cos ax|)+C.$$

109. 解 原式$=\int\frac{\frac{\sin ax+\cos ax}{2}-\frac{\sin ax-\cos ax}{2}}{\sin ax\pm\cos ax}\mathrm{d}x=\int\left(\pm\frac{1}{2}\mp\frac{1}{2}\frac{\sin ax\mp\cos ax}{\sin ax\pm\cos ax}\right)\mathrm{d}x$

$$=\pm\frac{x}{2}+\frac{1}{2a}\int\frac{1}{\sin ax\pm\cos ax}\mathrm{d}(\sin ax\pm\cos ax)$$

$$=\pm\frac{x}{2}+\frac{1}{2a}\ln|\sin ax\pm\cos ax|+C$$

$$=\frac{1}{2a}(\ln|\sin ax\pm\cos ax|\pm ax)+C.$$

110. 解 原式$=\int\frac{1}{\sqrt{a^2+b^2}\sin\left(x+\arctan\frac{a}{b}\right)}\mathrm{d}\left(x+\arctan\frac{a}{b}\right)$

$$=\frac{1}{\sqrt{a^2+b^2}}\int\csc\left(x+\arctan\frac{a}{b}\right)\mathrm{d}\left(x+\arctan\frac{a}{b}\right)$$

$$=\frac{1}{\sqrt{a^2+b^2}}\ln\left|\tan\frac{x+\arctan\frac{a}{b}}{2}\right|+C.$$

111. 解 原式$=\int\frac{\sec^2 x}{a^2+b^2\tan^2 x}\mathrm{d}x=\frac{1}{b}\int\frac{1}{a^2+(b\tan x)^2}\mathrm{d}(b\tan x)=\frac{1}{ab}\arctan\frac{b\tan x}{a}+C.$

112. 解 原式$=\int\frac{\sec^2 x}{a^2-b^2\tan^2 x}\mathrm{d}x=\frac{1}{b}\int\frac{1}{a^2-(b\tan x)^2}\mathrm{d}(b\tan x)$

$$=\frac{1}{b}\cdot\frac{1}{2a}\int\left(\frac{1}{a-b\tan x}+\frac{1}{a+b\tan x}\right)\mathrm{d}(b\tan x)$$

$$=\frac{1}{2ab}\ln\left|\frac{b\tan x+a}{b\tan x-a}\right|+C.$$

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113. 解　原式 $=-\frac{1}{a}\int x\mathrm{d}(\cos ax)=-\frac{1}{a}\left[x\cos ax-\frac{1}{a}\int\cos ax\mathrm{d}(ax)\right]=-\frac{x}{a}\cos ax+\frac{1}{a^2}\sin ax+C$.

114. 解　原式 $=-\frac{1}{a}\int x^2\mathrm{d}(\cos ax)=-\frac{1}{a}\left(x^2\cos ax-\int\cos ax\cdot 2x\mathrm{d}x\right)$，其中

$$\int\cos ax\cdot 2x\mathrm{d}x=\frac{1}{a}\int 2x\mathrm{d}(\sin ax)=\frac{1}{a}2x\sin ax-\frac{2}{a^2}\int\sin ax\mathrm{d}(ax)=\frac{2x}{a}\sin ax+\frac{2}{a^2}\cos ax+C_0,$$

则　原式 $=-\frac{x^2}{a}\cos ax+\frac{2x}{a^2}\sin ax+\frac{2}{a^3}\cos ax+C$.

115. 解　原式 $=\frac{1}{a}\int x\mathrm{d}(\sin ax)=\frac{1}{a}x\sin ax-\frac{1}{a^2}\int\sin ax\mathrm{d}(ax)=\frac{x}{a}\sin ax+\frac{1}{a^2}\cos ax+C$.

116. 解　原式 $=\frac{1}{a}\int x^2\mathrm{d}(\sin ax)=\frac{1}{a}\left(x^2\sin ax-\int\sin ax\cdot 2x\mathrm{d}x\right)$，其中

$$\int\sin ax\cdot 2x\mathrm{d}x=-\frac{1}{a}\int 2x\mathrm{d}(\cos ax)=-\frac{1}{a}\left[2x\cos ax-\frac{2}{a}\int\cos ax\mathrm{d}(ax)\right]=-\frac{2x}{a}\cos ax+\frac{2}{a^2}\sin ax+C_0,$$

则　原式 $=\frac{x^2}{a}\sin ax+\frac{2x}{a^2}\cos ax-\frac{2}{a^3}\sin ax+C$.

第四组

1. 解　原式 $=\int\sqrt{\left(\sin\frac{a}{2}x+\cos\frac{a}{2}x\right)^2}\,\mathrm{d}x=\int\pm\left(\sin\frac{a}{2}x+\cos\frac{a}{2}x\right)\mathrm{d}x=\pm\frac{2}{a}\int\sqrt{2}\sin\left(\frac{a}{2}x+\frac{\pi}{4}\right)\mathrm{d}\left(\frac{a}{2}x+\frac{\pi}{4}\right)$

$$=\mp\frac{2\sqrt{2}}{a}\cos\left(\frac{a}{2}x+\frac{\pi}{4}\right)+C.$$

2. 解　原式 $=\int\pm\left(\sin\frac{a}{2}x-\cos\frac{a}{2}x\right)\mathrm{d}x=\mp\int\frac{2}{a}\sqrt{2}\cos\left(\frac{a}{2}x+\frac{\pi}{4}\right)\mathrm{d}\left(\frac{a}{2}x+\frac{\pi}{4}\right)=\mp\frac{2\sqrt{2}}{a}\sin\left(\frac{a}{2}x+\frac{\pi}{4}\right)+C.$

3. 解　原式 $=\frac{2}{a}\int\frac{1}{\pm(\sin\frac{a}{2}x+\cos\frac{a}{2}x)}\mathrm{d}\left(\frac{a}{2}x\right)=\frac{2}{a}\int\pm\frac{1}{\sqrt{2}\sin\left(\frac{a}{2}x+\frac{\pi}{4}\right)}\mathrm{d}\left(\frac{a}{2}x\right)$

$$=\pm\frac{\sqrt{2}}{a}\int\csc\left(\frac{a}{2}x+\frac{\pi}{4}\right)\mathrm{d}\left(\frac{a}{2}x+\frac{\pi}{4}\right)=\pm\frac{\sqrt{2}}{a}\ln\left|\tan\frac{\frac{a}{2}x+\frac{\pi}{4}}{2}\right|+C.$$

4. 解　原式 $=\frac{2}{a}\int\frac{1}{\pm(\sin\frac{a}{2}x-\cos\frac{a}{2}x)}\mathrm{d}\left(\frac{a}{2}x\right)=\frac{2}{a}\int\frac{1}{\pm\sqrt{2}\sin\left(\frac{a}{2}x-\frac{\pi}{4}\right)}\mathrm{d}\left(\frac{a}{2}x\right)$

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$$=\pm\frac{\sqrt{2}}{a}\int\csc\left(\frac{a}{2}x-\frac{\pi}{4}\right)\mathrm{d}\left(\frac{a}{2}x-\frac{\pi}{4}\right)=\pm\frac{\sqrt{2}}{a}\ln\left|\tan\frac{\frac{a}{2}x-\frac{\pi}{4}}{2}\right|+C.$$

第五组

1. 解 $\int\arcsin\frac{x}{a}\mathrm{d}x=x\arcsin\frac{x}{a}-\int\left(\arcsin\frac{x}{a}\right)'x\,\mathrm{d}x=x\arcsin\frac{x}{a}-\frac{1}{a}\int\frac{x}{\sqrt{1-\frac{x^2}{a^2}}}\mathrm{d}x$

$$=x\arcsin\frac{x}{a}+\frac{1}{2}\int\frac{1}{\sqrt{a^2-x^2}}\mathrm{d}(-x^2)=x\arcsin\frac{x}{a}+\sqrt{a^2-x^2}+C.$$

2. 解 令 $\arcsin\frac{x}{a}=t$，则 $x=a\sin t$，

$$\text{原式}=\frac{a^2}{2}\int t\sin 2t\mathrm{d}t=-\frac{a^2}{4}\int t\mathrm{d}(\cos 2t)=-\frac{a^2}{4}\left(t\cos 2t-\int\cos 2t\mathrm{d}t\right)=-\frac{a^2}{4}t\cos 2t+\frac{a^2}{8}\sin 2t+C.$$

回代 $t=\arcsin\frac{x}{a}$，$\sin t=\frac{x}{a}$，可得

$$\text{原式}=\left(\frac{x^2}{2}-\frac{a^2}{4}\right)\arcsin\frac{x}{a}+\frac{x}{4}\sqrt{a^2-x^2}+C.$$

3. 解 令 $\arccos\frac{x}{a}=t$，则 $x=a\cos t$，

$$\text{原式}=a\int t\mathrm{d}(\cos t)=a\left(t\cos t-\int\cos t\mathrm{d}t\right)=at\cos t-a\sin t+C.$$

回代 $t=\arccos\frac{x}{a}$，$\cos t=\frac{x}{a}$，可得

$$\text{原式}=x\arccos\frac{x}{a}-\sqrt{a^2-x^2}+C.$$

4. 解 令 $\arccos\frac{x}{a}=t$，则 $x=a\cos t$，

$$\text{原式}=a^2\int t\cos t\mathrm{d}(\cos t)=\frac{a^2}{4}\int t\mathrm{d}(\cos 2t)=\frac{a^2}{4}\left(t\cos 2t-\int\cos 2t\mathrm{d}t\right)=\frac{a^2}{4}t\cos 2t-\frac{a^2}{8}\sin 2t+C.$$

回代 $t=\arccos\frac{x}{a}$，$\cos t=\frac{x}{a}$，可得

$$\text{原式}=\left(\frac{x^2}{2}-\frac{a^2}{4}\right)\arccos\frac{x}{a}-\frac{x}{4}\sqrt{a^2-x^2}+C.$$

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5. 解 $\int \arctan\frac{x}{a}\,dx = x\arctan\frac{x}{a} - \int \frac{\frac{x}{a}}{1+\frac{x^2}{a^2}}\,dx = x\arctan\frac{x}{a} - \frac{1}{2}\int \frac{a}{x^2+a^2}\,d(x^2)$

$$= x\arctan\frac{x}{a} - \frac{a}{2}\ln\left(a^2+x^2\right) + C.$$

6. 解 $\int x\arctan\frac{x}{a}\,dx = \frac{1}{2}\int \arctan\frac{x}{a}\,d(x^2) = \frac{1}{2}\left[x^2\arctan\frac{x}{a} - \int \frac{\frac{x^2}{a}}{1+\left(\frac{x}{a}\right)^2}\,dx\right]$

$$= \frac{1}{2}\left[x^2\arctan\frac{x}{a} - a\int \frac{\frac{x^2}{a^2}+1}{1+\left(\frac{x}{a}\right)^2}\,dx + a^2\int \frac{1}{1+\left(\frac{x}{a}\right)^2}\,d\left(\frac{x}{a}\right)\right]$$

$$= \frac{x^2}{2}\arctan\frac{x}{a} - \frac{a}{2}x + \frac{a^2}{2}\arctan\frac{x}{a} + C$$

$$= \frac{1}{2}(a^2+x^2)\arctan\frac{x}{a} - \frac{a}{2}x + C.$$

7. 解 $\int \arcsin ax\,dx = x\arcsin ax - \int (\arcsin ax)' x\,dx = x\arcsin ax - \int \frac{ax}{\sqrt{1-a^2x^2}}\,dx$

$$= x\arcsin ax + \frac{1}{2a}\int \frac{1}{\sqrt{1-a^2x^2}}\,d(-a^2x^2) = x\arcsin ax + \frac{\sqrt{1-a^2x^2}}{a} + C.$$

8. 解 令 $\arccos ax = t$，$x = \frac{\cos t}{a}$，

$$\text{原式} = \frac{1}{a}\int t\,d(\cos t) = \frac{1}{a}\left(t\cos t - \int \cos t\,dt\right) = \frac{1}{a}t\cos t - \frac{1}{a}\sin t + C.$$

回代 $t = \arccos ax$，$\cos t = ax$，可得

$$\text{原式} = x\arccos ax - \frac{\sqrt{1-a^2x^2}}{a} + C.$$

9. 解 $\int \arctan ax\,dx = x\arctan ax - \int \frac{ax}{1+a^2x^2}\,dx = x\arctan ax - \frac{1}{2a}\int \frac{d(a^2x^2)}{1+a^2x^2}$

$$= x\arctan ax - \frac{1}{2a}\ln\left(1+a^2x^2\right) + C.$$

10. 解 $\int \operatorname{arccot} ax\,dx = x\operatorname{arccot} ax + \int \frac{ax}{1+a^2x^2}\,dx = x\operatorname{arccot} ax + \frac{1}{2a}\int \frac{1}{1+a^2x^2}\,d(a^2x^2)$

$$= x\operatorname{arccot} ax + \frac{1}{2a}\ln\left(1+a^2x^2\right) + C.$$

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11. 解　**方法一**　换元法.

令 $\operatorname{arcsec} ax=t$，$ax=\sec t$，

$$\int \operatorname{arcsec} ax\mathrm{d}x=\frac{1}{a}\int t\mathrm{d}(\sec t)=\frac{1}{a}\left(t\sec t-\int \sec t\mathrm{d}t\right)=\frac{t\sec t}{a}-\frac{\ln|\sec t+\tan t|}{a}+C.$$

回代 $t=\operatorname{arcsec} ax$，$\sec t=ax$，可得

$$\text{原式}=x\operatorname{arcsec} ax-\frac{1}{a}\ln\left|ax+\sqrt{a^2x^2-1}\right|+C.$$

方法二　分部积分法.

$$\int \operatorname{arcsec} ax\mathrm{d}x=x\operatorname{arcsec} ax-\int(\operatorname{arcsec} ax)'x\mathrm{d}x=x\operatorname{arcsec} ax-\int\frac{x\mathrm{d}x}{x\sqrt{a^2x^2-1}}$$

$$=x\operatorname{arcsec} ax-\frac{1}{a}\ln\left|ax+\sqrt{a^2x^2-1}\right|+C.$$

12. 解　**方法一**　换元法.

令 $\operatorname{arccsc} ax=t$，$ax=\csc t$，

$$\text{原式}=\frac{1}{a}\int t\mathrm{d}(\csc t)=\frac{1}{a}\left(t\csc t-\int \csc t\mathrm{d}t\right)=\frac{1}{a}t\csc t-\frac{\ln|\csc t-\cot t|}{a}+C.$$

回代，$\text{原式}=x\operatorname{arccsc} ax+\frac{1}{a}\ln\left|ax+\sqrt{a^2x^2-1}\right|+C$.

方法二　分部积分法.

$$\text{原式}=x\operatorname{arccsc} ax-\int(\operatorname{arccsc} ax)'x\mathrm{d}x=x\operatorname{arccsc} ax+\int\frac{x}{x\sqrt{a^2x^2-1}}\mathrm{d}x$$

$$=x\operatorname{arccsc} ax+\frac{1}{a}\ln\left|ax+\sqrt{a^2x^2-1}\right|+C.$$

13. 解　$\int x\arcsin ax\mathrm{d}x=\frac{1}{2}\int \arcsin ax\mathrm{d}(x^2)=\frac{1}{2}\left(x^2\arcsin ax-\frac{1}{a}\int\frac{a^2x^2}{\sqrt{1-a^2x^2}}\mathrm{d}x\right)$

$$=\frac{1}{2}\left(x^2\arcsin ax-\frac{1}{a}\int\frac{a^2x^2-1}{\sqrt{1-a^2x^2}}\mathrm{d}x-\frac{1}{a}\int\frac{1}{\sqrt{1-a^2x^2}}\mathrm{d}x\right)$$

$$=\frac{x^2}{2}\arcsin ax+\frac{1}{2a}\int\sqrt{1-a^2x^2}\mathrm{d}x-\frac{1}{2a^2}\arcsin ax.$$

而 $\int\sqrt{1-a^2x^2}\mathrm{d}x\overset{\text{令}ax=\sin t}{=\!=\!=}\int\frac{\cos^2 t}{a}\mathrm{d}t=\frac{1}{2a}\int(1+\cos 2t)\,\mathrm{d}t=\frac{t}{2a}+\frac{\sin 2t}{4a}+C_0=\frac{\arcsin ax}{2a}+\frac{x\sqrt{1-a^2x^2}}{2}+C_0$，

回代，$\text{原式}=\frac{x^2}{2}\arcsin ax+\frac{\arcsin ax}{4a^2}+\frac{x\sqrt{1-a^2x^2}}{4a}-\frac{\arcsin ax}{2a^2}+C$

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$$=\frac{1}{4a^2}\left[(2a^2x^2-1)\arcsin ax+ax\sqrt{1-a^2x^2}\right]+C.$$

14. 解 $\int x\arccos ax\mathrm{d}x=\frac{1}{2}\int\arccos ax\mathrm{d}(x^2)=\frac{1}{2}\left(x^2\arccos ax+\frac{1}{a}\int\frac{a^2x^2}{\sqrt{1-a^2x^2}}\mathrm{d}x\right)$

$$=\frac{1}{2}\left(x^2\arccos ax+\frac{1}{a}\int\frac{a^2x^2-1}{\sqrt{1-a^2x^2}}\mathrm{d}x+\frac{1}{a}\int\frac{1}{\sqrt{1-a^2x^2}}\mathrm{d}x\right)$$

$$=\frac{x^2\arccos ax}{2}-\frac{1}{2a}\int\sqrt{1-a^2x^2}\mathrm{d}x-\frac{1}{2a^2}\arccos ax.$$

而 $\int\sqrt{1-a^2x^2}\mathrm{d}x\overset{令ax=\cos t}{=\!=\!=}\frac{-1}{a}\int\frac{1-\cos 2t}{2}\mathrm{d}t=-\frac{t}{2a}+\frac{\sin 2t}{4a}+C_0=-\frac{\arccos ax}{2a}+\frac{x\sqrt{1-a^2x^2}}{2}+C_0$，回代，

$$原式=\frac{x^2}{2}\arccos ax+\frac{\arccos ax}{4a^2}-\frac{x\sqrt{1-a^2x^2}}{4a}-\frac{\arccos ax}{2a^2}+C$$

$$=\frac{1}{4a^2}\left[(2a^2x^2-1)\arccos ax-ax\sqrt{1-a^2x^2}\right]+C.$$

15. 解 $\int x\arctan ax\mathrm{d}x=\frac{1}{2}\int\arctan ax\mathrm{d}(x^2)=\frac{1}{2}\left(x^2\arctan ax-\frac{1}{a}\int\frac{a^2x^2}{1+a^2x^2}\mathrm{d}x\right)$

$$=\frac{x^2}{2}\arctan ax-\frac{1}{2a}\int\frac{a^2x^2+1}{1+a^2x^2}\mathrm{d}x+\frac{1}{2a^2}\int\frac{1}{1+a^2x^2}\mathrm{d}(ax)$$

$$=\frac{x^2}{2}\arctan ax-\frac{x}{2a}+\frac{\arctan ax}{2a^2}+C$$

$$=\frac{1+a^2x^2}{2a^2}\arctan ax-\frac{x}{2a}+C.$$

16. 解 $\int x\operatorname{arccot} ax\mathrm{d}x=\frac{1}{2}\int\operatorname{arccot} ax\mathrm{d}(x^2)=\frac{1}{2}\left(x^2\operatorname{arccot} ax+\frac{1}{a}\int\frac{a^2x^2}{1+a^2x^2}\mathrm{d}x\right)$

$$=\frac{x^2}{2}\operatorname{arccot} ax+\frac{1}{2a}\int\frac{a^2x^2+1}{1+a^2x^2}\mathrm{d}x-\frac{1}{2a^2}\int\frac{1}{1+a^2x^2}\mathrm{d}(ax)$$

$$=\frac{a^2x^2+1}{2a^2}\operatorname{arccot} ax+\frac{x}{2a}+C.$$

17. 解 $\int x\operatorname{arcsec} ax\mathrm{d}x=\frac{1}{2}\int\operatorname{arcsec} ax\mathrm{d}(x^2)=\frac{1}{2}\left(x^2\operatorname{arcsec} ax-\int\frac{ax^2}{ax\sqrt{a^2x^2-1}}\mathrm{d}x\right)$

$$=\frac{x^2\operatorname{arcsec} ax}{2}-\frac{1}{4a^2}\int\frac{1}{\sqrt{a^2x^2-1}}\mathrm{d}(a^2x^2)=\frac{x^2\operatorname{arcsec} ax}{2}-\frac{1}{2a^2}\sqrt{a^2x^2-1}+C.$$

18. 解 $\int x\operatorname{arccsc} ax\mathrm{d}x=\frac{1}{2}\int\operatorname{arccsc} ax\mathrm{d}(x^2)=\frac{1}{2}\left(x^2\operatorname{arccsc} ax+\int\frac{ax^2}{ax\sqrt{a^2x^2-1}}\mathrm{d}x\right)$

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$$=\frac{x^2}{2}\operatorname{arccsc}ax+\frac{1}{4a^2}\int\frac{1}{\sqrt{a^2x^2-1}}\mathrm{d}(a^2x^2)$$

$$=\frac{x^2}{2}\operatorname{arccsc}ax+\frac{1}{2a^2}\sqrt{a^2x^2-1}+C.$$

19. 解　令 $\arcsin ax=t$，$\sin t=ax$，

$$\text{原式}=\frac{1}{a}\int t^2\mathrm{d}(\sin t)=\frac{1}{a}\left[t^2\sin t+2\int t\mathrm{d}(\cos t)\right]=\frac{1}{a}\left[t^2\sin t+2\left(t\cos t-\int\cos t\mathrm{d}t\right)\right]$$

$$=\frac{1}{a}t^2\sin t+\frac{2t}{a}\cos t-\frac{2\sin t}{a}+C.$$

回代 $\arcsin ax=t$，可得

$$\text{原式}=x(\arcsin ax)^2+\frac{2\arcsin ax}{a}\sqrt{1-a^2x^2}-2x+C.$$

20. 解　令 $t=\arccos ax$，$ax=\cos t$，

$$\text{原式}=\frac{1}{a}\int t^2\mathrm{d}(\cos t)=\frac{1}{a}\left[t^2\cos t-2\int t\mathrm{d}(\sin t)\right]=\frac{1}{a}\left[t^2\cos t-2\left(t\sin t-\int\sin t\mathrm{d}t\right)\right]$$

$$=\frac{1}{a}t^2\cos t-\frac{2}{a}t\sin t-\frac{2}{a}\cos t+C.$$

回代 $\arccos ax=t$，$\cos t=ax$，可得

$$\text{原式}=x(\arccos ax)^2-2x-\frac{2\sqrt{1-a^2x^2}}{a}\arccos ax+C.$$

21. 解　令 $\arcsin ax=t$，$ax=\sin t$，

$$\text{原式}=\int\frac{ta^2}{(\sin t)^2}\mathrm{d}\left(\frac{\sin t}{a}\right)=-a\int t\mathrm{d}\left(\frac{1}{\sin t}\right)=-a\left(\frac{t}{\sin t}-\int\frac{1}{\sin t}\mathrm{d}t\right)$$

$$=-\frac{at}{\sin t}+a\ln\left|\csc t-\cot t\right|+C.$$

回代可得

$$\text{原式}=-\frac{\arcsin ax}{x}+a\ln\left|\frac{1-\sqrt{1-a^2x^2}}{ax}\right|+C.$$

22. 解　令 $\arccos ax=t$，$ax=\cos t$，

$$\text{原式}=\int\frac{ta^2}{\cos^2 t}\cdot\frac{1}{a}\mathrm{d}(\cos t)=-a\int t\mathrm{d}(\sec t)=-at\sec t+a\int\sec t\mathrm{d}t=-at\sec t+a\ln\left|\frac{1+\sin t}{\cos t}\right|+C.$$

回代可得

$$\text{原式}=-\frac{\arccos ax}{x}+a\ln\left|\frac{1+\sqrt{1-a^2x^2}}{ax}\right|+C.$$

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23. 解　令 $\arctan ax=t$，$ax=\tan t$，

$$原式=\int\frac{ta^2}{\tan^2 t}\mathrm{d}\left(\frac{\tan t}{a}\right)=-a\int t\mathrm{d}\left(\frac{1}{\tan t}\right)=-a\left(\frac{t}{\tan t}-\int\frac{1}{\tan t}\mathrm{d}t\right)=\frac{-at}{\tan t}+a\ln|\sin t|+C.$$

回代 $t=\arctan ax$，可得

$$原式=-\frac{\arctan ax}{x}-\frac{a}{2}\ln\frac{1+a^2x^2}{a^2x^2}+C.$$

24. 解　令 $\operatorname{arccot} ax=t$，$ax=\cot t$，

$$原式=\int\frac{ta^2}{\cot^2 t}\mathrm{d}\left(\frac{\cot t}{a}\right)=-a\int t\mathrm{d}(\tan t)=-a\left(\frac{t}{\cot t}-\int\frac{\sin t}{\cos t}\mathrm{d}t\right)=-at\tan t-a\ln|\cos t|+C.$$

回代 $t=\operatorname{arccot} ax$，$\cot t=ax$，可得

$$原式=-\frac{\operatorname{arccot} ax}{x}-\frac{a}{2}\ln\frac{a^2x^2}{1+a^2x^2}+C.$$

25. 解　令 $\operatorname{arcsec} ax=t$，$ax=\sec t$，

$$原式=\int\frac{ta^2}{\sec^2 t}\mathrm{d}\left(\frac{\sec t}{a}\right)=-a\int t\mathrm{d}(\cos t)=-a\left(t\cos t-\int\cos t\mathrm{d}t\right)=-at\cos t+a\sin t+C.$$

回代 $t=\operatorname{arcsec} ax$，可得

$$原式=-\frac{a\operatorname{arcsec} ax}{ax}+a\sqrt{1-\frac{1}{a^2x^2}}+C=-\frac{\operatorname{arcsec} ax}{x}+\frac{\sqrt{a^2x^2-1}}{x}+C.$$

26. 解　令 $\operatorname{arccsc} ax=t$，$\csc t=ax$.

$$原式=\int\frac{ta^2}{\csc^2 t}\mathrm{d}\left(\frac{\csc t}{a}\right)=-a\int\frac{t}{\csc^2 t}\cdot\csc t\cdot\cot t\mathrm{d}t$$

$$=-a\int t\cos t\mathrm{d}t=-a(t\sin t+\cos t)+C.$$

回代 $t=\operatorname{arccsc} ax$，得

$$原式=-a\left(\operatorname{arccsc} ax\cdot\frac{1}{ax}+\frac{\sqrt{a^2x^2-1}}{ax}\right)+C=-\frac{\operatorname{arccsc} ax}{x}-\frac{\sqrt{a^2x^2-1}}{x}+C.$$

27. 解　凑微分 $\mathrm{d}(\arcsin ax)=\frac{a}{\sqrt{1-(ax)^2}}\mathrm{d}x$，故

$$原式=\frac{1}{a}\int\arcsin ax\mathrm{d}(\arcsin ax)=\frac{1}{2a}(\arcsin ax)^2+C.$$

28. 解　凑微分 $\mathrm{d}(\arccos ax)=\frac{a}{-\sqrt{1-(ax)^2}}\mathrm{d}x$，

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$$原式=-\frac{1}{a}\int \arccos ax\mathrm{d}(\arccos ax)=-\frac{1}{2a}(\arccos ax)^2+C.$$

29. 解　凑微分 $\mathrm{d}(\arctan ax)=\frac{a}{1+a^2x^2}\mathrm{d}x$，故

$$原式=\frac{1}{a}\int \arctan ax\mathrm{d}(\arctan ax)=\frac{1}{2a}(\arctan ax)^2+C.$$

30. 解　凑微分 $\mathrm{d}(\operatorname{arccot} ax)=-\frac{a}{1+a^2x^2}\mathrm{d}x$，故

$$原式=\frac{1}{-a}\int \operatorname{arccot} ax\mathrm{d}(\operatorname{arccot} ax)=-\frac{1}{2a}(\operatorname{arccot} ax)^2+C.$$

31. 解　分部积分法.

$$原式=x\cdot\arcsin\frac{x}{a}-\int x\mathrm{d}\left(\arcsin\frac{x}{a}\right)=x\cdot\arcsin\frac{x}{a}-\int x\frac{\frac{1}{a}}{\sqrt{1-\left(\frac{x}{a}\right)^2}}\mathrm{d}x=x\cdot\arcsin\frac{x}{a}-\int\frac{x}{\sqrt{a^2-x^2}}\mathrm{d}x$$

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$$=x\arcsin\frac{x}{a}+\sqrt{a^2-x^2}+C.$$

32. 解　分部积分法.

$$原式=x\left(\arcsin\frac{x}{a}\right)^2-\int x\mathrm{d}\left[\left(\arcsin\frac{x}{a}\right)^2\right]=x\left(\arcsin\frac{x}{a}\right)^2-\int x\cdot 2\arcsin\frac{x}{a}\cdot\frac{\frac{1}{a}}{\sqrt{1-\left(\frac{x}{a}\right)^2}}\mathrm{d}x$$

$$=x\left(\arcsin\frac{x}{a}\right)^2-\int\arcsin\frac{x}{a}\cdot\frac{\mathrm{d}(x^2)}{\sqrt{a^2-x^2}}=x\left(\arcsin\frac{x}{a}\right)^2+2\int\arcsin\frac{x}{a}\mathrm{d}\left(\sqrt{a^2-x^2}\right)$$

$$=x\left(\arcsin\frac{x}{a}\right)^2+2\sqrt{a^2-x^2}\cdot\arcsin\frac{x}{a}-2\int\sqrt{a^2-x^2}\cdot\frac{\frac{1}{a}}{\sqrt{1-\left(\frac{x}{a}\right)^2}}\mathrm{d}x$$

$$=x\left(\arcsin\frac{x}{a}\right)^2+2\sqrt{a^2-x^2}\cdot\arcsin\frac{x}{a}-2\int 1\mathrm{d}x$$

$$=x\left(\arcsin\frac{x}{a}\right)^2+2\sqrt{a^2-x^2}\arcsin\frac{x}{a}-2x+C.$$

33. 解　分部积分法.

$$原式=x\left(\arcsin\frac{x}{a}\right)^3-\int x\mathrm{d}\left(\arcsin\frac{x}{a}\right)^3,$$

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其中 $\int x\mathrm{d}\left(\arcsin\frac{x}{a}\right)^3=\int x\cdot 3\cdot\left(\arcsin\frac{x}{a}\right)^2\cdot\frac{\frac{1}{a}}{\sqrt{1-\left(\frac{x}{a}\right)^2}}\mathrm{d}x=\int\left(\arcsin\frac{x}{a}\right)^2\cdot 3x\cdot\frac{1}{\sqrt{a^2-x^2}}\mathrm{d}x$

$$=-\frac{3}{2}\int\frac{\left(\arcsin\frac{x}{a}\right)^2}{\sqrt{a^2-x^2}}\mathrm{d}(a^2-x^2)=-\frac{3}{2}\cdot 2\int\left(\arcsin\frac{x}{a}\right)^2\mathrm{d}\left(\sqrt{a^2-x^2}\right)$$

$$=-3\left[\sqrt{a^2-x^2}\left(\arcsin\frac{x}{a}\right)^2-\int\sqrt{a^2-x^2}\mathrm{d}\left(\arcsin\frac{x}{a}\right)^2\right]$$

$$=-3\sqrt{a^2-x^2}\cdot\left(\arcsin\frac{x}{a}\right)^2+3\int\sqrt{a^2-x^2}\cdot 2\arcsin\frac{x}{a}\frac{\frac{1}{a}}{\sqrt{1-\left(\frac{x}{a}\right)^2}}\mathrm{d}x$$

$$=-3\sqrt{a^2-x^2}\cdot\left(\arcsin\frac{x}{a}\right)^2+6\int\arcsin\frac{x}{a}\mathrm{d}x.$$

再利用本组31式，代入可得

$$\text{原式}=x\left(\arcsin\frac{x}{a}\right)^3+3\sqrt{a^2-x^2}\left(\arcsin\frac{x}{a}\right)^2-6x\arcsin\frac{x}{a}-6\sqrt{a^2-x^2}+C.$$

34. 解　分部积分法.

$$\text{原式}=\frac{1}{2}\int\arcsin\frac{x}{a}\mathrm{d}(x^2)=\frac{1}{2}x^2\cdot\arcsin\frac{x}{a}-\frac{1}{2}\int x^2\cdot\frac{\frac{1}{a}}{\sqrt{1-\left(\frac{x}{a}\right)^2}}\mathrm{d}x=\frac{1}{2}x^2\arcsin\frac{x}{a}-\frac{1}{2}\int\frac{x^2}{\sqrt{a^2-x^2}}\mathrm{d}x$$

$$=\frac{1}{2}x^2\arcsin\frac{x}{a}+\frac{1}{2}\int\frac{a^2-x^2-a^2}{\sqrt{a^2-x^2}}\mathrm{d}x=\frac{1}{2}x^2\arcsin\frac{x}{a}+\frac{1}{2}\int\sqrt{a^2-x^2}\,\mathrm{d}x-\frac{a^2}{2}\int\frac{1}{\sqrt{a^2-x^2}}\mathrm{d}x$$

$$=\frac{1}{2}x^2\arcsin\frac{x}{a}+\frac{1}{2}\left(\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\arcsin\frac{x}{a}\right)-\frac{a^2}{2}\arcsin\frac{x}{a}+C$$

$$=\left(\frac{x^2}{2}-\frac{a^2}{4}\right)\arcsin\frac{x}{a}+\frac{x}{4}\sqrt{a^2-x^2}+C.$$

35. 解　$\text{原式}=\frac{1}{3}\int\arcsin\frac{x}{a}\mathrm{d}(x^3)=\frac{1}{3}x^3\arcsin\frac{x}{a}-\frac{1}{3}\int x^3\mathrm{d}\left(\arcsin\frac{x}{a}\right)$

$$=\frac{1}{3}x^3\cdot\arcsin\frac{x}{a}-\frac{1}{3}\int x^3\cdot\frac{1}{\sqrt{a^2-x^2}}\mathrm{d}x,$$

其中对于 $\int x^3\frac{1}{\sqrt{a^2-x^2}}\mathrm{d}x$，令 $x=a\sin t$，$t\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$，有

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$$\int x^3\frac{1}{\sqrt{a^2-x^2}}\mathrm{d}x=\int a^3\sin^3 t\cdot\frac{1}{a\cdot\cos t}a\cos t\mathrm{d}t=a^3\int\sin^3 t\mathrm{d}t=-a^3\int\sin^2 t\mathrm{d}(\cos t)=-a^3\int(1-\cos^2 t)\,\mathrm{d}(\cos t)$$

$$=-a^3\int\mathrm{d}(\cos t)+a^3\int\cos^2 t\mathrm{d}(\cos t)=-a^3\cos t+\frac{a^3}{3}\cos^3 t+C_0.$$

对于 $x=a\sin t$，即 $\sin t=\frac{x}{a}$，则 $\cos t=\frac{\sqrt{a^2-x^2}}{a}$，故

$$\text{原式}=\frac{1}{3}x^3\arcsin\frac{x}{a}+\frac{1}{3}a^3\frac{\sqrt{a^2-x^2}}{a}-\frac{1}{9}a^3\frac{(a^2-x^2)^{\frac{3}{2}}}{a^3}+C=\frac{x^3}{3}\arcsin\frac{x}{a}+\frac{x^2+2a^2}{9}\sqrt{a^2-x^2}+C.$$

36. 解　令 $\arcsin x=t$，$t\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$，有 $x=\sin t$，$\sqrt{1-x^2}=\sqrt{1-\sin^2 t}=\cos t$，故

$$\text{原式}=\int\frac{\sin t\cdot t}{\cos t}\cdot\cos t\mathrm{d}t=\int t\sin t\mathrm{d}t=-\int t\mathrm{d}(\cos t)=-t\cos t+\int\cos t\mathrm{d}t=-t\cos t+\sin t+C$$

$$=x-\sqrt{1-x^2}\arcsin x+C.$$

37. 解　令 $\arcsin x=t$，$t\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$，则 $x=\sin t$，$\sqrt{1-x^2}=\cos t$，故

$$\text{原式}=\int\frac{\sin^2 t\cdot t}{\cos t}\cdot\cos t\mathrm{d}t=\int t\sin^2 t\mathrm{d}t=\int t\frac{1-\cos 2t}{2}\mathrm{d}t=\frac{1}{4}t^2-\frac{1}{2}\int t\cdot\cos 2t\mathrm{d}t$$

$$=\frac{1}{4}t^2-\frac{1}{4}\int t\mathrm{d}(\sin 2t)=\frac{1}{4}t^2-\frac{1}{4}t\cdot\sin 2t+\frac{1}{4}\int\sin 2t\mathrm{d}t$$

$$=\frac{1}{4}t^2-\frac{1}{4}t\cdot\sin 2t-\frac{1}{8}\cos 2t+C_1=\frac{1}{4}t^2-\frac{1}{2}t\cdot\sin t\cos t-\frac{1}{8}(1-2\sin^2 t)+C_1$$

$$=\frac{1}{4}(\arcsin x)^2-\frac{1}{2}\arcsin x\cdot x\cdot\sqrt{1-x^2}-\frac{1}{8}(1-2x^2)+C_1$$

$$=\frac{x^2}{4}-\frac{x}{2}\sqrt{1-x^2}\arcsin x+\frac{1}{4}(\arcsin x)^2+C.$$

38. 解　令 $\arcsin x=t$，$t\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$，则 $x=\sin t$，$\sqrt{1-x^2}=\cos t$，因此

$$\text{原式}=\int\frac{\sin^3 t\cdot t}{\cos t}\cdot\cos t\mathrm{d}t=\int t\sin^3 t\mathrm{d}t,$$

其中　$I=\int x\sin^3 x\mathrm{d}x=-\int x(1-\cos^2 x)\mathrm{d}(\cos x)=-\int x\mathrm{d}(\cos x)+\int x\cos^2 x\mathrm{d}(\cos x)$

$$=-x\cos x+\sin x+x\cos^3 x-\int\cos x(\cos^2 x-2x\cos x\sin x)\mathrm{d}x,$$

$$\int\cos x(\cos^2 x-2x\cos x\sin x)\mathrm{d}x=\int(1-\sin^2 x)\mathrm{d}(\sin x)-\int 2x(1-\sin^2 x)\sin x\mathrm{d}x$$

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$$=\sin x-\frac{\sin^3 x}{3}-\int 2x\sin x\mathrm{d}x+2I$$

$$=\sin x-\frac{\sin^3 x}{3}+\int 2x\mathrm{d}(\cos x)+2I$$

$$=\sin x-\frac{\sin^3 x}{3}+2x\cos x-2\sin x+2I,$$

故

$$I=-x\cos x+\sin x+x\cos^3 x-\left[\sin x-\frac{\sin^3 x}{3}-(-2x\cos x+2\sin x)+2I\right],$$

解得

$$I=\frac{x\cos^3 x}{3}+\frac{\sin^3 x}{9}-x\cos x+\frac{2}{3}\sin x+C,$$

于是，原式$=\frac{t\cos^3 t}{3}+\frac{\sin^3 t}{9}-t\cos t+\frac{2}{3}\sin t+C=\frac{\arcsin x\cdot(1-x^2)^{\frac{3}{2}}}{3}+\frac{x^3}{9}-\arcsin x\cdot\sqrt{1-x^2}+\frac{2}{3}x+C$.

39. 解　令$\arcsin x=t$，$t\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$，则$x=\sin t$，$1-x^2=\cos^2 t$，故

$$\text{原式}=\int\frac{t}{\cos^3 t}\cdot\cos t\mathrm{d}t=\int\frac{t}{\cos^2 t}\mathrm{d}t=\int t\sec^2 t\mathrm{d}t=\int t\mathrm{d}(\tan t)=t\tan t-\int\tan t\mathrm{d}t$$

$$=t\tan t+\ln|\cos t|+C=\frac{x\arcsin x}{\sqrt{1-x^2}}+\frac{1}{2}\ln|1-x^2|+C.$$

40. 解　令$\arcsin x=t$，$t\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$，则$x=\sin t$，$1-x^2=\cos^2 t$，故

$$\text{原式}=\int\frac{\sin t\cdot t}{\cos^3 t}\cdot\cos t\mathrm{d}t=\int\frac{t\sin t}{\cos^2 t}\mathrm{d}t=\int t\mathrm{d}(\cos t)^{-1}=\frac{t}{\cos t}-\int\frac{1}{\cos t}\mathrm{d}t=\frac{t}{\cos t}-\int\frac{\cos t}{\cos^2 t}\mathrm{d}t$$

$$=\frac{t}{\cos t}-\int\frac{\mathrm{d}(\sin t)}{1-\sin^2 t}=\frac{t}{\cos t}-\frac{1}{2}\int\left(\frac{1}{1-\sin t}+\frac{1}{1+\sin t}\right)\mathrm{d}(\sin t)$$

$$=\frac{t}{\cos t}+\frac{1}{2}\ln\left|\frac{1-\sin t}{1+\sin t}\right|=\frac{\arcsin x}{\sqrt{1-x^2}}+\frac{1}{2}\ln\left|\frac{1-x}{1+x}\right|+C.$$

41. 解　$\text{原式}=x\arccos\frac{x}{a}-\int x\left[\frac{\frac{1}{a}}{-\sqrt{1-\left(\frac{x}{a}\right)^2}}\right]\mathrm{d}x=x\arccos\frac{x}{a}+\int\frac{x}{\sqrt{a^2-x^2}}\mathrm{d}x$　（由第二组14式）

$$=x\arccos\frac{x}{a}-\sqrt{a^2-x^2}+C.$$

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42. 解 原式 $=\frac{1}{2}\int \arccos\frac{x}{a}\mathrm{d}(x^2)=\frac{1}{2}x^2\left(\arccos\frac{x}{a}\right)-\frac{1}{2}\int x^2\mathrm{d}\left(\arccos\frac{x}{a}\right)$

$=\frac{x^2}{2}\arccos\frac{x}{a}+\frac{1}{2}\int x^2\frac{1}{\sqrt{a^2-x^2}}\mathrm{d}x$ （由第二组16式）

$=\frac{x^2}{2}\arccos\frac{x}{a}-\frac{1}{2}\left(\frac{x}{2}\sqrt{a^2-x^2}-\frac{a^2}{2}\arcsin\frac{x}{a}\right)+C$

$=\frac{x^2}{2}\arccos\frac{x}{a}-\frac{x}{4}\sqrt{a^2-x^2}+\frac{a^2}{4}\arcsin\frac{x}{a}+C.$

43. 解 令 $t=\arccos\frac{x}{a}$，$t\in(0,\pi)$，则 $\cos t=\frac{x}{a}$，$x=a\cos t$，故

$$\text{原式}=\int a^2\cos^2 t\cdot t\mathrm{d}(a\cos t)=-a^3\int t\cos^2 t\cdot\sin t\mathrm{d}t=\frac{a^3}{3}\int t\mathrm{d}(\cos^3 t)=\frac{a^3}{3}t\cos^3 t-\frac{a^3}{3}\int\cos^3 t\mathrm{d}t$$

$$=\frac{a^3}{3}t\cos^3 t-\frac{a^3}{3}\left(\sin t-\frac{\sin^3 t}{3}\right)+C=\frac{a^3}{3}\cdot\arccos\frac{x}{a}\cdot\left(\frac{x}{a}\right)^3-\frac{a^3}{3}\cdot\frac{\sqrt{a^2-x^2}}{a}+\frac{a^3}{9}\frac{(a^2-x^2)^{\frac{3}{2}}}{a^3}+C$$

$$=\frac{x^3}{3}\arccos\frac{x}{a}-\frac{x^2+2a^2}{9}\sqrt{a^2-x^2}+C.$$

44. 解 原式 $=\frac{1}{2}\int\arctan\frac{x}{a}\mathrm{d}(x^2)=\frac{x^2}{2}\arctan\frac{x}{a}-\frac{1}{2}\int x^2\frac{\frac{1}{a}}{1+\left(\frac{x}{a}\right)^2}\mathrm{d}x=\frac{x^2}{2}\arctan\frac{x}{a}-\frac{1}{2}\int\frac{ax^2}{a^2+x^2}\mathrm{d}x$，

其中 $\int\frac{ax^2}{a^2+x^2}\mathrm{d}x=\int\frac{a(x^2+a^2)-a^3}{a^2+x^2}\mathrm{d}x=\int a\mathrm{d}x-a^3\int\frac{1}{a^2+x^2}\mathrm{d}x=ax-a^3\cdot\frac{1}{a}\cdot\arctan\frac{x}{a}+C_0$，故

$$\text{原式}=\frac{x^2}{2}\arctan\frac{x}{a}-\frac{ax}{2}+\frac{a^2}{2}\arctan\frac{x}{a}+C=\frac{x^2+a^2}{2}\arctan\frac{x}{a}-\frac{ax}{2}+C.$$

45. 解 原式 $=\frac{1}{3}\int\arctan\frac{x}{a}\mathrm{d}(x^3)=\frac{1}{3}x^3\arctan\frac{x}{a}-\frac{1}{3}\int x^3\mathrm{d}\left(\arctan\frac{x}{a}\right)=\frac{1}{3}x^3\arctan\frac{x}{a}-\frac{1}{3}\int x^3\cdot\frac{\frac{1}{a}}{1+\left(\frac{x}{a}\right)^2}\mathrm{d}x$

$$=\frac{1}{3}x^3\arctan\frac{x}{a}-\frac{1}{3}\int x^3\frac{a}{a^2+x^2}\mathrm{d}x=\frac{1}{3}x^3\arctan\frac{x}{a}-\frac{a}{3}\int\frac{x(x^2+a^2)-a^2x}{a^2+x^2}\mathrm{d}x$$

$$=\frac{1}{3}x^3\arctan\frac{x}{a}-\frac{a}{3}\int x\mathrm{d}x+\frac{a^3}{3}\int\frac{x}{a^2+x^2}\mathrm{d}x=\frac{1}{3}x^3\arctan\frac{x}{a}-\frac{a}{6}x^2+\frac{a^3}{6}\ln(a^2+x^2)+C.$$

46. 解 原式 $=-\int\arctan\frac{x}{a}\mathrm{d}\left(\frac{1}{x}\right)=-\frac{\arctan\frac{x}{a}}{x}+\int\frac{1}{x}\cdot\frac{\frac{1}{a}}{1+\left(\frac{x}{a}\right)^2}\mathrm{d}x=-\frac{\arctan\frac{x}{a}}{x}+\int\frac{1}{x}\cdot\frac{a}{a^2+x^2}\mathrm{d}x$

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$$=-\frac{\arctan\frac{x}{a}}{x}+\frac{1}{a}\int\left(\frac{1}{x}-\frac{x}{a^2+x^2}\right)\mathrm{d}x=-\frac{\arctan\frac{x}{a}}{x}+\frac{1}{a}\ln|x|-\frac{1}{2a}\ln(a^2+x^2)+C$$

$$=-\frac{1}{x}\arctan\frac{x}{a}-\frac{1}{a}\ln\frac{\sqrt{a^2+x^2}}{|x|}+C=-\frac{1}{x}\arctan\frac{x}{a}-\frac{1}{a}\ln\sqrt{\frac{a^2}{x^2}+1}+C$$

$$=-\frac{1}{x}\arctan\frac{x}{a}-\frac{1}{a}\ln\left|\frac{a}{x}\sqrt{1+\frac{x^2}{a^2}}\right|+C.$$

47. 解 $$\text{原式}=x\cdot\operatorname{arccot}\frac{x}{a}+\int x\cdot\frac{\frac{1}{a}}{1+\left(\frac{x}{a}\right)^2}\mathrm{d}x=x\cdot\operatorname{arccot}\frac{x}{a}+\int\frac{ax}{a^2+x^2}\mathrm{d}x=x\operatorname{arccot}\frac{x}{a}+\frac{a}{2}\int\frac{\mathrm{d}(x^2)}{a^2+x^2}$$

$$=x\operatorname{arccot}\frac{x}{a}+\frac{a}{2}\ln\left(a^2+x^2\right)+C.$$

48. 解 $$\text{原式}=\frac{1}{2}\int\operatorname{arccot}\frac{x}{a}\mathrm{d}(x^2)=\frac{x^2}{2}\operatorname{arccot}\frac{x}{a}-\frac{1}{2}\int x^2\mathrm{d}\left(\operatorname{arccot}\frac{x}{a}\right)=\frac{x^2}{2}\operatorname{arccot}\frac{x}{a}+\frac{1}{2}\int x^2\frac{\frac{1}{a}}{1+\left(\frac{x}{a}\right)^2}\mathrm{d}x$$

$$=\frac{x^2}{2}\operatorname{arccot}\frac{x}{a}+\frac{a}{2}\int\frac{x^2}{a^2+x^2}\mathrm{d}x=\frac{x^2}{2}\operatorname{arccot}\frac{x}{a}+\frac{a}{2}\int\frac{x^2+a^2-a^2}{a^2+x^2}\mathrm{d}x$$

$$=\frac{x^2}{2}\operatorname{arccot}\frac{x}{a}+\frac{ax}{2}-\frac{a^3}{2}\int\frac{\mathrm{d}x}{a^2+x^2}=\frac{x^2}{2}\operatorname{arccot}\frac{x}{a}+\frac{ax}{2}-\frac{a^3}{2}\cdot\frac{1}{a}\arctan\frac{x}{a}+C_1$$

$$=\frac{x^2}{2}\operatorname{arccot}\frac{x}{a}+\frac{ax}{2}-\frac{a^2}{2}\arctan\frac{x}{a}+C_1,$$

又 $$\arctan x+\operatorname{arccot}x=\frac{\pi}{2},$$

故 $$\text{原式}=\frac{x^2+a^2}{2}\operatorname{arccot}\frac{x}{a}+\frac{ax}{2}+C.$$

49. 解 $$\text{原式}=\frac{1}{3}\int\operatorname{arccot}\frac{x}{a}\mathrm{d}(x^3)=\frac{1}{3}x^3\operatorname{arccot}\frac{x}{a}-\frac{1}{3}\int x^3\mathrm{d}\left(\operatorname{arccot}\frac{x}{a}\right)=\frac{1}{3}x^3\operatorname{arccot}\frac{x}{a}+\frac{1}{3}\int x^3\frac{\frac{1}{a}}{1+\left(\frac{x}{a}\right)^2}\mathrm{d}x$$

$$=\frac{1}{3}x^3\operatorname{arccot}\frac{x}{a}+\frac{a}{3}\int\frac{x^3}{a^2+x^2}\mathrm{d}x=\frac{1}{3}x^3\operatorname{arccot}\frac{x}{a}+\frac{a}{3}\int\frac{x(x^2+a^2)-a^2x}{a^2+x^2}\mathrm{d}x$$

$$=\frac{1}{3}x^3\operatorname{arccot}\frac{x}{a}+\frac{a}{3}\int x\mathrm{d}x-\frac{a^3}{6}\int\frac{\mathrm{d}(x^2)}{a^2+x^2}=\frac{x^3}{3}\operatorname{arccot}\frac{x}{a}-\frac{a^3}{6}\ln(a^2+x^2)+\frac{ax^2}{6}+C.$$

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50. 解 $$\text{原式}=-\int \operatorname{arccot}\frac{x}{a}\mathrm{d}\left(\frac{1}{x}\right)=-\frac{1}{x}\operatorname{arccot}\frac{x}{a}+\int\frac{1}{x}\mathrm{d}\left(\operatorname{arccot}\frac{x}{a}\right)=-\frac{1}{x}\operatorname{arccot}\frac{x}{a}-\int\frac{1}{x}\cdot\frac{\frac{1}{a}}{1+\left(\frac{x}{a}\right)^2}\mathrm{d}x$$

$$=-\frac{1}{x}\operatorname{arccot}\frac{x}{a}-\int\frac{1}{x}\cdot\frac{a}{a^2+x^2}\mathrm{d}x=-\frac{1}{x}\operatorname{arccot}\frac{x}{a}-\frac{1}{a}\int\left(\frac{1}{x}-\frac{x}{a^2+x^2}\right)\mathrm{d}x$$

$$=-\frac{1}{x}\operatorname{arccot}\frac{x}{a}-\frac{1}{a}\ln|x|+\frac{1}{2a}\int\frac{\mathrm{d}(x^2)}{a^2+x^2}=-\frac{1}{x}\operatorname{arccot}\frac{x}{a}-\frac{1}{a}\ln|x|+\frac{1}{2a}\ln(a^2+x^2)+C$$

$$=-\frac{1}{x}\operatorname{arccot}\frac{x}{a}+\frac{1}{a}\ln\left(\frac{\sqrt{a^2+x^2}}{|x|}\right)+C=-\frac{1}{x}\operatorname{arccot}\frac{x}{a}+\frac{1}{a}\ln\left|\frac{a}{x}\sqrt{1+\frac{x^2}{a^2}}\right|+C.$$

第六组

1. 解 $$\int \mathrm{e}^{ax}\mathrm{d}x=\frac{1}{a}\int \mathrm{e}^{ax}\mathrm{d}(ax)=\frac{1}{a}\mathrm{e}^{ax}+C.$$

2. 解 $$\int x\mathrm{e}^{ax}\mathrm{d}x=\frac{1}{a}\int x\mathrm{d}(\mathrm{e}^{ax})=\frac{1}{a}\left(x\mathrm{e}^{ax}-\int \mathrm{e}^{ax}\mathrm{d}x\right)=\frac{x\mathrm{e}^{ax}}{a}-\frac{1}{a^2}\mathrm{e}^{ax}+C=\frac{(ax-1)\mathrm{e}^{ax}}{a^2}+C.$$

3. 解 $$\int xa^x\mathrm{d}x=\frac{1}{\ln a}\int x\mathrm{d}(a^x)=\frac{1}{\ln a}\left(xa^x-\int a^x\mathrm{d}x\right)=\frac{1}{\ln a}\left(xa^x-\frac{a^x}{\ln a}+C_1\right)=\frac{x}{\ln a}a^x-\frac{a^x}{(\ln a)^2}+C.$$

4. 解 $$\int\frac{1}{1+\mathrm{e}^x}\mathrm{d}x=\int\frac{\mathrm{e}^x}{\mathrm{e}^x(1+\mathrm{e}^x)}\mathrm{d}x=\int\frac{1+\mathrm{e}^x-\mathrm{e}^x}{(1+\mathrm{e}^x)\mathrm{e}^x}\mathrm{d}(\mathrm{e}^x)=\int\left(\frac{1}{\mathrm{e}^x}-\frac{1}{1+\mathrm{e}^x}\right)\mathrm{d}(\mathrm{e}^x)=\ln\mathrm{e}^x-\ln(1+\mathrm{e}^x)+C$$

$$=\ln\frac{\mathrm{e}^x}{1+\mathrm{e}^x}+C.$$

5. 解 $$\int\frac{\mathrm{d}x}{a+b\mathrm{e}^{px}}=\int\frac{b\mathrm{e}^{px}\mathrm{d}x}{b\mathrm{e}^{px}(a+b\mathrm{e}^{px})}=\frac{1}{p}\int\frac{\mathrm{d}(b\mathrm{e}^{px})}{b\mathrm{e}^{px}(a+b\mathrm{e}^{px})}=\frac{1}{ap}\int\left(\frac{1}{b\mathrm{e}^{px}}-\frac{1}{a+b\mathrm{e}^{px}}\right)\mathrm{d}(b\mathrm{e}^{px})$$

$$=\frac{1}{ap}\ln b\mathrm{e}^{px}-\frac{1}{ap}\ln(a+b\mathrm{e}^{px})+C_1=\frac{1}{ap}(\ln b+\ln\mathrm{e}^{px})-\frac{1}{ap}\ln(a+b\mathrm{e}^{px})+C_1$$

$$=\frac{x}{a}-\frac{1}{ap}\ln(a+b\mathrm{e}^{px})+C.$$

6. 解 令 $\sqrt{a+b\mathrm{e}^{\beta x}}=t$，则 $x=\dfrac{\ln\left(\dfrac{t^2-a}{b}\right)}{\beta}$，$\displaystyle\int\frac{\mathrm{d}x}{\sqrt{a+b\mathrm{e}^{\beta x}}}=\int\frac{1}{t}\mathrm{d}\left[\frac{\ln\left(\dfrac{t^2-a}{b}\right)}{\beta}\right]=\frac{2}{\beta}\int\frac{1}{t^2-a}\mathrm{d}t$.

①当 $a>0$，$b>0$ 时，

$$\text{原式}=\frac{2}{\beta}\int\frac{1}{t^2-(\sqrt{a})^2}\mathrm{d}t=\frac{1}{\beta\sqrt{a}}\ln\frac{\sqrt{a+b\mathrm{e}^{\beta x}}-\sqrt{a}}{\sqrt{a+b\mathrm{e}^{\beta x}}+\sqrt{a}}+C.$$

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②当 $a<0$，$b>0$ 时，

$$原式=\frac{2}{\beta}\int\frac{1}{t^2-a}\mathrm{d}t=\frac{2}{\beta}\int\frac{1}{t^2+(\sqrt{-a})^2}\mathrm{d}t=\frac{2}{\beta}\cdot\frac{1}{\sqrt{-a}}\arctan\frac{t}{\sqrt{-a}}+C$$

$$=\frac{2}{\beta\sqrt{-a}}\arctan\frac{\sqrt{a+b\mathrm{e}^{\beta x}}}{\sqrt{-a}}+C.$$

7. 解 $$\int\frac{\mathrm{d}x}{a\mathrm{e}^{mx}+b\mathrm{e}^{-mx}}=\int\frac{\mathrm{e}^{mx}}{b+a(\mathrm{e}^{mx})^2}\mathrm{d}x=\frac{1}{m}\int\frac{\mathrm{d}(\mathrm{e}^{mx})}{b+a(\mathrm{e}^{mx})^2}=\frac{1}{am}\int\frac{\mathrm{d}(\mathrm{e}^{mx})}{\frac{b}{a}+(\mathrm{e}^{mx})^2}$$

$$=\frac{1}{am}\cdot\sqrt{\frac{a}{b}}\cdot\arctan\frac{\mathrm{e}^{mx}}{\sqrt{\frac{b}{a}}}+C=\frac{1}{m\sqrt{ab}}\arctan\left(\mathrm{e}^{mx}\sqrt{\frac{a}{b}}\right)+C.$$

8. 解 $$\int(a^x-a^{-x})\mathrm{d}x=\int a^x\mathrm{d}x-\int a^{-x}\mathrm{d}x=\frac{a^x}{\ln a}+\int a^{-x}\mathrm{d}(-x)=\frac{a^x}{\ln a}+\frac{a^{-x}}{\ln a}+C=\frac{a^x+a^{-x}}{\ln a}+C.$$

9. 解 $$\int a^{px}\mathrm{d}x=\frac{1}{p}\int a^{px}\mathrm{d}(px)=\frac{1}{p}\cdot\frac{a^{px}}{\ln a}+C=\frac{a^{px}}{p\ln a}+C.$$

第七组

1. 解 $$\int\ln x\mathrm{d}x=x\ln x-\int x\mathrm{d}(\ln x)=x\ln x-x+C.$$

2. 解 $$\int\frac{\mathrm{d}x}{x\ln x}=\int\frac{\mathrm{d}(\ln x)}{\ln x}=\ln|\ln x|+C.$$

3. 解 $$\int x^n\ln x\mathrm{d}x=\frac{1}{n+1}\int\ln x\mathrm{d}(x^{n+1})=\frac{1}{n+1}x^{n+1}\ln x-\frac{1}{n+1}\int x^n\mathrm{d}x$$

$$=\frac{1}{n+1}x^{n+1}\ln x-\frac{1}{(n+1)^2}x^{n+1}+C=\frac{1}{n+1}x^{n+1}\left(\ln x-\frac{1}{n+1}\right)+C.$$

4. 解 令 $\ln x=t$，$x=\mathrm{e}^t$，于是

$$原式=\int t^2\mathrm{d}(\mathrm{e}^t)=t^2\mathrm{e}^t-\int\mathrm{e}^t\mathrm{d}(t^2)=t^2\mathrm{e}^t-2\int t\mathrm{e}^t\mathrm{d}t=t^2\mathrm{e}^t-2\int t\mathrm{d}(\mathrm{e}^t)=t^2\mathrm{e}^t-2t\mathrm{e}^t+2\int\mathrm{e}^t\mathrm{d}t$$

$$=t^2\mathrm{e}^t-2t\mathrm{e}^t+2\mathrm{e}^t+C=x(\ln x)^2-2x\ln x+2x+C.$$

5. 解 $$\int x\ln x\mathrm{d}x=\frac{1}{2}\int\ln x\mathrm{d}(x^2)=\frac{x^2}{2}\ln x-\frac{1}{2}\int x\mathrm{d}x=\frac{x^2}{2}\ln x-\frac{x^2}{4}+C.$$

6. 解 $$\int x^2\ln x\mathrm{d}x=\frac{1}{3}\int\ln x\mathrm{d}(x^3)=\frac{x^3}{3}\ln x-\frac{1}{3}\int x^2\mathrm{d}x=\frac{x^3}{3}\ln x-\frac{x^3}{9}+C.$$

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7. 解 $\int\frac{\ln x}{(ax+b)^2}\mathrm{d}x=\frac{1}{a}\int\frac{\ln x}{(ax+b)^2}\mathrm{d}(ax+b)=-\frac{1}{a}\int\ln x\mathrm{d}\left(\frac{1}{ax+b}\right)=-\frac{\ln x}{a(ax+b)}+\frac{1}{a}\int\frac{1}{(ax+b)x}\mathrm{d}x$

$=-\frac{\ln x}{a(ax+b)}+\frac{1}{a}\int\frac{-\frac{a}{b}}{ax+b}\mathrm{d}x+\frac{1}{a}\int\frac{\frac{1}{b}}{x}\mathrm{d}x=-\frac{\ln x}{a(ax+b)}-\frac{1}{b}\int\frac{1}{ax+b}\mathrm{d}x+\frac{1}{ab}\int\frac{1}{x}\mathrm{d}x$

$=-\frac{\ln x}{a(ax+b)}-\frac{1}{ab}\int\frac{1}{ax+b}\mathrm{d}(ax+b)+\frac{\ln x}{ab}=-\frac{\ln x}{a(ax+b)}+\frac{1}{ab}\ln\left|\frac{x}{ax+b}\right|+C$.

8. 解 用分部积分法.

$$\text{原式}=x\ln\frac{x+a}{x-a}-\int x\cdot\frac{x-a}{x+a}\cdot\frac{-2a}{(x-a)^2}\mathrm{d}x=x\ln\frac{x+a}{x-a}+a\int\frac{\mathrm{d}(x^2)}{x^2-a^2}$$

$$=x\ln\frac{x+a}{x-a}+a\ln|x^2-a^2|+C$$

$$=x\ln|x+a|-x\ln|x-a|+a\ln|x+a|+a\ln|x-a|+C$$

$$=(x+a)\ln|x+a|-(x-a)\ln|x-a|+C.$$

9. 解 用分部积分法.

$$\text{原式}=-\int\ln\frac{x+a}{x-a}\mathrm{d}\left(\frac{1}{x}\right)=-\frac{1}{x}\ln\frac{x+a}{x-a}+\int\frac{1}{x}\cdot\frac{x-a}{x+a}\cdot\frac{-2a}{(x-a)^2}\mathrm{d}x$$

$$=-\frac{1}{x}\ln\frac{x+a}{x-a}+\int\frac{-2a}{x(x+a)(x-a)}\mathrm{d}x$$

$$=-\frac{1}{x}\ln\frac{x+a}{x-a}-2a\left[\int\left(-\frac{1}{a^2}\cdot\frac{1}{x}\right)\mathrm{d}x+\frac{1}{2a^2}\int\frac{1}{x+a}\mathrm{d}x+\frac{1}{2a^2}\int\frac{1}{x-a}\mathrm{d}x\right]$$

$$=-\frac{1}{x}\ln\frac{x+a}{x-a}-2a\left(-\frac{1}{a^2}\ln|x|+\frac{1}{2a^2}\ln|x+a|+\frac{1}{2a^2}\ln|x-a|+C_1\right)$$

$$=-\frac{1}{x}\ln\frac{x+a}{x-a}+\frac{2}{a}\ln|x|-\frac{1}{a}\ln|x+a|-\frac{1}{a}\ln|x-a|+C$$

$$=-\frac{1}{x}\ln\frac{x+a}{x-a}-\frac{1}{a}\left(\ln|x+a|+\ln|x-a|-2\ln|x|\right)+C$$

$$=\frac{1}{x}\ln\frac{x-a}{x+a}-\frac{1}{a}\ln\frac{x^2-a^2}{x^2}+C.$$

10. 解 $\int\ln(x^2+a^2)\mathrm{d}x=x\ln(x^2+a^2)-2\int\frac{x^2+a^2-a^2}{x^2+a^2}\mathrm{d}x$

$=x\ln(x^2+a^2)-2x+2a^2\int\frac{1}{x^2+a^2}\mathrm{d}x=x\ln(x^2+a^2)-2x+2a\arctan\frac{x}{a}+C$.

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11. 解 $\int x\ln(x^2+a^2)\,dx=\frac{1}{2}\int\ln(x^2+a^2)\,d(x^2)=\frac{1}{2}x^2\ln(x^2+a^2)-\frac{1}{2}\int\frac{x^2\cdot 2x}{x^2+a^2}\,dx$

$=\frac{1}{2}x^2\ln(x^2+a^2)-\frac{1}{2}\int\frac{x^2}{x^2+a^2}\,d(x^2)=\frac{1}{2}x^2\ln(x^2+a^2)-\frac{1}{2}\int\frac{x^2+a^2-a^2}{x^2+a^2}\,d(x^2)$

$=\frac{1}{2}x^2\ln(x^2+a^2)-\frac{1}{2}x^2+\frac{a^2}{2}\int\frac{1}{x^2+a^2}\,d(x^2)$

$=\frac{1}{2}x^2\ln(x^2+a^2)-\frac{x^2}{2}+\frac{a^2}{2}\ln(x^2+a^2)+C$

$=\frac{1}{2}(x^2+a^2)\ln(x^2+a^2)-\frac{1}{2}x^2+C$.

12. 解 $\int x^2\ln(x^2+a^2)\,dx=\frac{1}{3}\int\ln(x^2+a^2)\,d(x^3)=\frac{1}{3}x^3\ln(x^2+a^2)-\frac{1}{3}\int\frac{x^3\cdot 2x}{x^2+a^2}\,dx$

$=\frac{1}{3}x^3\ln(x^2+a^2)-\frac{2}{3}\int\frac{x^4-a^4+a^4}{x^2+a^2}\,dx$

$=\frac{1}{3}x^3\ln(x^2+a^2)-\frac{2}{3}\int\frac{(x^2-a^2)(x^2+a^2)}{x^2+a^2}\,dx-\frac{2}{3}a^4\int\frac{1}{x^2+a^2}\,dx$

$=\frac{1}{3}x^3\ln(x^2+a^2)-\frac{2}{3}\int(x^2-a^2)\,dx-\frac{2}{3}a^3\arctan\frac{x}{a}$

$=\frac{1}{3}x^3\ln(x^2+a^2)-\frac{2}{9}x^3+\frac{2}{3}a^2x-\frac{2}{3}a^3\arctan\frac{x}{a}+C$

$=\frac{1}{3}\left[x^3\ln(x^2+a^2)-\frac{2}{3}x^3+2a^2x-2a^3\arctan\frac{x}{a}\right]+C$.

13. 解 $\int x^{2n}\ln(x^2+a^2)\,dx=\frac{1}{2n+1}\int\ln(x^2+a^2)\,d(x^{2n+1})=\frac{1}{2n+1}x^{2n+1}\ln(x^2+a^2)-\frac{2}{2n+1}\int\frac{x^{2n+2}}{x^2+a^2}\,dx$

$=\frac{1}{2n+1}x^{2n+1}\ln(x^2+a^2)-\frac{2}{2n+1}\int\left[x^{2n}-a^2x^{2n-2}+a^4x^{2n-4}+\cdots+(-1)^{n-1}a^{2n}+(-1)^n\frac{a^{2n+2}}{x^2+a^2}\right]dx$

$=\frac{1}{2n+1}\left[x^{2n+1}\ln(x^2+a^2)+(-1)^n2a^{2n+1}\arctan\frac{x}{a}-2\sum_{k=0}^{n}\frac{(-1)^{n-k}}{2k+1}a^{2n-2k}x^{2k+1}\right]+C$.

14. 解 $\int\ln(x^2-a^2)\,dx=x\ln(x^2-a^2)-2\int\frac{x^2-a^2+a^2}{x^2-a^2}\,dx$

$=x\ln(x^2-a^2)-2x-2a^2\int\frac{1}{x^2-a^2}\,dx=x\ln(x^2-a^2)-2x+a\ln\left|\frac{x+a}{x-a}\right|+C$.

15. 解 当$|x|>a$时，$\int\ln|x^2-a^2|\,dx=\int\ln(x^2-a^2)\,dx$（由14式）$=x\ln(x^2-a^2)-2x+a\ln\left|\frac{x+a}{x-a}\right|+C$；

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当$|x|<a$时，$\int \ln|x^2-a^2|\,\mathrm{d}x=\int \ln(a^2-x^2)\,\mathrm{d}x=x\ln(a^2-x^2)+2\int \frac{x^2}{a^2-x^2}\,\mathrm{d}x$

$$=x\ln(a^2-x^2)-2\int \frac{a^2-x^2-a^2}{a^2-x^2}\,\mathrm{d}x=x\ln(a^2-x^2)-2x+2a^2\int \frac{1}{a^2-x^2}\,\mathrm{d}x$$

$$=x\ln(a^2-x^2)-2x+a\ln\left|\frac{a+x}{a-x}\right|+C.$$

故$\int \ln|x^2-a^2|\,\mathrm{d}x=x\ln|x^2-a^2|-2x+a\ln\left|\frac{x+a}{x-a}\right|+C.$

16. 解　当$|x|>a$时，

$$\int x\ln(x^2-a^2)\,\mathrm{d}x=\frac{1}{2}\int \ln(x^2-a^2)\mathrm{d}(x^2)=\frac{1}{2}x^2\ln(x^2-a^2)-\int \frac{x^3}{x^2-a^2}\,\mathrm{d}x$$

$$=\frac{1}{2}x^2\ln(x^2-a^2)-\frac{1}{2}\int \frac{x^2-a^2+a^2}{x^2-a^2}\,\mathrm{d}(x^2)$$

$$=\frac{1}{2}x^2\ln(x^2-a^2)-\frac{1}{2}x^2-\frac{a^2}{2}\int \frac{1}{x^2-a^2}\,\mathrm{d}(x^2)$$

$$=\frac{1}{2}x^2\ln(x^2-a^2)-\frac{1}{2}x^2-\frac{a^2}{2}\ln|x^2-a^2|+C;$$

当$|x|<a$时，

$$\int x\ln(a^2-x^2)\,\mathrm{d}x=\frac{1}{2}\int \ln(a^2-x^2)\,\mathrm{d}(x^2)$$

$$=\frac{1}{2}x^2\ln(a^2-x^2)+\int \frac{x^2\cdot x}{a^2-x^2}\,\mathrm{d}x$$

$$=\frac{1}{2}x^2\ln(a^2-x^2)-\frac{1}{2}\int \frac{a^2-x^2-a^2}{a^2-x^2}\,\mathrm{d}(x^2)$$

$$=\frac{1}{2}x^2\ln(a^2-x^2)-\frac{1}{2}x^2-\frac{a^2}{2}\int \frac{1}{a^2-x^2}\,\mathrm{d}(-x^2)$$

$$=\frac{1}{2}x^2\ln(a^2-x^2)-\frac{1}{2}x^2-\frac{a^2}{2}\ln|x^2-a^2|+C.$$

故　$$\int x\ln|x^2-a^2|\,\mathrm{d}x=\frac{1}{2}\left[(x^2-a^2)\ln|x^2-a^2|-x^2\right]+C.$$

17. 解　当$|x|>a$时，

$$\int x^2\ln(x^2-a^2)\,\mathrm{d}x=\frac{1}{3}\int \ln(x^2-a^2)\,\mathrm{d}(x^3)=\frac{1}{3}x^3\ln(x^2-a^2)-\frac{2}{3}\int \frac{x^4}{x^2-a^2}\,\mathrm{d}x$$

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$$=\frac{1}{3}x^3\ln(x^2-a^2)-\frac{2}{3}\int\left(\frac{x^4-a^4}{x^2-a^2}+\frac{a^4}{x^2-a^2}\right)\mathrm{d}x$$

$$=\frac{1}{3}x^3\ln(x^2-a^2)-\frac{2}{3}\int(x^2+a^2)\,\mathrm{d}x-\frac{2}{3}\int\frac{a^4}{x^2-a^2}\,\mathrm{d}x$$

$$=\frac{1}{3}x^3\ln(x^2-a^2)-\frac{2}{9}x^3-\frac{2a^2}{3}x+\frac{a^3}{3}\ln\left|\frac{x+a}{x-a}\right|+C;$$

当$|x|<a$时，

$$\int x^2\ln(a^2-x^2)\,\mathrm{d}x=\frac{1}{3}\int\ln(a^2-x^2)\,\mathrm{d}(x^3)=\frac{1}{3}x^3\ln(a^2-x^2)+\frac{2}{3}\int\frac{x^3\cdot x}{a^2-x^2}\,\mathrm{d}x$$

$$=\frac{1}{3}x^3\ln(a^2-x^2)-\frac{2}{3}\int\left(\frac{a^4-x^4}{a^2-x^2}-\frac{a^4}{a^2-x^2}\right)\mathrm{d}x$$

$$=\frac{1}{3}x^3\ln(a^2-x^2)-\frac{2}{3}\int(a^2+x^2)\,\mathrm{d}x+\frac{2}{3}a^4\int\frac{1}{a^2-x^2}\,\mathrm{d}x$$

$$=\frac{1}{3}x^3\ln(a^2-x^2)-\frac{2}{3}a^2x-\frac{2}{9}x^3+\frac{a^3}{3}\ln\left|\frac{x+a}{x-a}\right|+C.$$

故 $$\int x^2\ln|x^2-a^2|\,\mathrm{d}x=\frac{1}{3}\left(x^3\ln|x^2-a^2|-\frac{2}{3}x^3-2a^2x+a^3\ln\left|\frac{x+a}{x-a}\right|\right)+C.$$

18. 解 令$\ln x=t$，$x=\mathrm{e}^t$，于是

$$\text{原式}=\int\sin t\mathrm{d}(\mathrm{e}^t)=\int\sin t\cdot\mathrm{e}^t\mathrm{d}t$$

$$=\frac{\begin{vmatrix}(\mathrm{e}^t)' & (\sin t)'\\ \mathrm{e}^t & \sin t\end{vmatrix}}{1^2+1^2}+C=\frac{\mathrm{e}^t\sin t-\mathrm{e}^t\cos t}{2}+C$$

$$=\frac{x}{2}[\sin(\ln x)-\cos(\ln x)]+C.$$

19. 解 令$\ln x=t$，$x=\mathrm{e}^t$，于是

$$\text{原式}=\int\cos t\mathrm{d}(\mathrm{e}^t)=\int\cos t\cdot\mathrm{e}^t\mathrm{d}t$$

$$=\frac{\begin{vmatrix}(\mathrm{e}^t)' & (\cos t)'\\ \mathrm{e}^t & \cos t\end{vmatrix}}{1^2+1^2}+C=\frac{\mathrm{e}^t\cos t+\mathrm{e}^t\sin t}{2}+C$$

$$=\frac{x}{2}[\cos(\ln x)+\sin(\ln x)]+C.$$

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20. 解　令 $b\ln x=t$，$\ln x=\dfrac{t}{b}$，$x=\mathrm{e}^{\frac{t}{b}}$，于是

$$\int x^{p}\cos(b\ln x)\,\mathrm{d}x=\int \mathrm{e}^{\frac{p}{b}t}\cos t\,\mathrm{d}(\mathrm{e}^{\frac{t}{b}})=\frac{1}{b}\int \mathrm{e}^{\frac{p+1}{b}t}\cos t\,\mathrm{d}t$$

$$=\frac{1}{b}\cdot\frac{1}{\left(\dfrac{p+1}{b}\right)^{2}+1^{2}}\begin{vmatrix}\left(\mathrm{e}^{\frac{p+1}{b}t}\right)' & (\cos t)'\\ \mathrm{e}^{\frac{p+1}{b}t} & \cos t\end{vmatrix}+C$$

$$=\frac{b}{(p+1)^{2}+b^{2}}\left(\frac{p+1}{b}\cdot\mathrm{e}^{\frac{p+1}{b}t}\cdot\cos t+\mathrm{e}^{\frac{p+1}{b}t}\cdot\sin t\right)+C$$

$$=\frac{x^{p+1}}{(p+1)^{2}+b^{2}}\left[(p+1)\cos(b\ln x)+b\sin(b\ln x)\right]+C.$$

21. 解　令 $b\ln x=t$，$\ln x=\dfrac{t}{b}$，$x=\mathrm{e}^{\frac{t}{b}}$，于是

$$\int x^{p}\sin(b\ln x)\,\mathrm{d}x=\int \mathrm{e}^{\frac{p}{b}t}\sin t\,\mathrm{d}(\mathrm{e}^{\frac{t}{b}})=\frac{1}{b}\int \mathrm{e}^{\frac{p+1}{b}t}\sin t\,\mathrm{d}t$$

$$=\frac{1}{b}\cdot\frac{1}{\left(\dfrac{p+1}{b}\right)^{2}+1^{2}}\begin{vmatrix}\left(\mathrm{e}^{\frac{p+1}{b}t}\right)' & (\sin t)'\\ \mathrm{e}^{\frac{p+1}{b}t} & \sin t\end{vmatrix}+C$$

$$=\frac{b}{(p+1)^{2}+b^{2}}\left(\frac{p+1}{b}\cdot\mathrm{e}^{\frac{p+1}{b}t}\cdot\sin t-\mathrm{e}^{\frac{p+1}{b}t}\cdot\cos t\right)+C$$

$$=\frac{x^{p+1}}{(p+1)^{2}+b^{2}}\left[(p+1)\sin(b\ln x)-b\cos(b\ln x)\right]+C.$$

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